

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
C	0000734528	PRODUCTION RELEASED	2009-06-04

K24 MLB SCHEMATIC

6/12/2009

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48	WELLSPRING 1	YUAN_MA	04/22/2008
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54	AUDIO: HEADPHONE FILTER	AUDIO	02/03/2009
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60	5V/3.3V SUPPLY	RAYMOND	02/08/2008
61	1.5V/0.75V DDR3 SUPPLY	RAYMOND	01/31/2008
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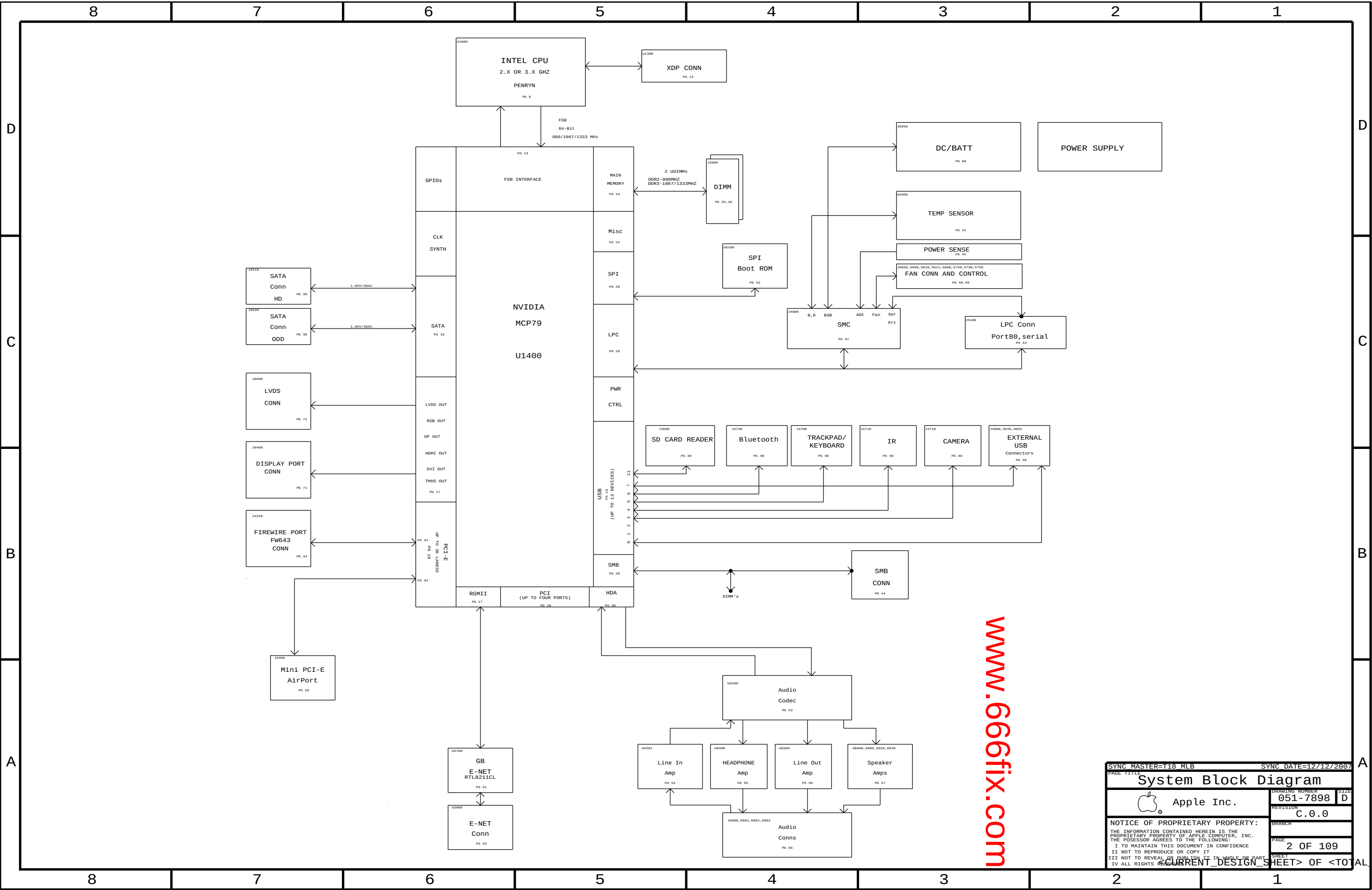
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Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-7898	1	SCHEM, MLB, K24	SCH	CRITICAL	
820-2530	1	PCBF, MLB, K24	PCB	CRITICAL	

DRAWING TITLE		SCHEM, MLB, K24	
	Apple Inc.		DRAWING NUMBER 051-7898
			SIZE D
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SYNC MASTER=T18 MLB		SYNC DATE=12/12/2007	
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System Block Diagram			
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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
630-9923	PCBA, MLB, BETTER, K24	K24_COMMON, CPU_2_26GHZ, EEE_6GC, KB_BL
630-9924	PCBA, MLB, BEST, K24	K24_COMMON, CPU_2_53GHZ, EEE_6GD, KB_BL

Bar Code Labels / EEE #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:6G4]	CRITICAL	EEE_6G4
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:6GC]	CRITICAL	EEE_6GC
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEE:6GD]	CRITICAL	EEE_6GD

BOM Groups

BOM GROUP	BOM OPTIONS
K24_COMMON	COMMON, ALTERNATE, K24_MCP, K24_MISC, K24_DEBUG_PROD, K24_PROGPARTS
K24_MCP	MCP_B03, BOOT_MODE_USER, MCPSEQ_SMC
K24_MISC	ONEWIRE_PU, DP_ESD, MIKEY, BKLT_PROD, SUPERCAP_NO, LDO_NO
K24_PROGPARTS	BOOTROM_PROD, SMC_PROD, IR_PROD, WELLSPRING_PROD
K24_DEBUG_ENG	DEVEL_BOM, SMC_DEBUG_YES, XDP
K24_DEBUG_PVT	DEVEL_BOM, BMON_PROD, SMC_DEBUG_YES, XDP, NO_VREFMRGN
K24_DEBUG_PROD	BMON_PROD, SMC_DEBUG_YES, XDP, LPCPLUS_NOT, NO_VREFMRGN
K24_DEVEL_ENG	BMON_ENG, XDP_CONN, LPCPLUS, VREFMRGN, FWPHY_WAKE_YES
K24_DEVEL_PVT	LPCPLUS

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S3646	1	PDC, SLGBE, PRQ, 2.0, 25W, 1066, M0, 3M, BGA	U1000	CRITICAL	CPU_2_0GHZ
337S3704	1	PDC, SLGE2, PRQ, 2.26, 25W, 1066, R0, 3M, BGA	U1000	CRITICAL	CPU_2_26GHZ
337S3039	1	PDC, SLBAN, PRQ, 2.4, 25W, 1066, M0, 3M, BGA	U1000	CRITICAL	CPU_2_4GHZ
337S3756	1	PDC, SLGF6, PRQ, 2.53, 25W, 1066, R0, 3M, BGA	U1000	CRITICAL	CPU_2_53GHZ
337S3761	1	PDC, SLGLA, PRQ, 2.66, 25W, 1066, E0, 3M, BGA	U1000	CRITICAL	CPU_2_66GHZ
338S0710	1	IC, GMCP, MCP79, 35X35MM, BGA1437, B03	U1400	CRITICAL	MCP_B03

Programmable Parts

338S0563	1	IC, SMC, HSB/2117, 9X9MM, TLP, HF	U4900	CRITICAL	SMC_BLANK
341S2445	1	IC, SMC, K24	U4900	CRITICAL	SMC_PROD
335S0610	1	IC, FLASH, SPI, 32MBIT, 3.3V, 86MHZ, 8-SOP	U6100	CRITICAL	BOOTROM_BLANK
341S2441	1	IC, PRGRM, EFI BOOTROM, UNLOCK, K24	U6100	CRITICAL	BOOTROM_PROD
338S0375	1	IC, CY7C63833, ENCORE II, USB CONTROLLER	U4800	CRITICAL	IR_BLANK
341S2093	1	IC, IR CONTROLLER, M97	U4800	CRITICAL	IR_PROD
337S2983	1	IC, P50C+ W/ USB, 56 PIN, MLP, CY9C24794	U5701	CRITICAL	WELLSPRING_BLANK
341S2503	1	IC, PRGRM, WELLSPRING CONTROLLER	U5701	CRITICAL	WELLSPRING_PROD

LOCKED BOOTROM APN IS 341S2443

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S0778	152S0693		ALL	CYNTEC AS ALTERNATE
152S0796	152S0685		ALL	CYNTEC AS ALTERNATE
157S0058	157S0055		ALL	DELTA AS ALTERNATE
104S0018	104S0023		ALL	DALE/VISHAY AS ALTERNATE
128S0093	128S0218		ALL	KEMET AS ALTERNATE
152S0874	152S0516		ALL	MAGLAYERS AS ALTERNATE
152S0847	152S0586		ALL	MAGLAYERS AS ALTERNATE
152S1025	152S1024		ALL	TOKO AS ALTERNATE
337S3769	337S3704		ALL	INTEL P7550 CPU AS ALTERNATE
353S2718	353S2310		ALL	INTERISL AS ALTERNATE

DEVELOPMENT BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
085-0741	1	K24 MLB DEVELOPMENT BOM	DEVEL	CRITICAL	DEVEL_BOM

K24 BOARD STACK-UP

Top

2

3

4

5

6

7

8

9

10

11

BOTTOM

SIGNAL

GROUND

SIGNAL(High Speed)

SIGNAL(High Speed)

GROUND

POWER

POWER

GROUND

SIGNAL(High Speed)

SIGNAL(High Speed)

GROUND

SIGNAL

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SYNC MASTER=M97 MLB

BOM Configuration

Apple Inc.

051-7898

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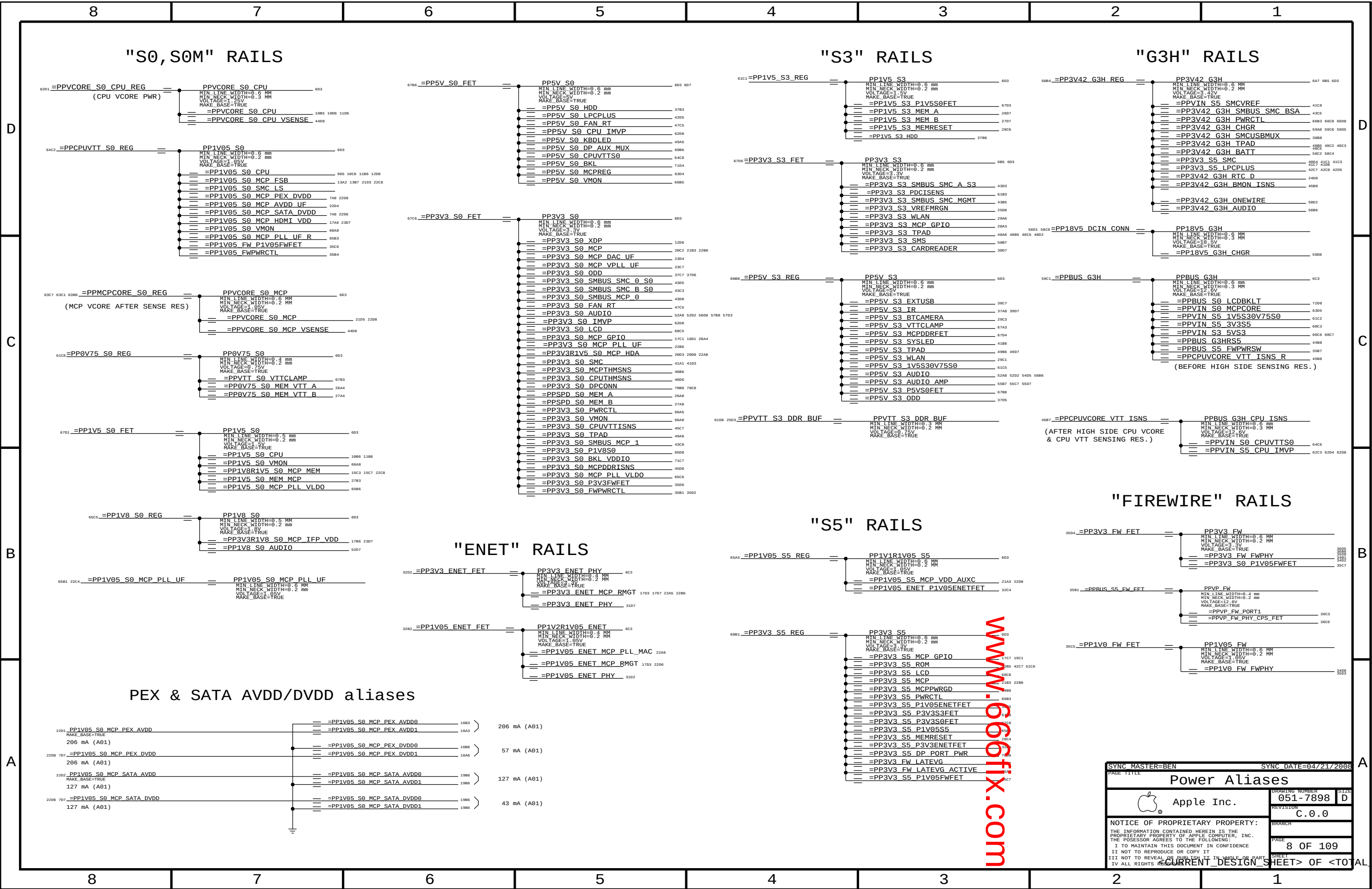
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Revision History	
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SYNC DATE=04/21/2008

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Power Aliases

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DRAWING NUMBER
051-7898

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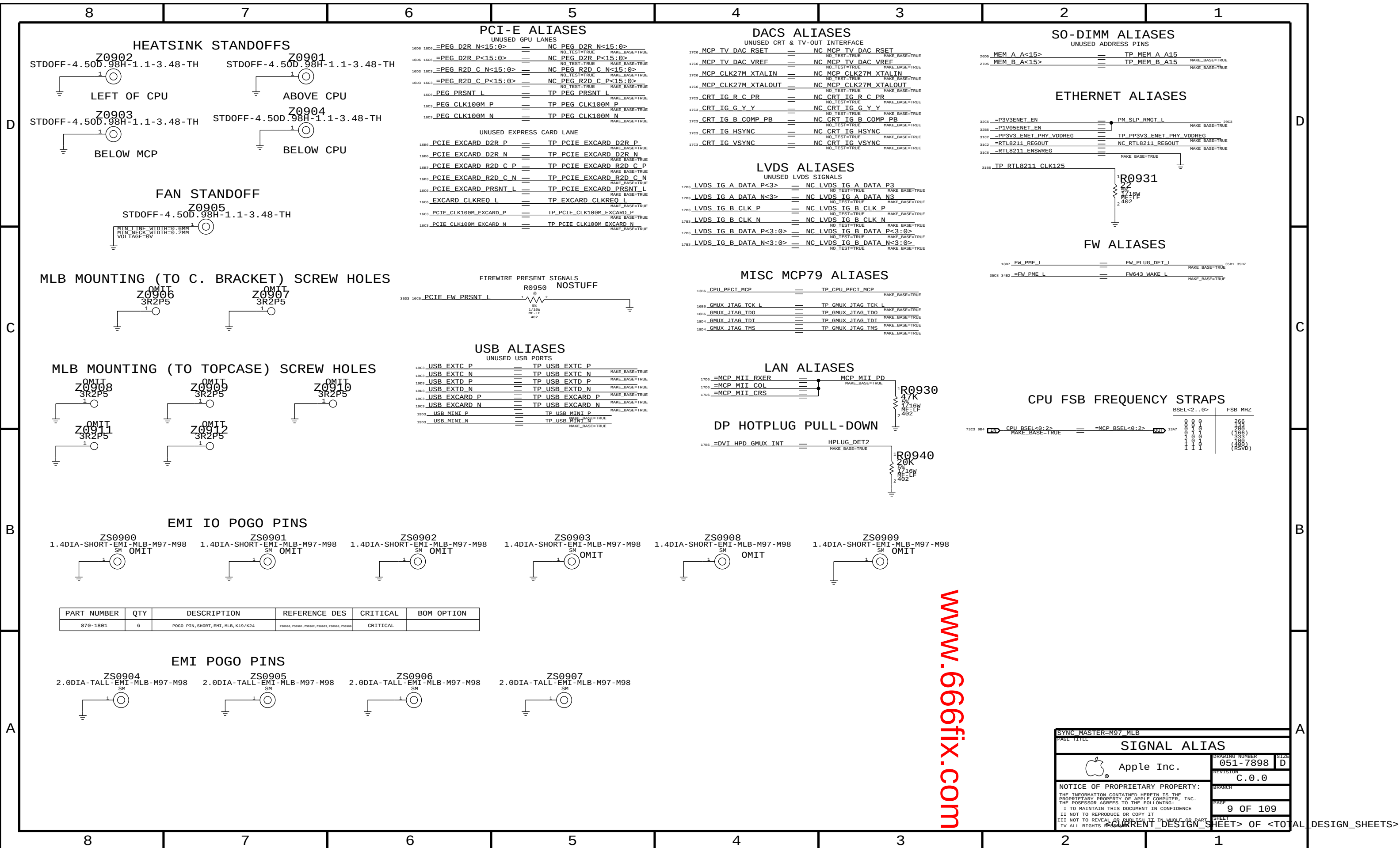
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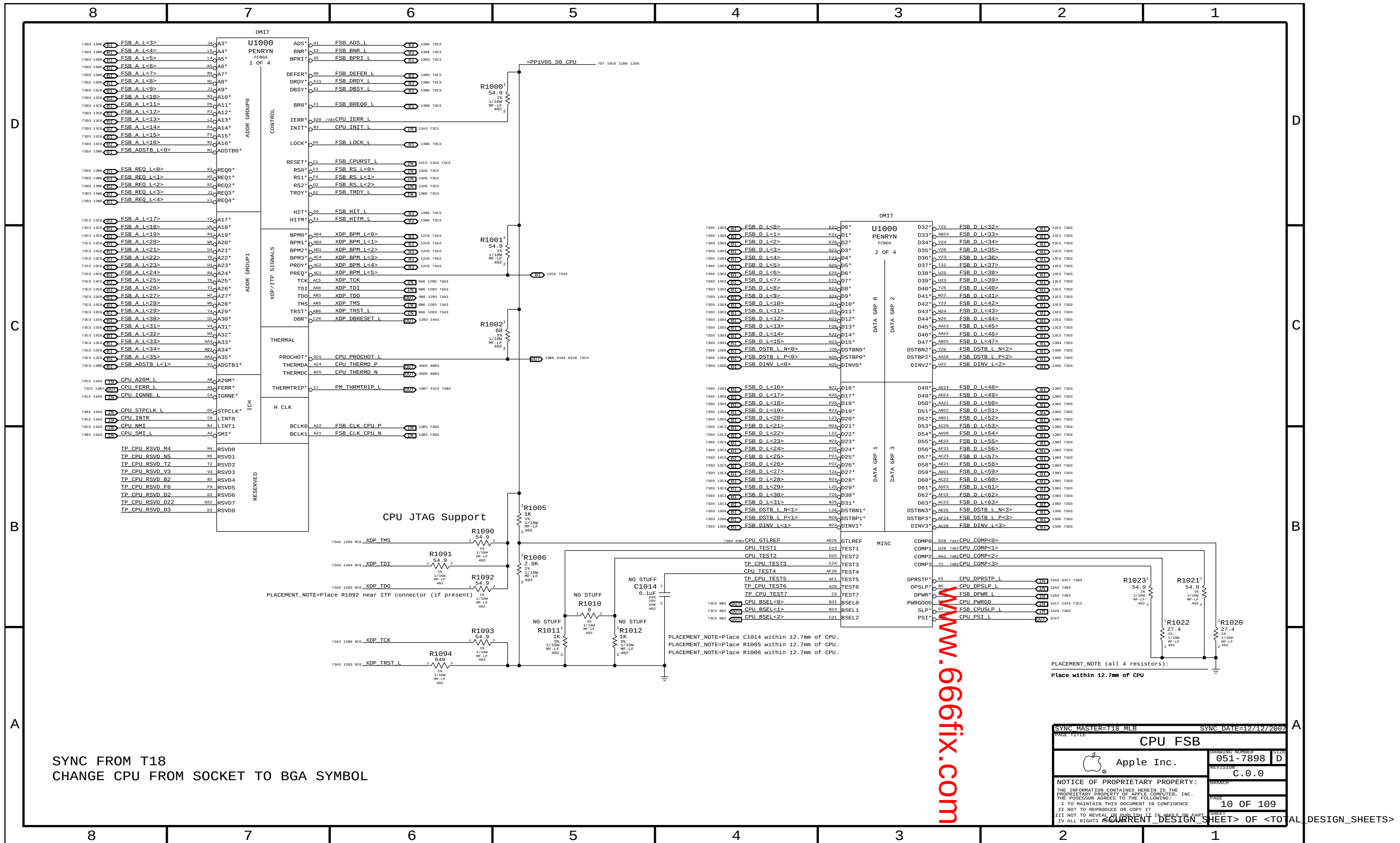
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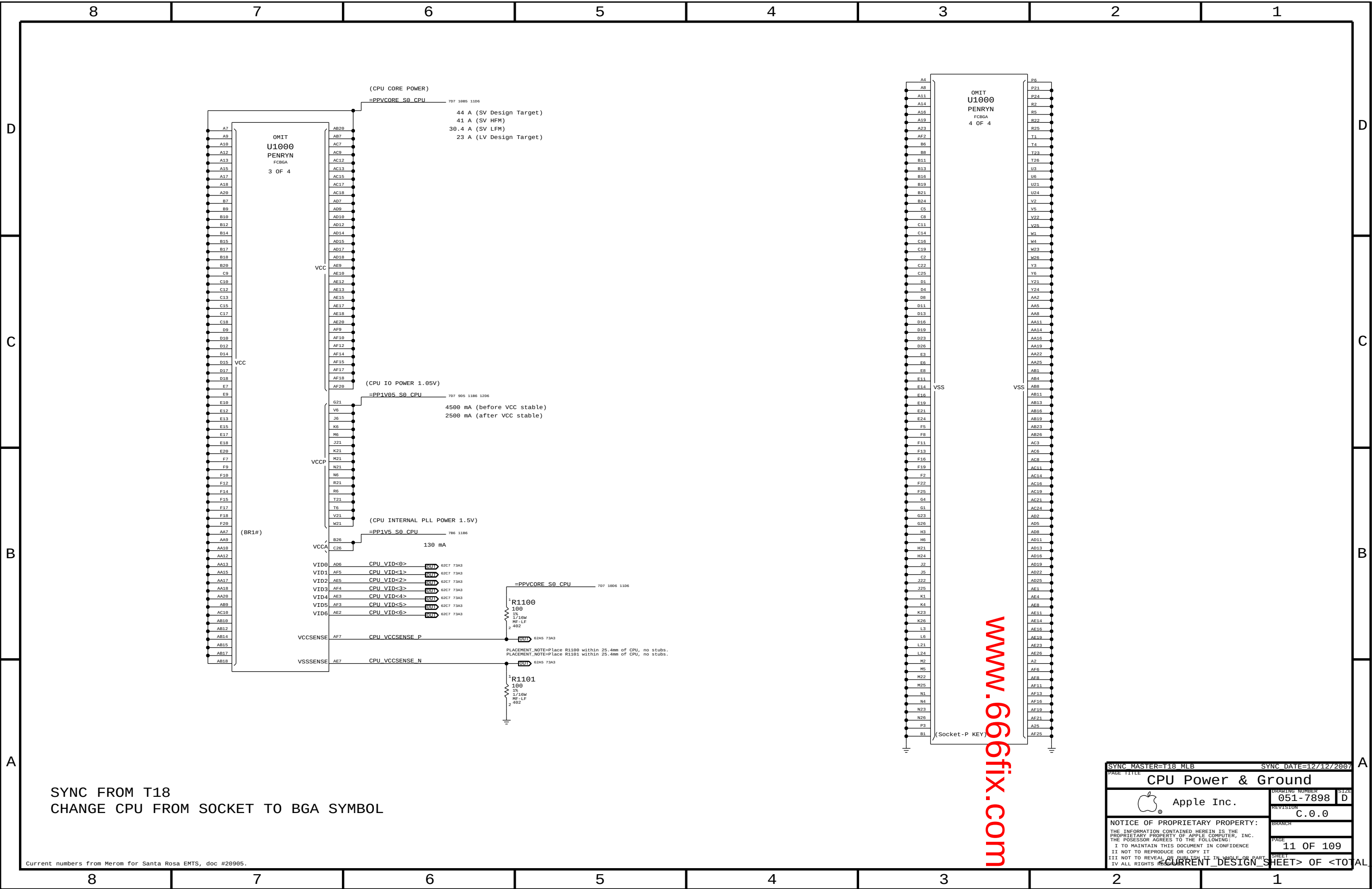
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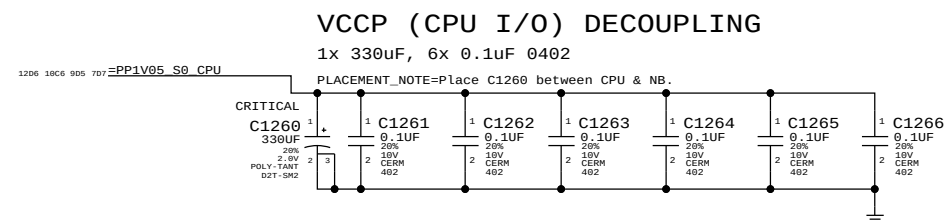
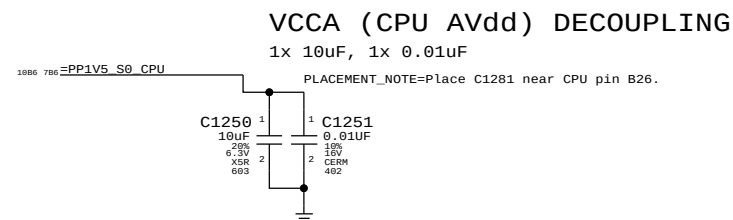
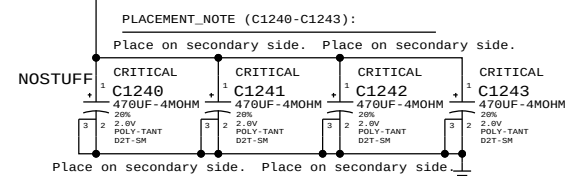
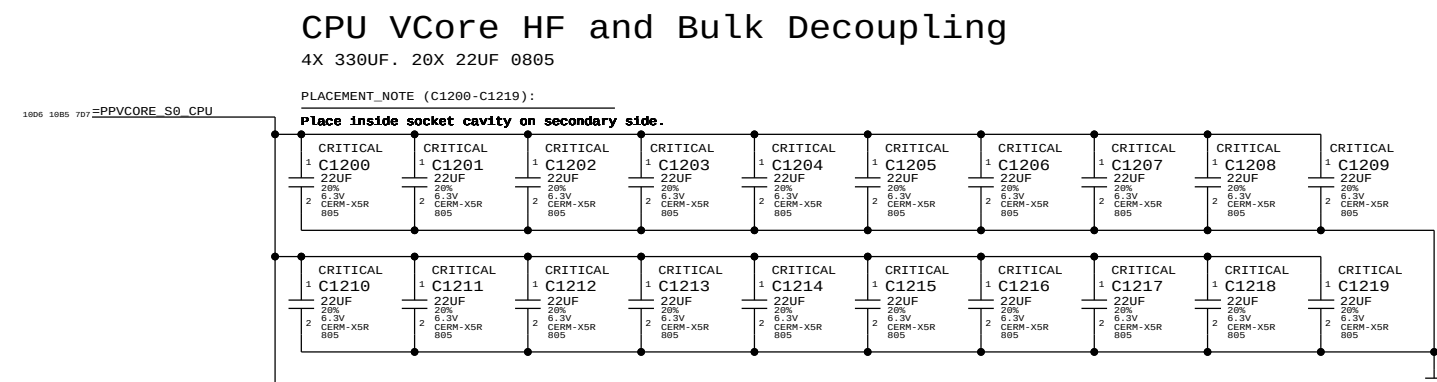




SYNC FROM T18
CHANGE CPU FROM SOCKET TO BGA SYMBOL

Current numbers from Merom for Santa Rosa EMTS, doc #209905.

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PAGE TITLE			
CPU Power & Ground			
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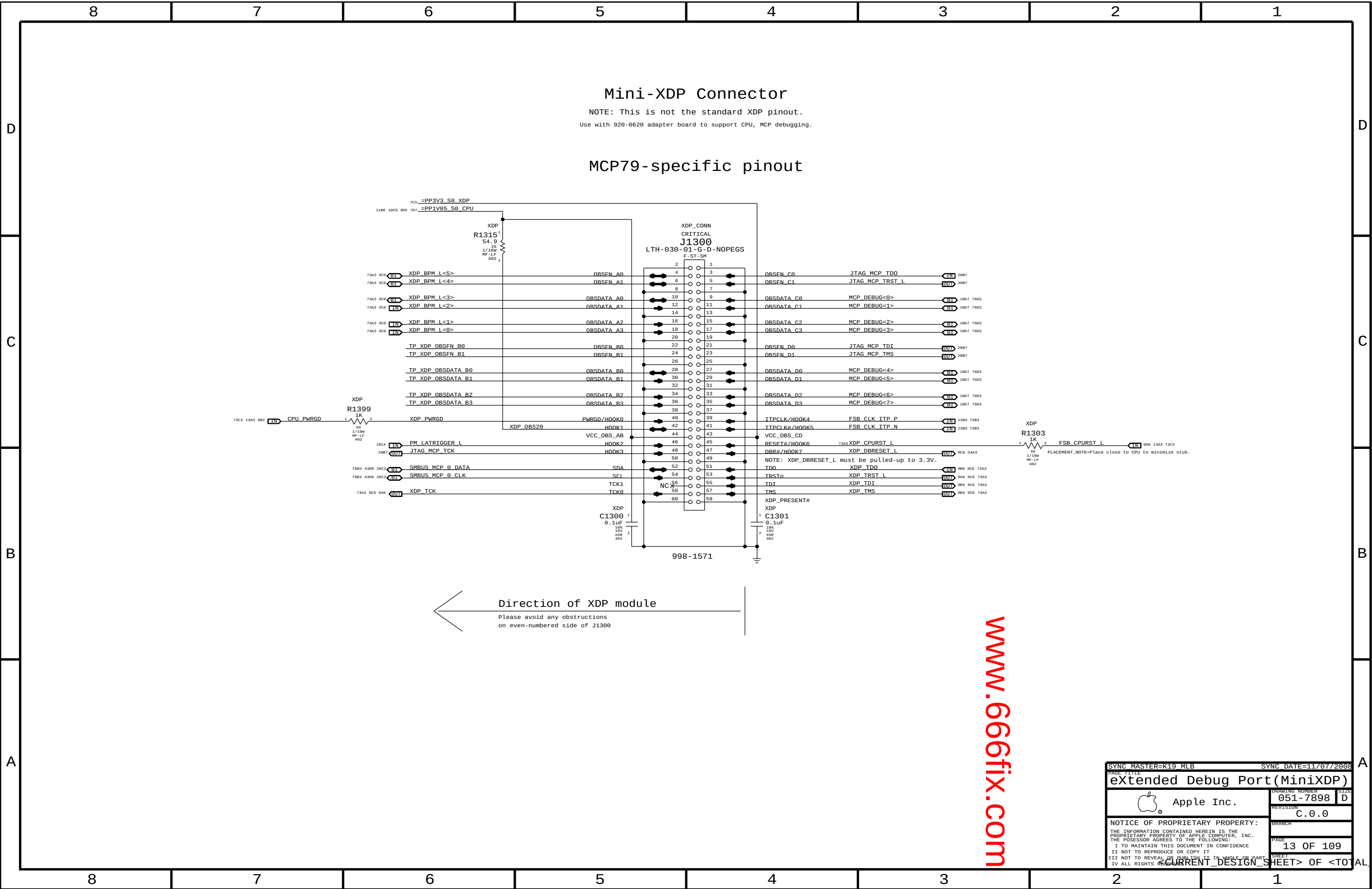


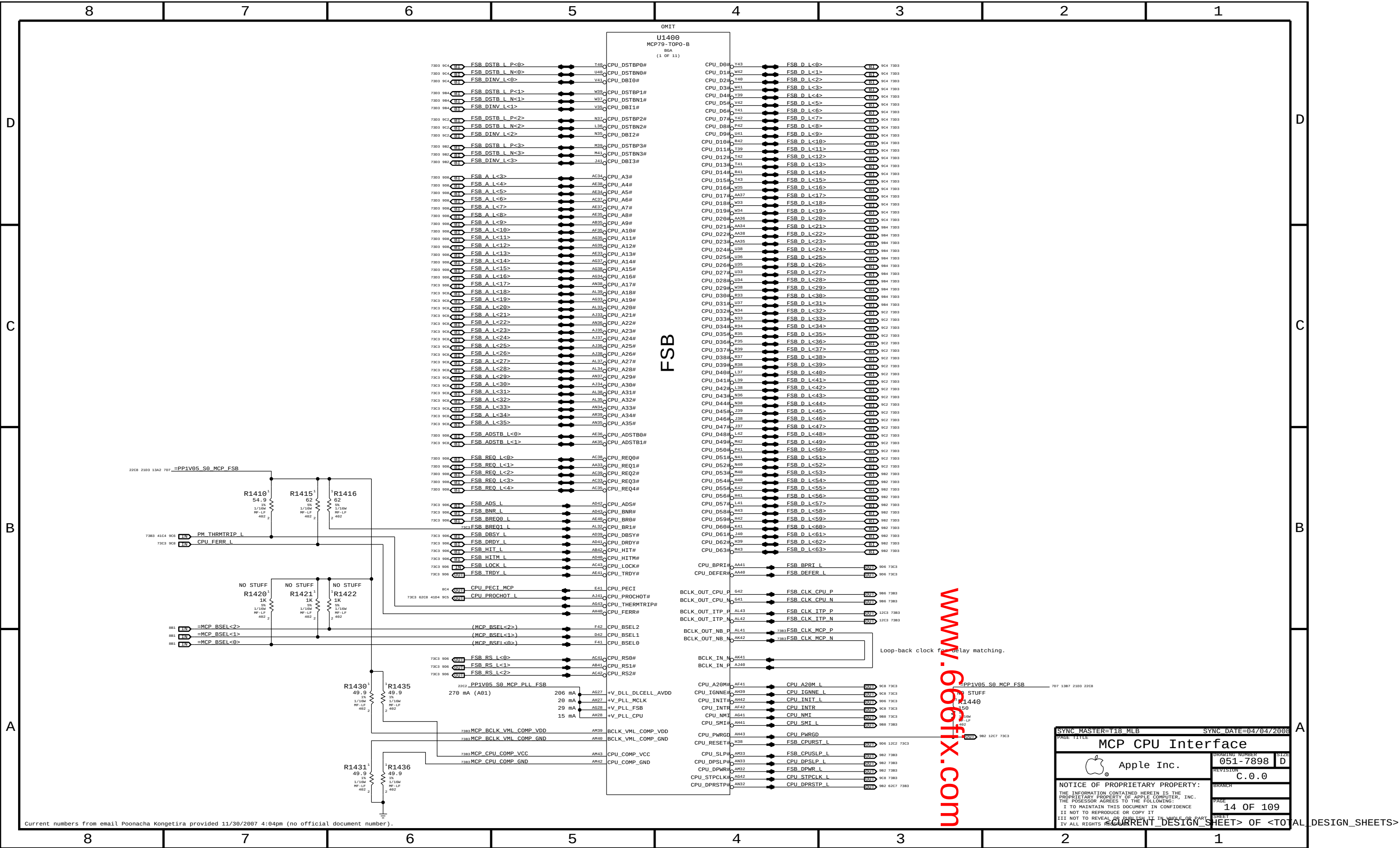
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SYNC FROM T18
REMOVE NO STUFF CAPS C1220 TO C1231
REMOVE C1244 & C1245
CHANGE C1240-C1243 AND C1260 FROM 128S0241(9 MILLI-OHM) TO 128S0231(6 MILLI-OHM)

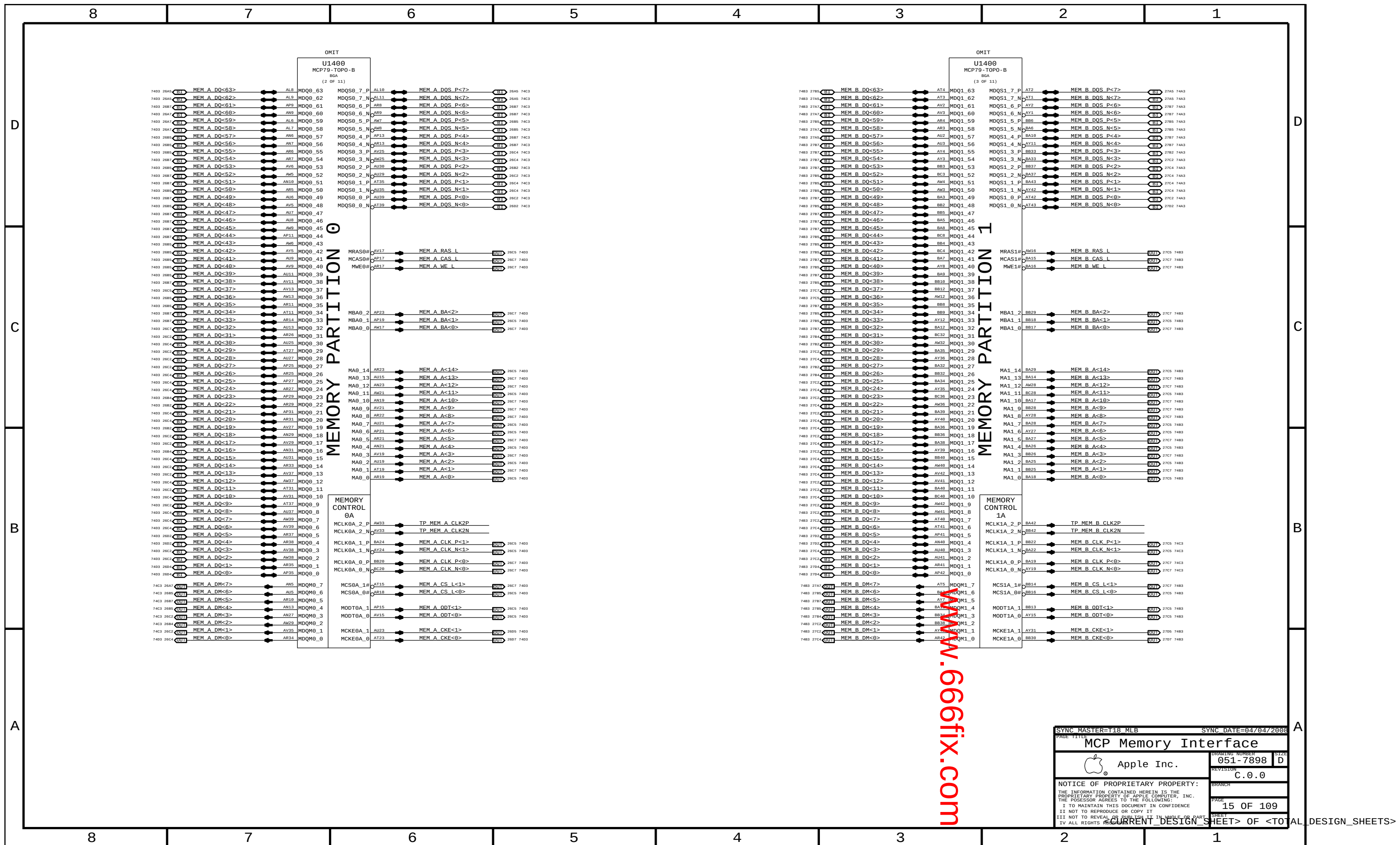
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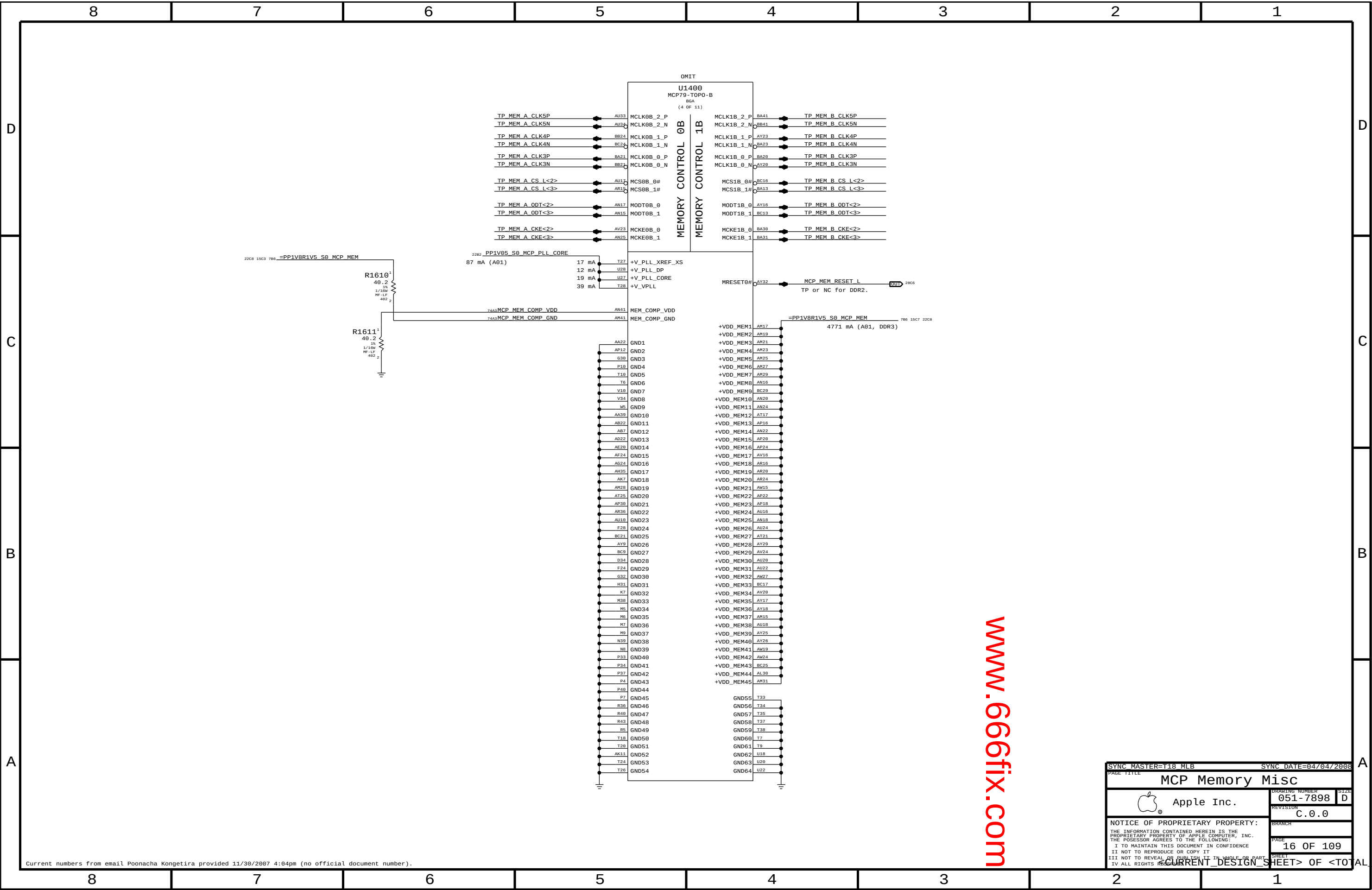
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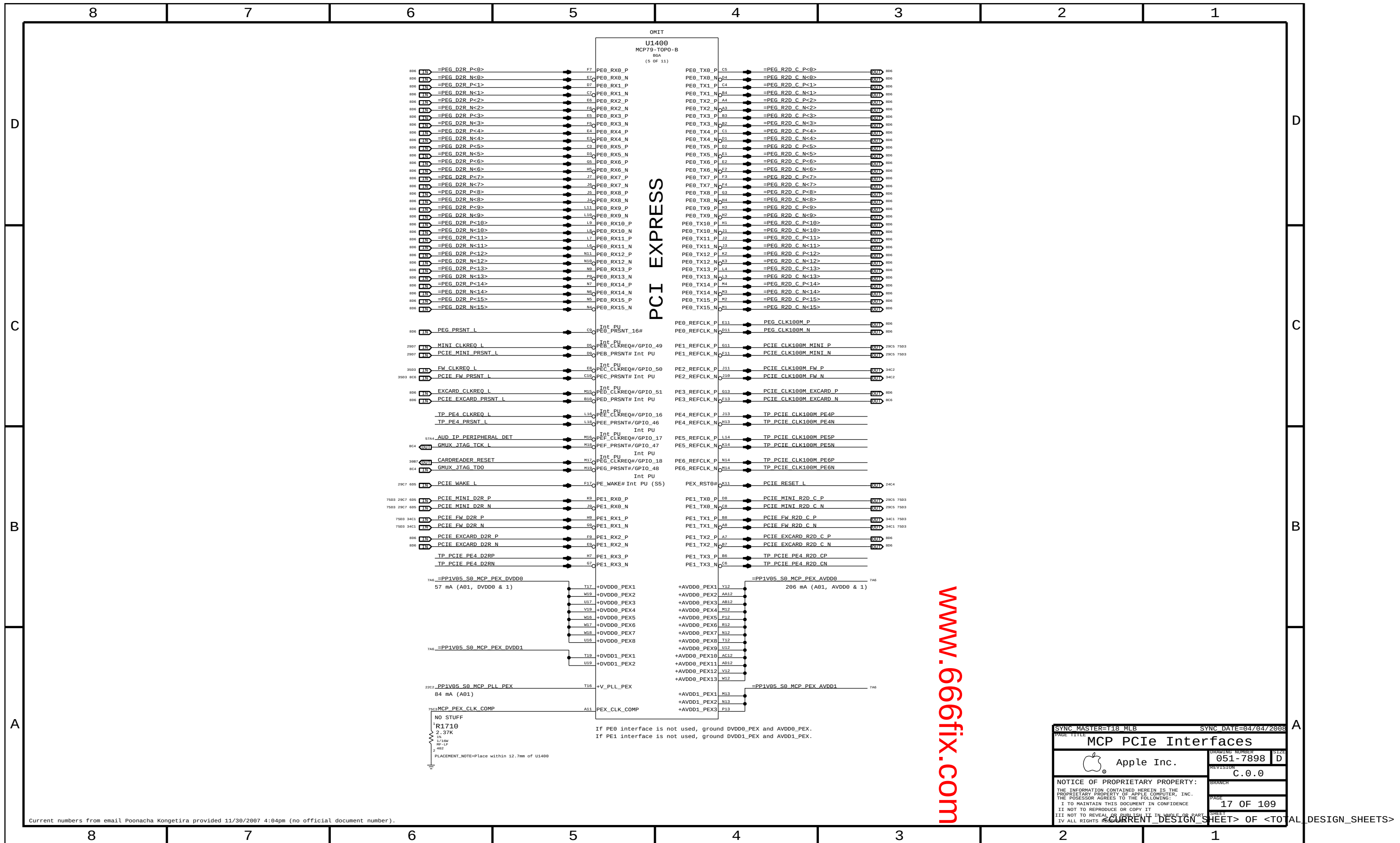
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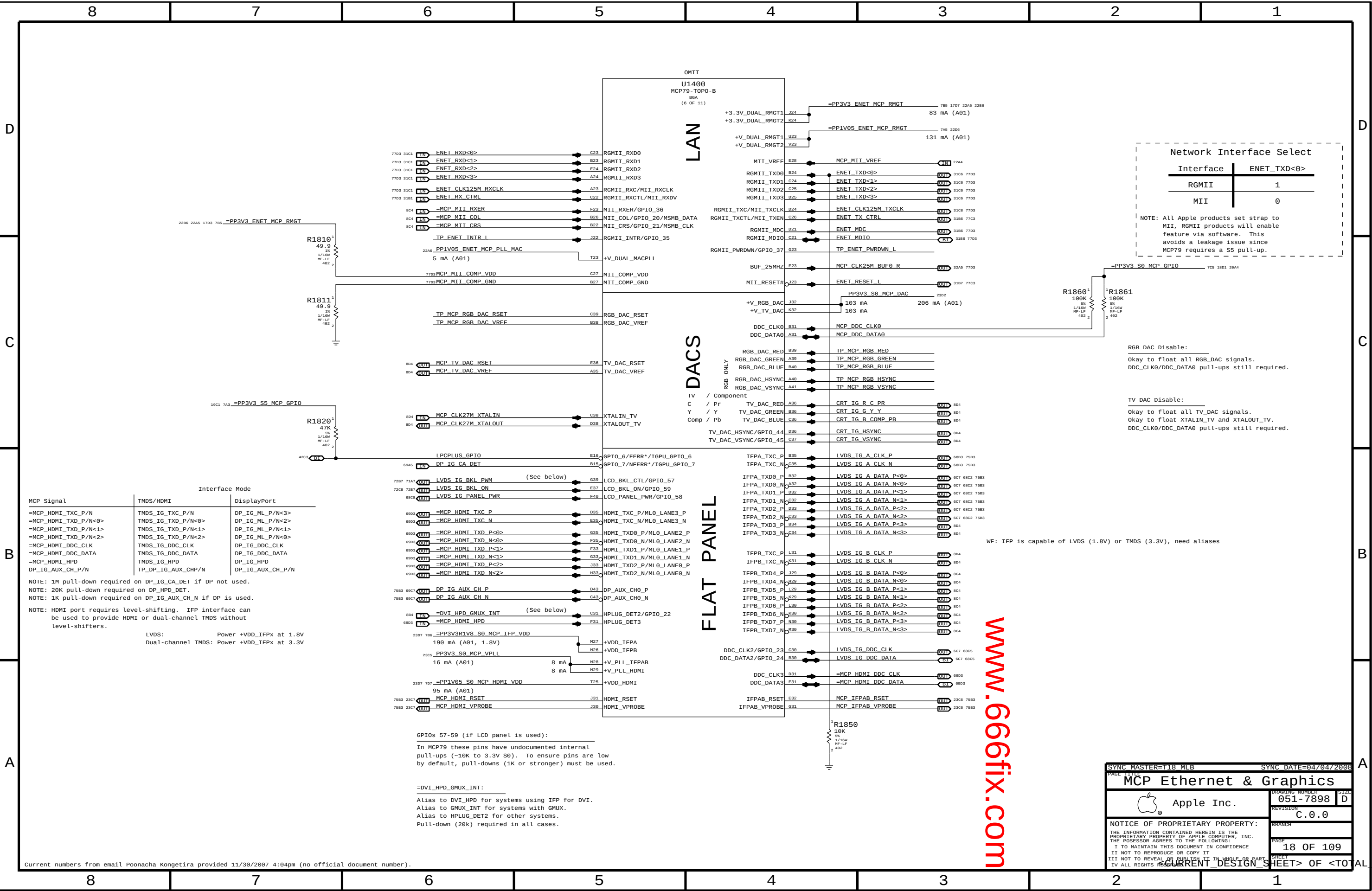


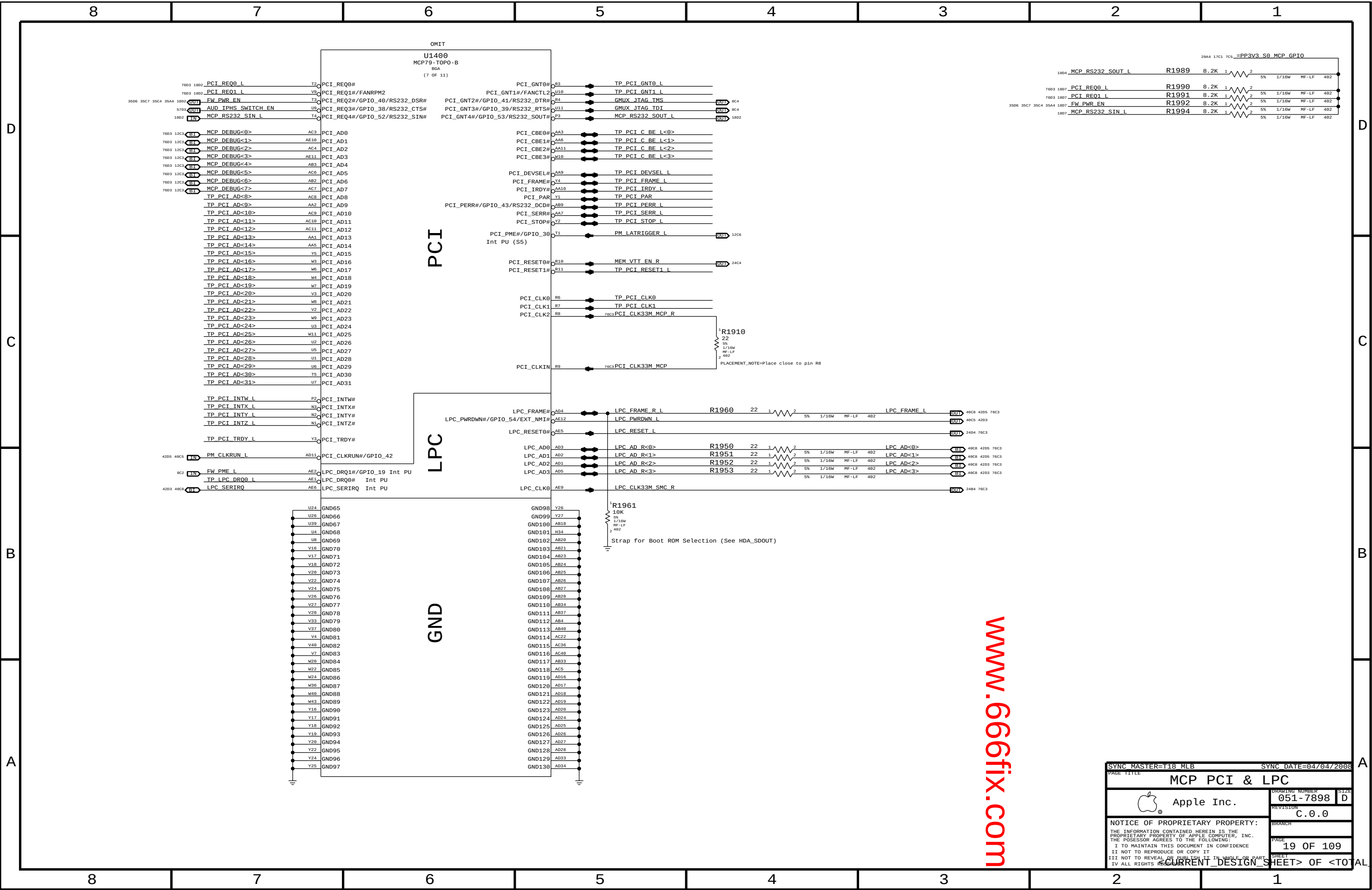


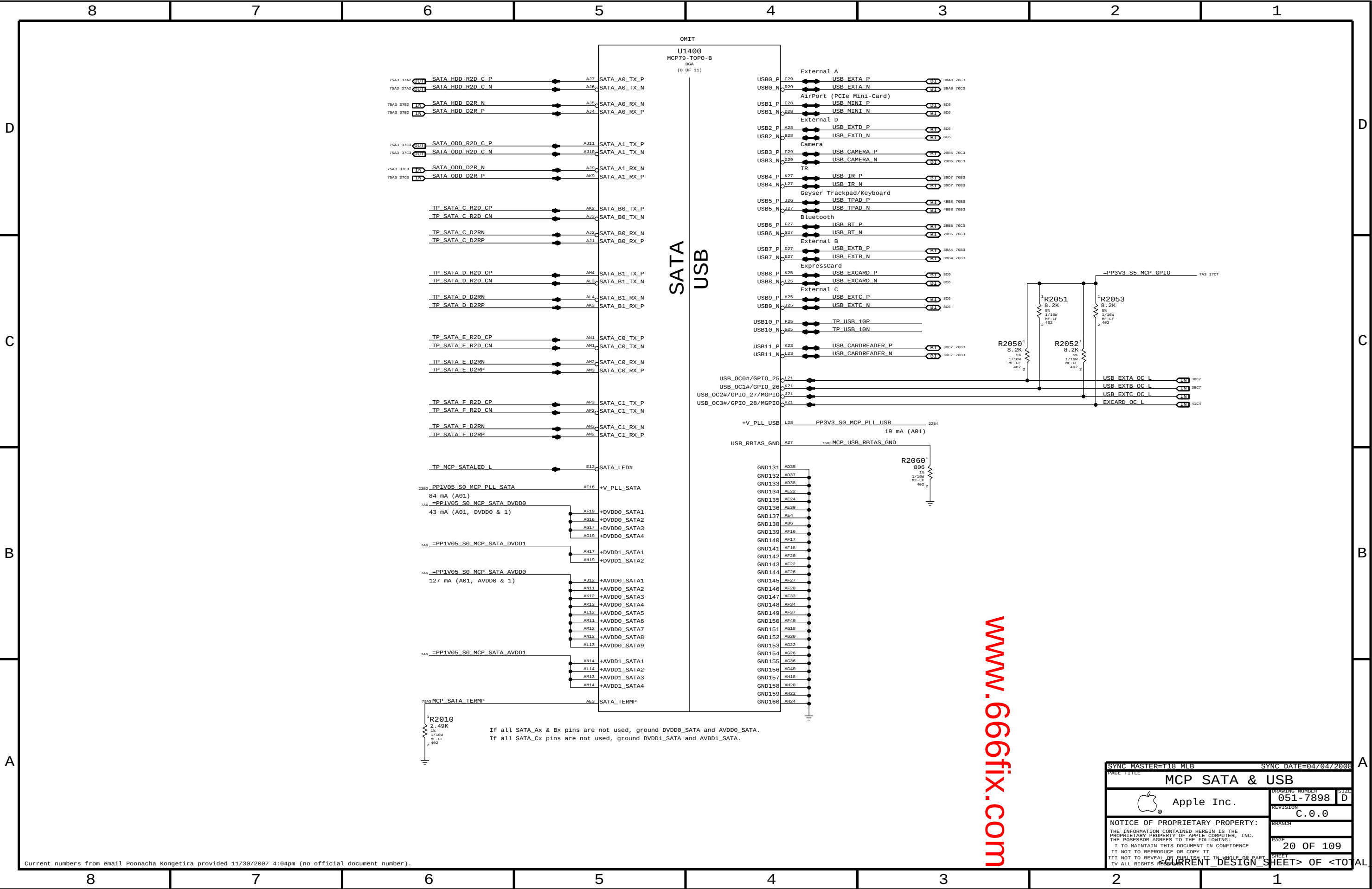
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MCP Memory		Misc	
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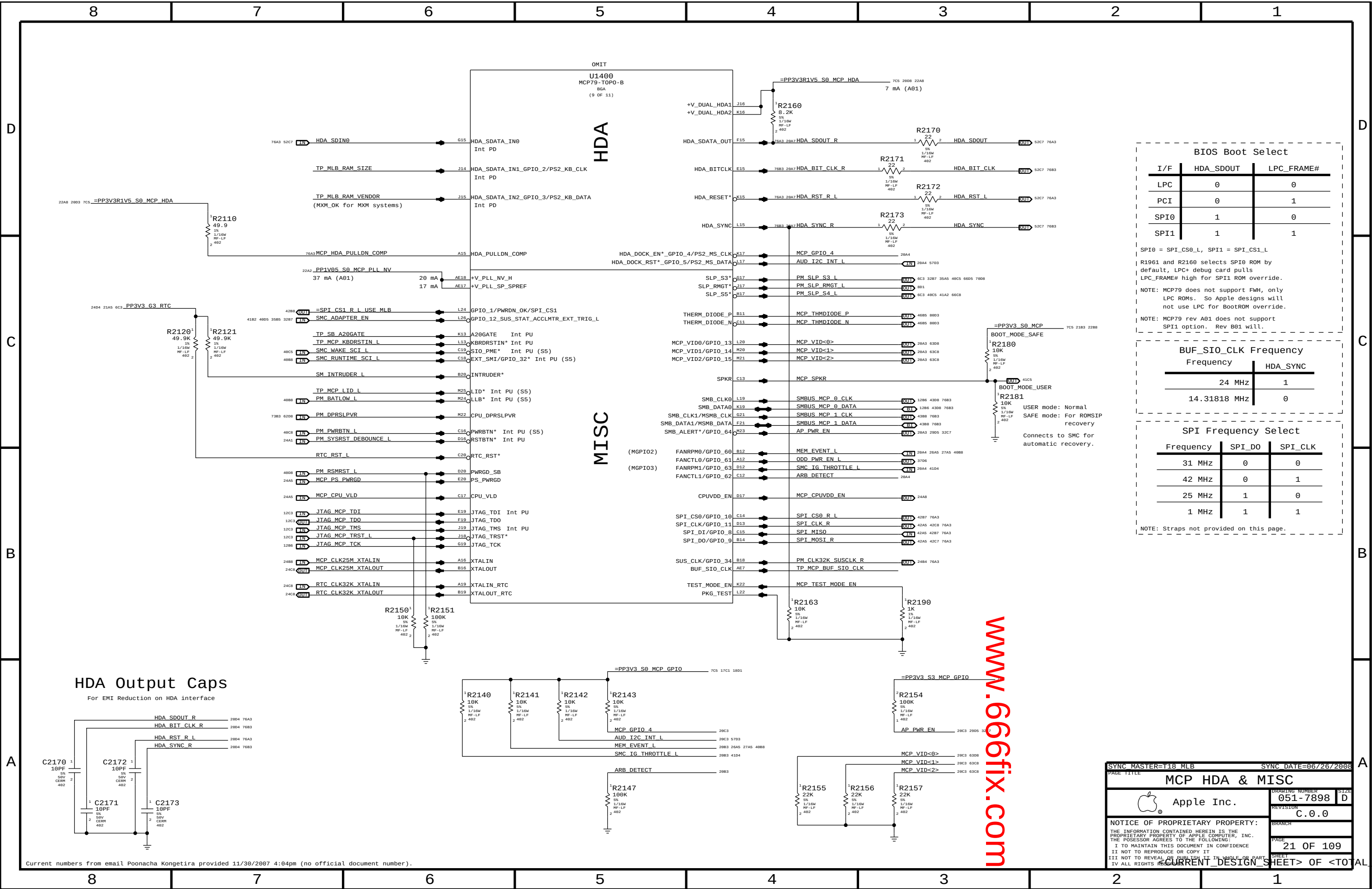




If all SATA_Ax & Bx pins are not used, ground DVDD0_SATA and AVDD0_SATA.
If all SATA_Cx pins are not used, ground DVDD1_SATA and AVDD1_SATA.

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SYNC MASTER=T18 MLB		SYNC DATE=04/04/2008	
PAGE TITLE			
MCP SATA & USB			
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BIOS Boot Select		
I/F	HDA_SDOUT	LPC_FRAME#
LPC	0	0
PCI	0	1
SPI0	1	0
SPI1	1	1

SPI0 = SPI_CS0_L, SPI1 = SPI_CS1_L
R1961 and R2160 selects SPI0 ROM by default, LPC+ debug card pulls LPC_FRAME# high for SPI1 ROM override.
NOTE: MCP79 does not support FWH, only LPC ROMs. So Apple designs will not use LPC for BootROM override.
NOTE: MCP79 rev A01 does not support SPI1 option. Rev B01 will.

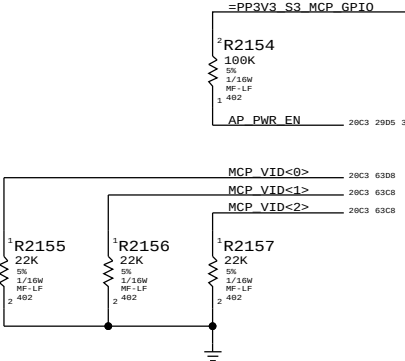
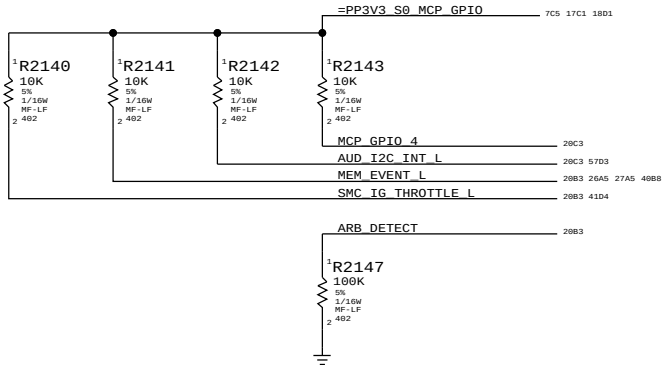
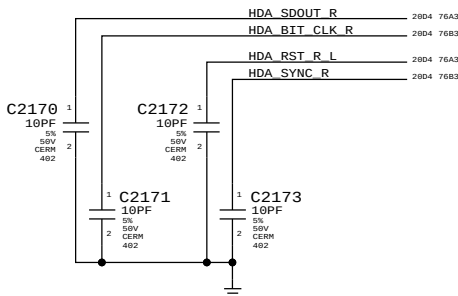
BUF_SIO_CLK Frequency	
Frequency	HDA_SYNC
24 MHz	1
14.31818 MHz	0

SPI Frequency Select		
Frequency	SPI_D0	SPI_CLK
31 MHz	0	0
42 MHz	0	1
25 MHz	1	0
1 MHz	1	1

NOTE: Straps not provided on this page.

HDA Output Caps

For EMI Reduction on HDA interface



SYNC MASTER=T18 MLB

SYNC DATE=06/26/2008

MCP HDA & MISC

Apple Inc.

051-7898

C.0.0

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21 OF 109

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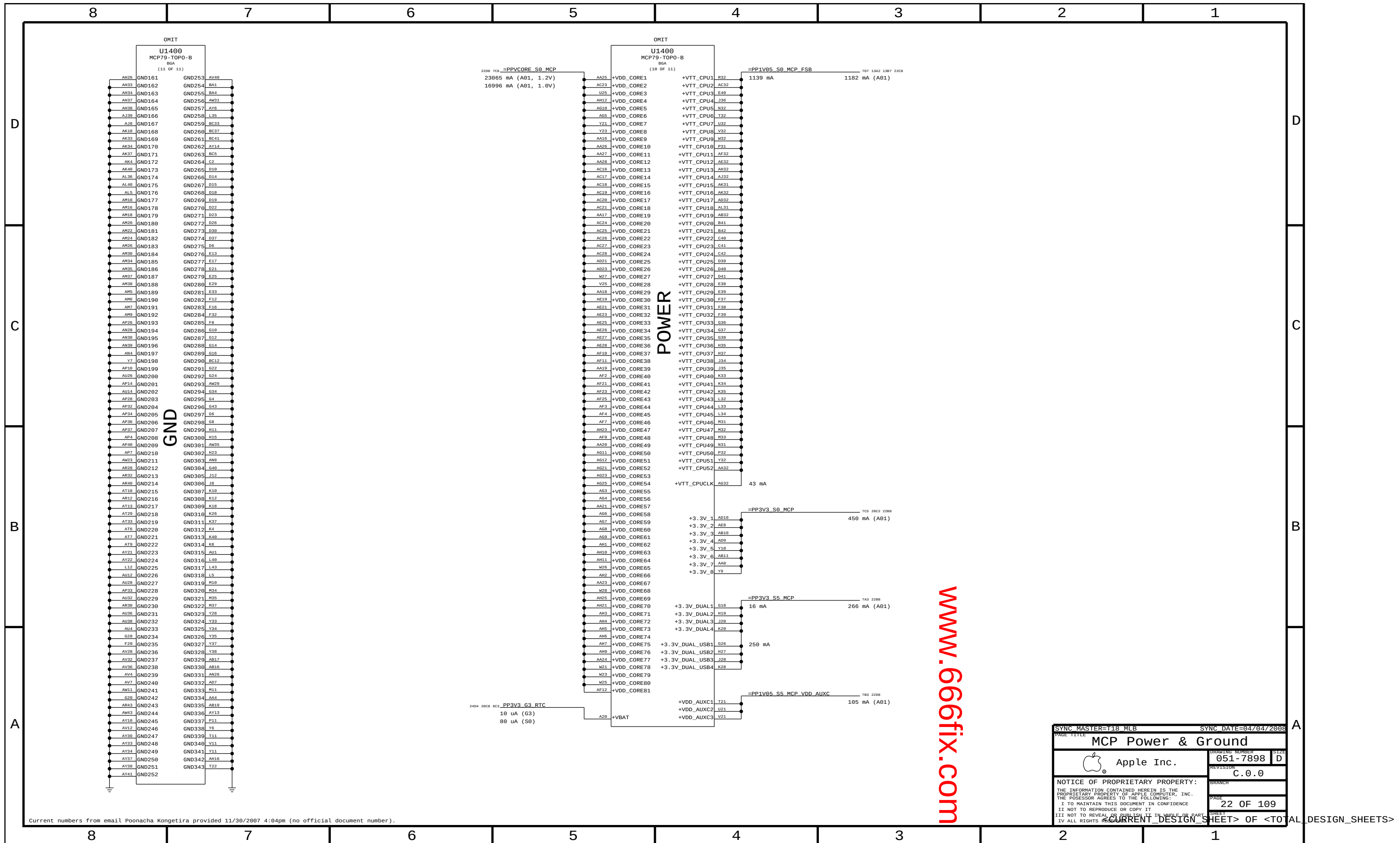
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

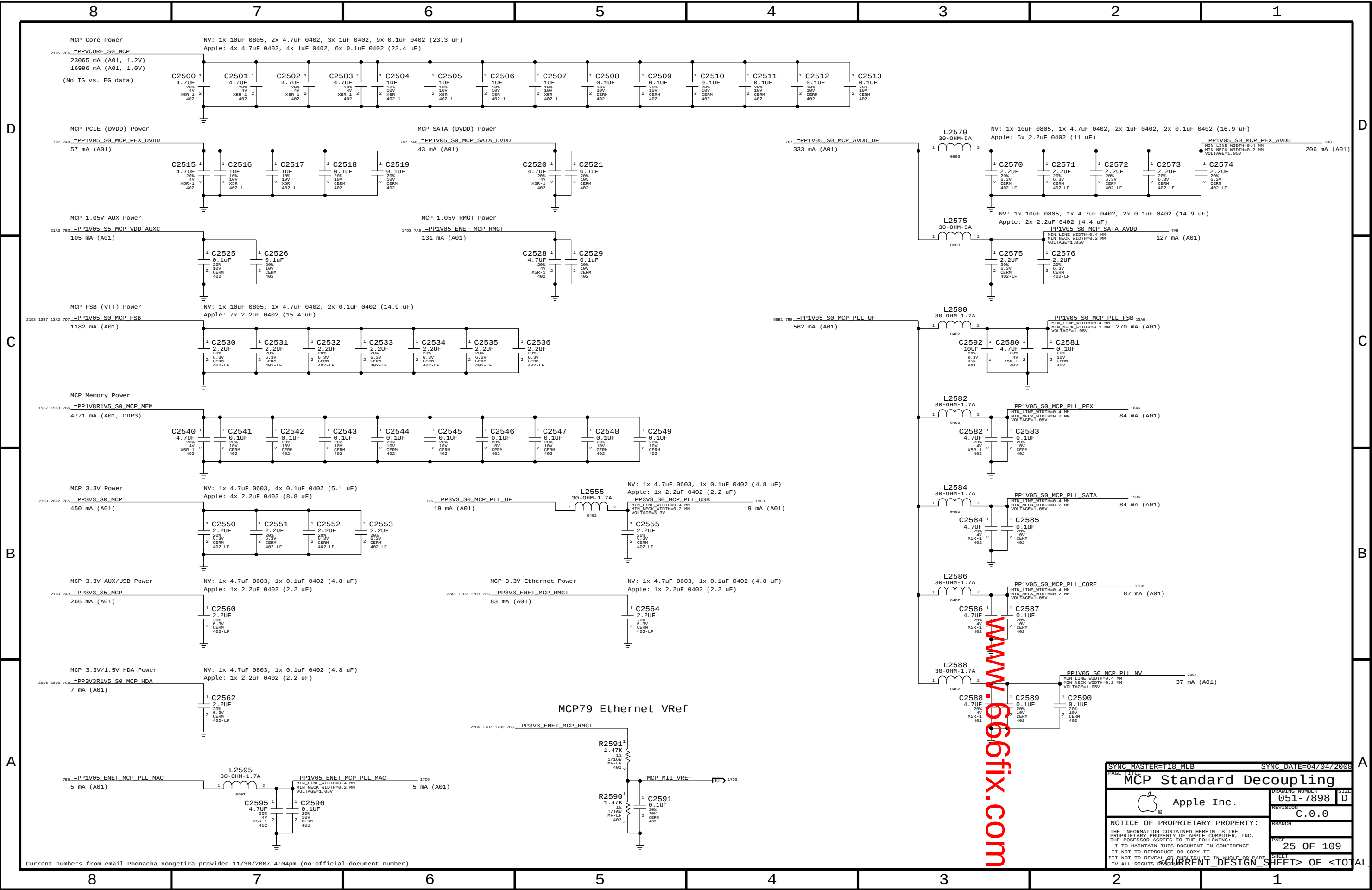
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DESIGN SHEET

OF

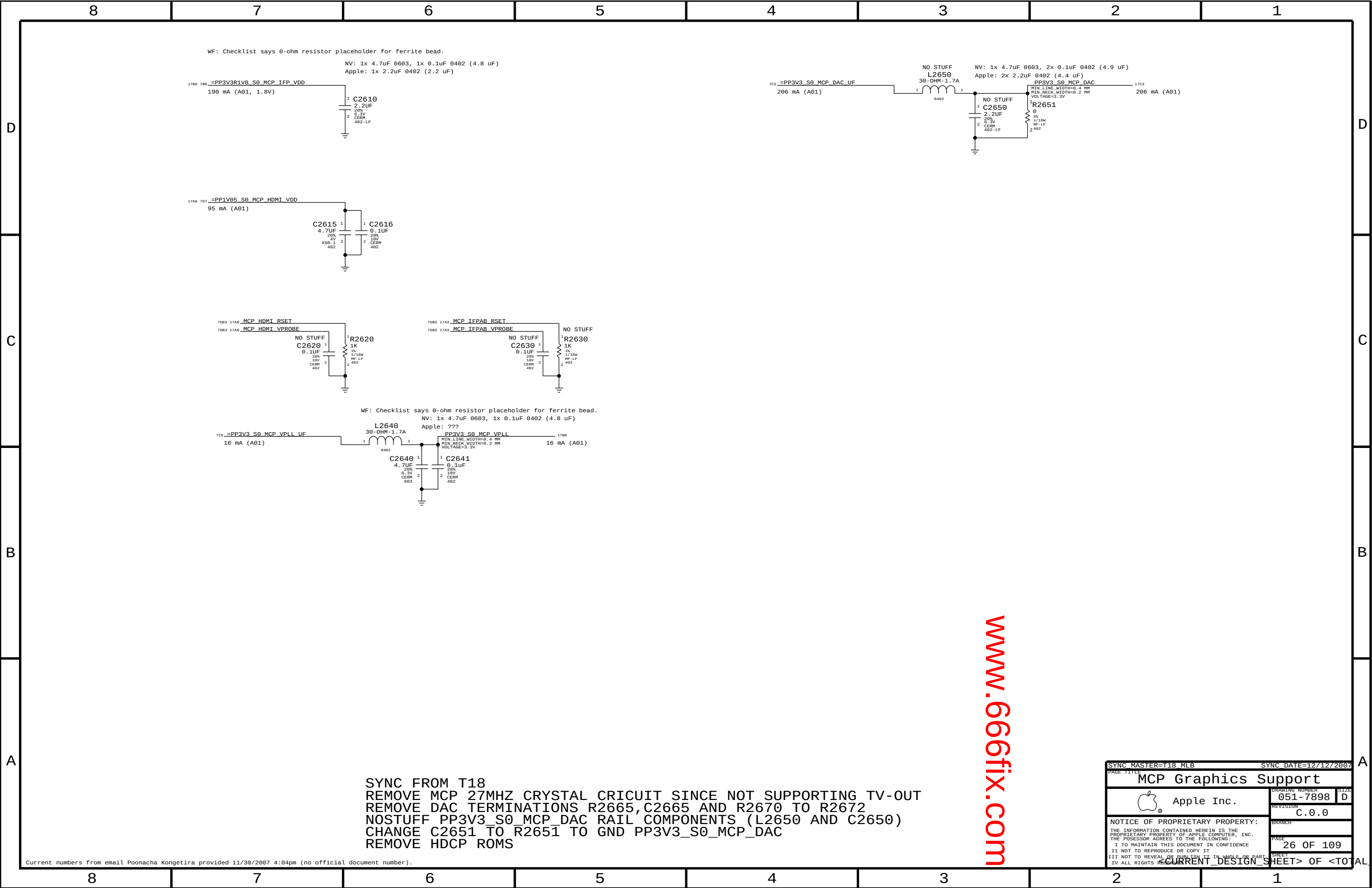
TOTAL DESIGN SHEETS>





Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

SYNC MASTER=T18 MLB		SYNC DATE=04/04/2008	
PAGE TITLE		MCP Standard Decoupling	
DRAWING NUMBER		051-7898	
REVISION		C.0.0	
BRANCH			
PAGE		25 OF 109	
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SYNC FROM T18
REMOVE MCP 27MHZ CRYSTAL CRICUIT SINCE NOT SUPPORTING TV-OUT
REMOVE DAC TERMINATIONS R2665,C2665 AND R2670 TO R2672
NOSTUFF PP3V3_S0_MCP_DAC RAIL COMPONENTS (L2650 AND C2650)
CHANGE C2651 TO R2651 TO GND PP3V3_S0_MCP_DAC
REMOVE HDCP ROMS

Current numbers from email Poonacha Kongetira provided 11/30/2007 4:04pm (no official document number).

SYNC MASTER=T18 MLB		SYNC DATE=12/12/2007	
PAGE TITLE			
MCP Graphics Support			
 Apple Inc.		DRAWING NUMBER	051-7898
		SIZE	D
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		BRANCH	
CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>		PAGE	26 OF 109
		SHEET	

Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PP3V3_S5_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- VREFMRGN
- NO_VREFMRGN

DAC channel
Min DAC code
Max DAC code
Max sink I
Max source I
Nominal Vref
Min Vref
Max Vref
Vref Stepping
(per DAC LSB)

MEM A VREF DQ
A
0x00
0x87
-3.75 mA
5 mA
0.75 V
0.375 V
1.250 V
6.5 mV

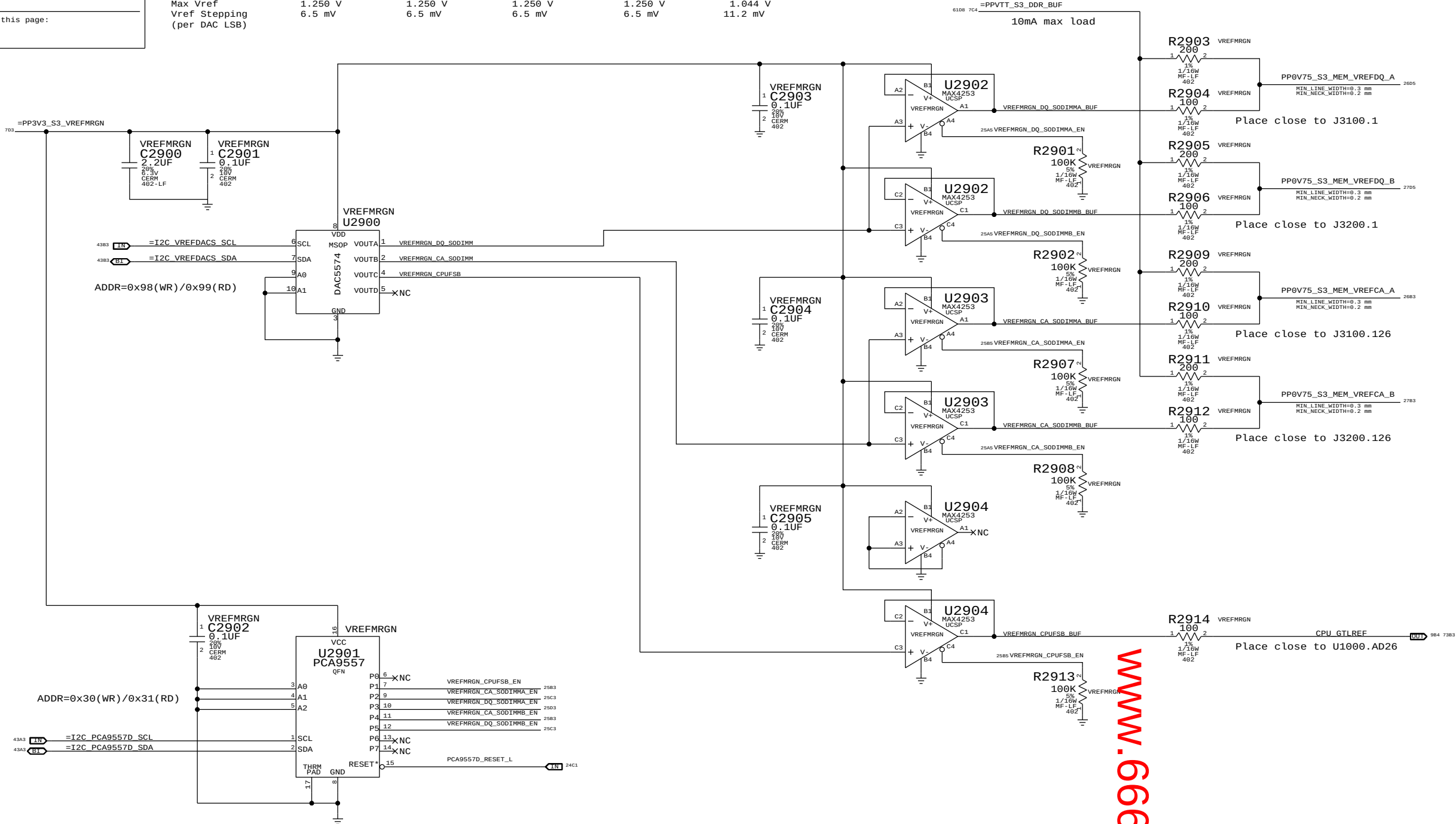
MEM A VREF CA
B
0x00
0x87
-3.75 mA
5 mA
0.75 V
0.375 V
1.250 V
6.5 mV

MEM B VREF DQ
A
0x00
0x87
-3.75 mA
5 mA
0.75 V
0.375 V
1.250 V
6.5 mV

MEM B VREF CA
B
0x00
0x87
-3.75 mA
5 mA
0.75 V
0.375 V
1.250 V
6.5 mV

CPU FSB VREF
C
0x00
0x55
-0.91 mA
0.52 mA
0.70 V
0.091 V
1.044 V
11.2 mV

SO-DIMM A and SO-DIMM B Vref settings should be margined separately
(i.e. not simultaneously) due to current limitation of TPS51116 regulator.



Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2903	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2905	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2909	CRITICAL	NO_VREFMRGN
116S0004	1	RES,MTL FILM,0,5%,0402,SM,LF	R2911	CRITICAL	NO_VREFMRGN

SYNC MASTER=BEN		SYNC DATE=03/31/2008	
PAGE TITLE			
FSB/DDR3 Vref Margining			
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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

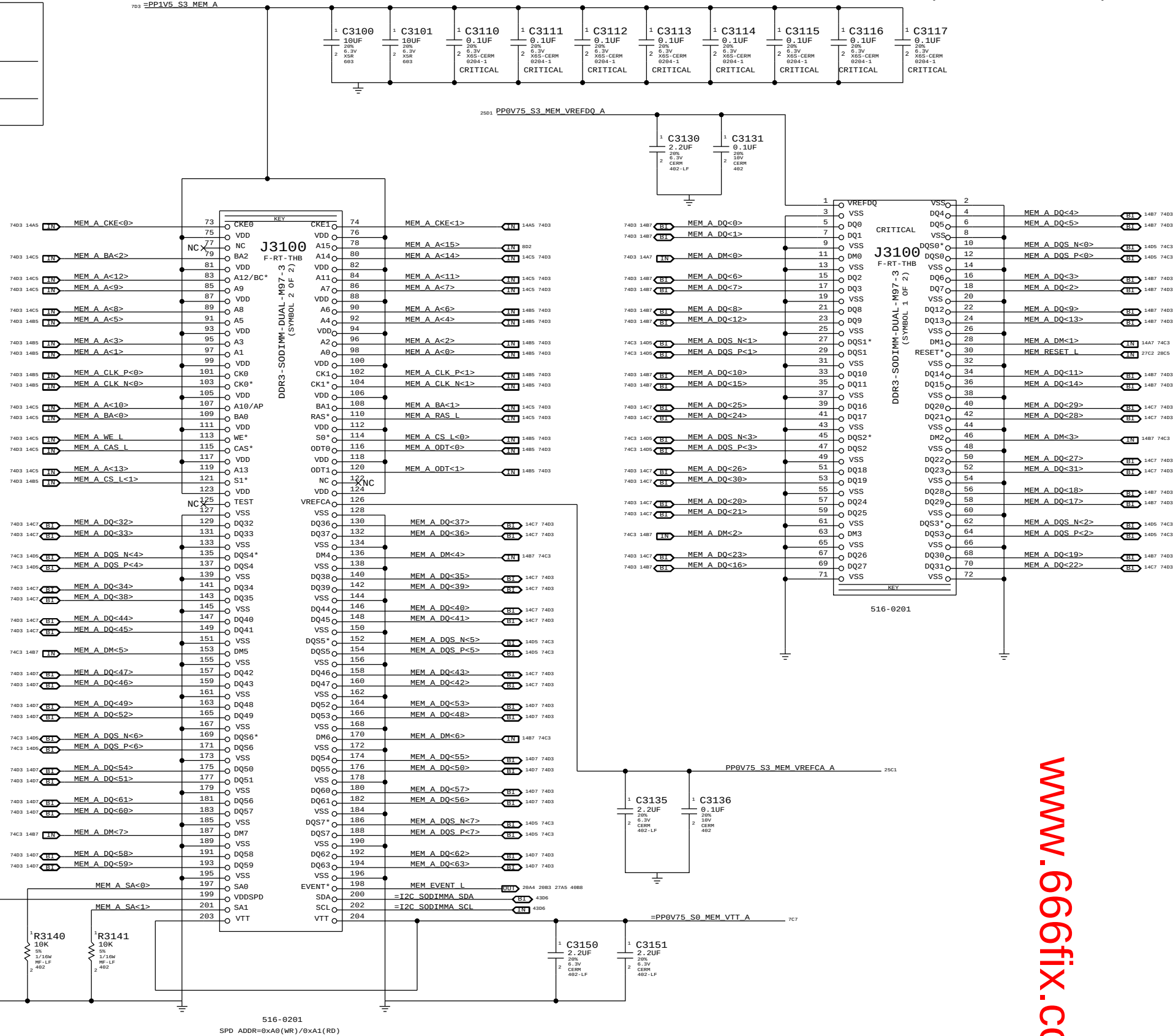
Signal aliases required by this page:

- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

BOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GROUND RETURN CAPS (CONNECTOR SIDE)



"Factory" (top) slot

SYNC MASTER=BEN		SYNC DATE=06/30/2008	
DDR3 S0-DIMM Connector A			
DRAWING NUMBER		SIZE	
051-7898		D	
REVISION		C.0.0	
BRANCH		PAGE	
31 OF 109		SHEET	
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Page Notes

Power aliases required by this page:

- #PP1V5_S0_MEM_B
- #PP1V5_S3_MEM_B
- #PP0V75_S0_MEM_VTT_B
- #PPSPD_S0_MEM_B (2.5 - 3.3V)

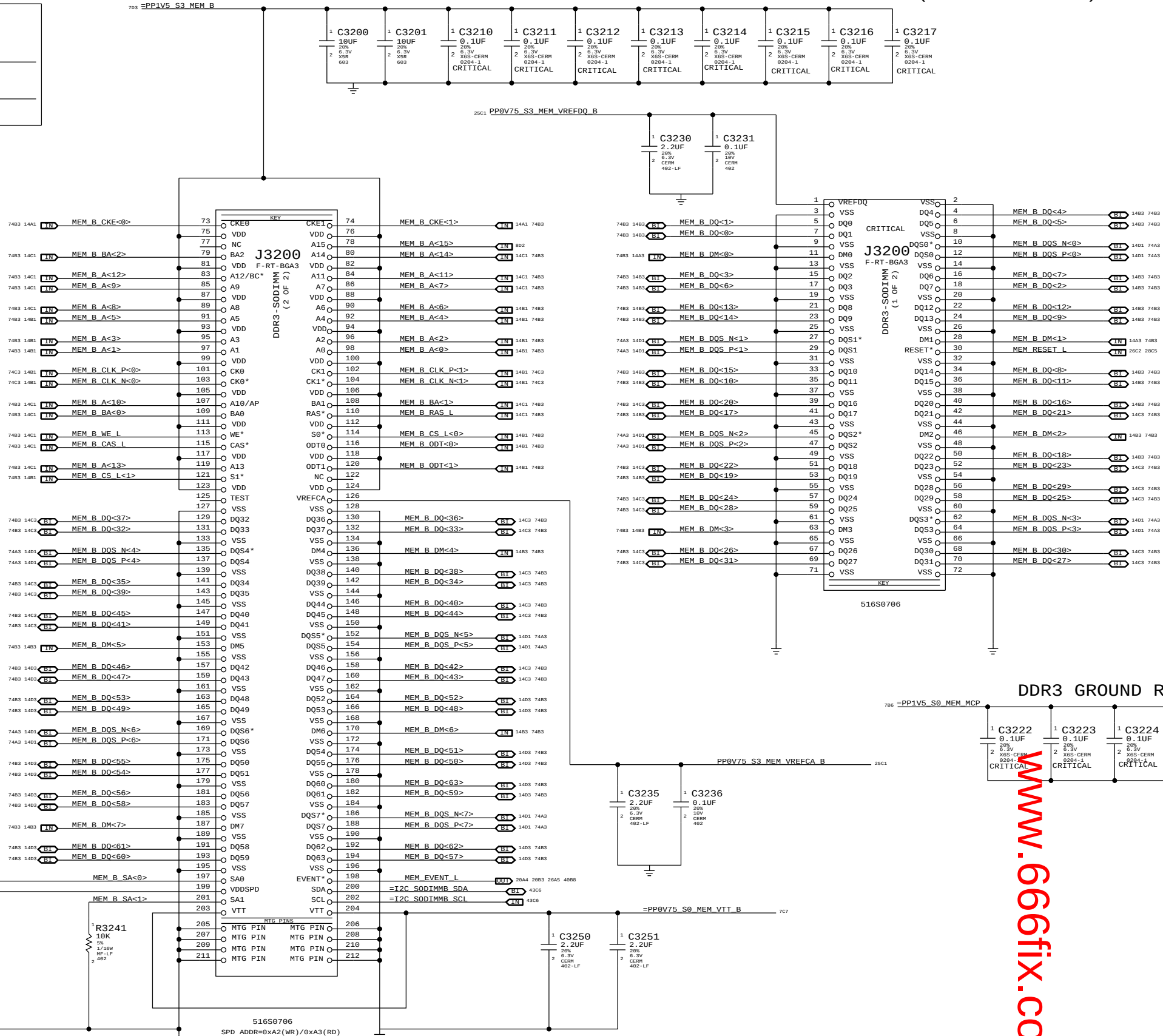
Signal aliases required by this page:

- #I2C_S0DIMMB_SCL
- #I2C_S0DIMMB_SDA

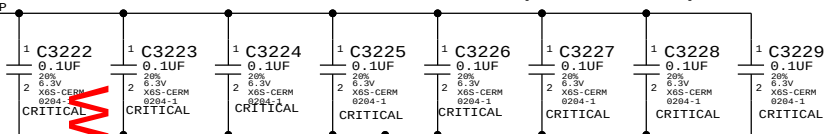
BOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GROUND RETURN CAPS (CONNECTOR SIDE)



DDR3 GROUND RETURN CAPS (MCP SIDE)

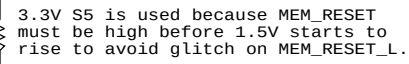


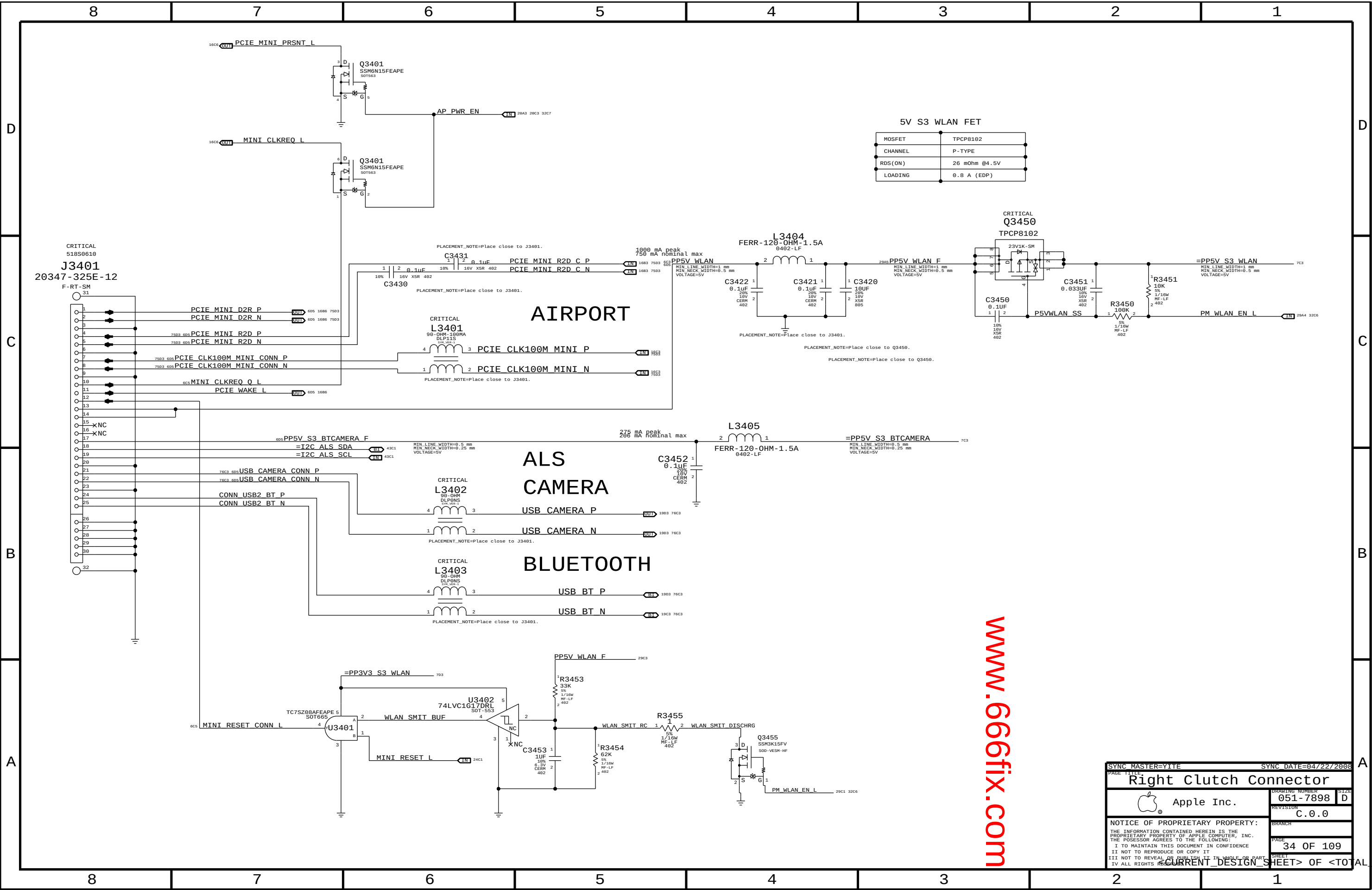
"Expansion" (bottom) slot

SYNC MASTER=BEN		SYNC DATE=05/09/2008	
PAGE TITLE			
DDR3 S0-DIMM Connector B			
DRAWING NUMBER		SIZE	
051-7898		D	
REVISION		C.0.0	
BRANCH		PAGE	
		32 OF 109	
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Required becaues MCP79 does not meet DDR3 spec power-up reset timing requirement.





5V S3 WLAN FET

MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	26 mOhm @4.5V
LOADING	0.8 A (EDP)

SYNC MASTER=YITE

SYNC DATE=04/22/2008

Right Clutch Connector

Apple Inc.

051-7898

C.0.0

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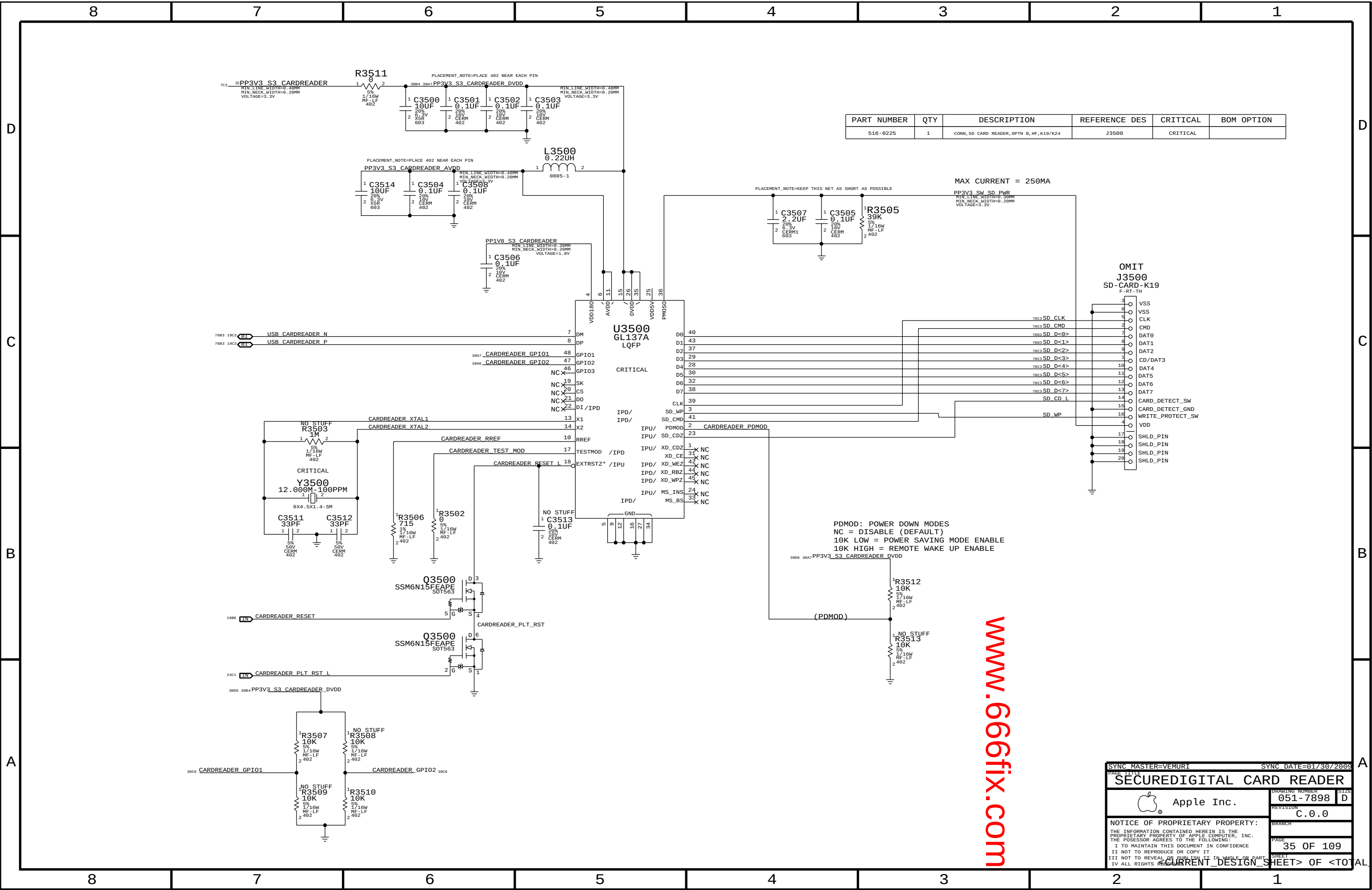
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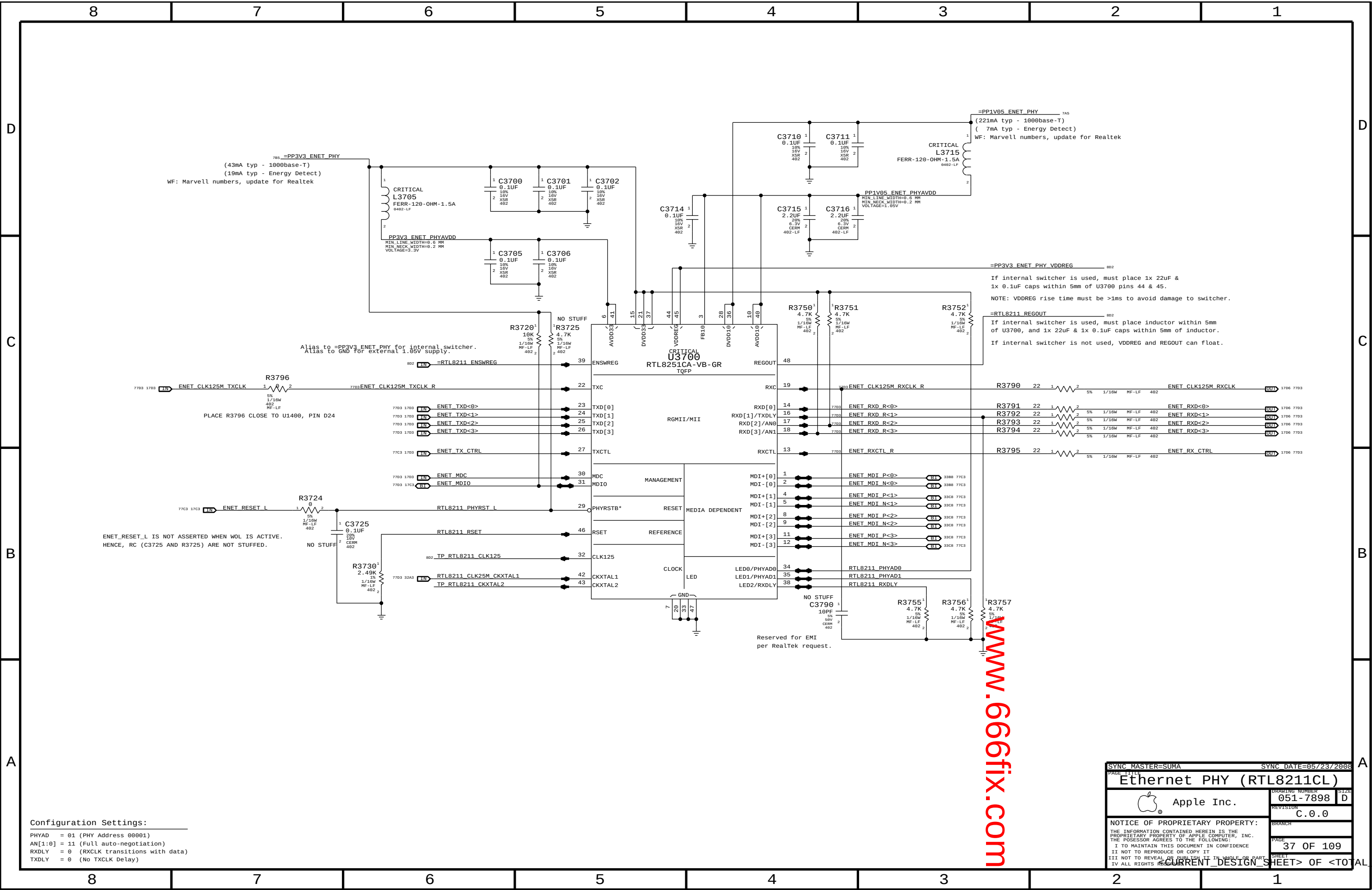
OF

TOTAL DESIGN SHEETS



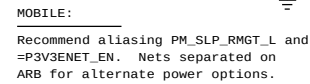
SYNC MASTER=VEMURI		SYNC DATE=01/30/2009	
PAGE TITLE			
SECUREDIGITAL CARD READER			
 Apple Inc.		DRAWING NUMBER	051-7898
		SIZE	D
		REVISION	C.0.0
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		PAGE	35 OF 109
CURRENT DESIGN		SHEET	
			SHEET# OF <TOT

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@ 2.5V Vgs:
Rds(on) = 90mOhm max
I(max) = 1.7A (85C)



```
"WLAN" = ("S3" && "AP_PWR_EN" && ("AC" || "S0"))
```

PM_WLAN_EN_L 2944 29C1
Pull-up is with power FET.

Q3805 SSM6N15FEAPE SOT563

2905 28C3 28A3 AP PwR_EN

Q3805 SSM6N15FEAPE SOT563

AC OR S0_L

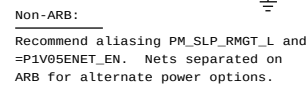
Q3801 SSM6N15FEAPE SOT563

4182 4805 3585 28C7 SMC_ADAPTER_EN

PM_SLP_S3_L 105 48C5 35A5 28C3 6C3

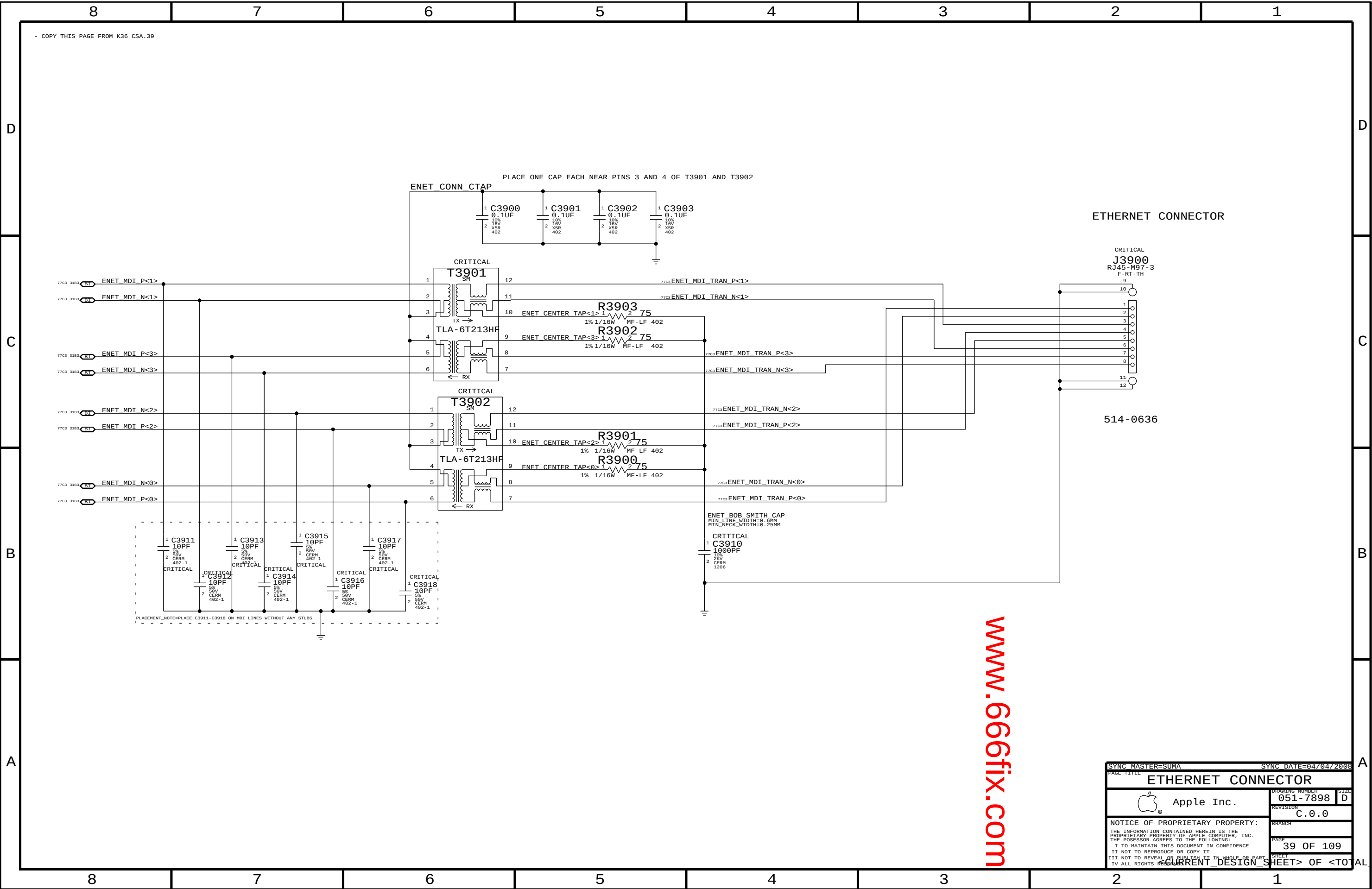
7B3 =PP1V05_ENET_P1V05ENETFET

1.8V Vgs



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SYNCH MASTER=SUMA		SYNCH DATE=07/01/2006	
PARTIAL			
Ethernet & AirPort Support			
 Apple Inc.		DRAWING NUMBER 051-7898	
		SIZE D	
		REVISION C.0.0	
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SYNC MASTER=SUMA		SYNC DATE=04/04/2008	
PAGE TITLE			
ETHERNET CONNECTOR			
	Apple Inc.		DRAWING NUMBER
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		BRANCH	
		PAGE	
		39 OF 109	
CURRENT DESIGN SHEET		OF TOTAL DESIGN SHEETS	

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0558	1	RES, 0.68 OHM, 1%, 0402, SMD	R4100	CRITICAL	

D

D

C

C

B

B

A

A

SYNC MASTER=K19 MLB		SYNC DATE=11/02/2008	
PAGE 1 TITLE		FireWire LLC/PHY (FW643)	
DRAWING NUMBER		051-7898	SIZE D
REVISION		C.0.0	
BRANCH			
PAGE		41 OF 109	
SHEET			
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CURRENT DESIGN SHEET		> OF <TOTAL DESIGN SHEETS>	

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D

C

B

A

```
Configures PHY for:
- 1-port Portable Power Class (0)
- Port "1" Bilingual (1394B)
```

R4380¹
10K
1%
1/16W
NF-LF
402

FWPHY_DS0 == =FW_PHY_DS0 34C6
MAKE_BASE=TRUE
FWPHY_DS2 == =FW_PHY_DS2 34C6
MAKE_BASE=TRUE
FWPHY_DS1 == =FW_PHY_DS1 34C6
MAKE_BASE=TRUE

R4381¹
10K
1%
1/16W
NF-LF
402

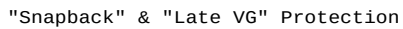
FW_P0 TPBIAS == NC FW0 TPBIAS MAKE_BASE=TRUE
FW_P2 TPBIAS == NC FW2 TPBIAS MAKE_BASE=TRUE
FW_P0 TPA_N == NC FW0 TPA_N MAKE_BASE=TRUE
FW_P0 TPA_P == NC FW0 TPAP MAKE_BASE=TRUE
FW_P2 TPA_N == NC FW2 TPA_N MAKE_BASE=TRUE
FW_P2 TPA_P == NC FW2 TPAP MAKE_BASE=TRUE
FW_P0 TPB_N == NC FW0 TPBN MAKE_BASE=TRUE
FW_P0 TPB_P == NC FW0 TPBP MAKE_BASE=TRUE
FW_P2 TPB_N == NC FW2 TPBN MAKE_BASE=TRUE
FW_P2 TPB_P == NC FW2 TPBP MAKE_BASE=TRUE

D

C

B

A

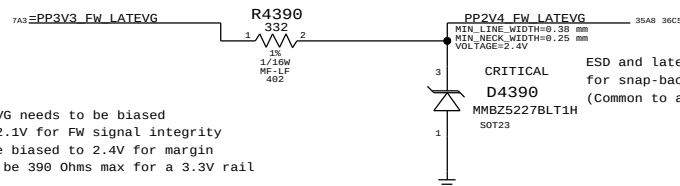


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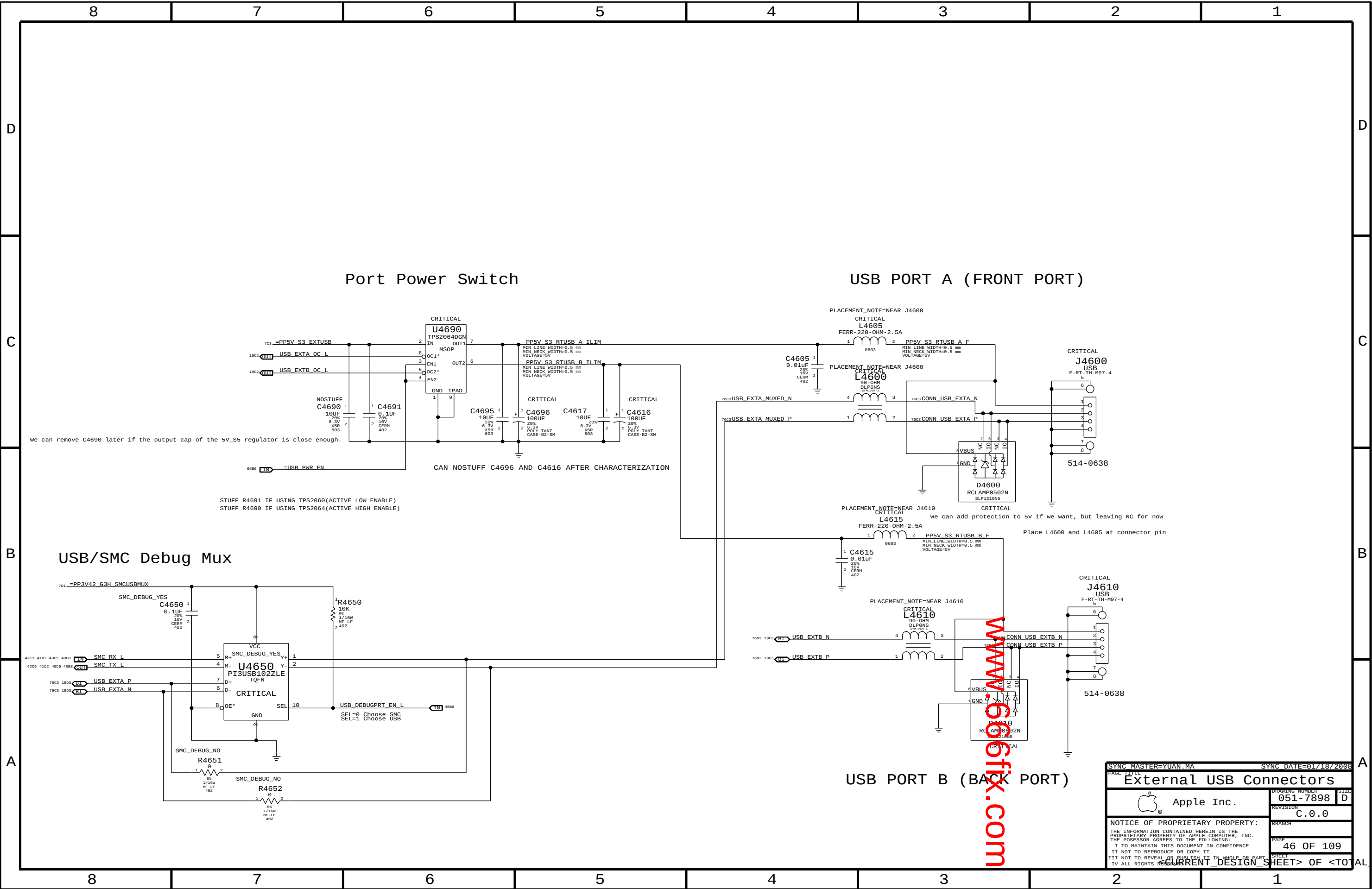
PORT 1
BILINGUAL



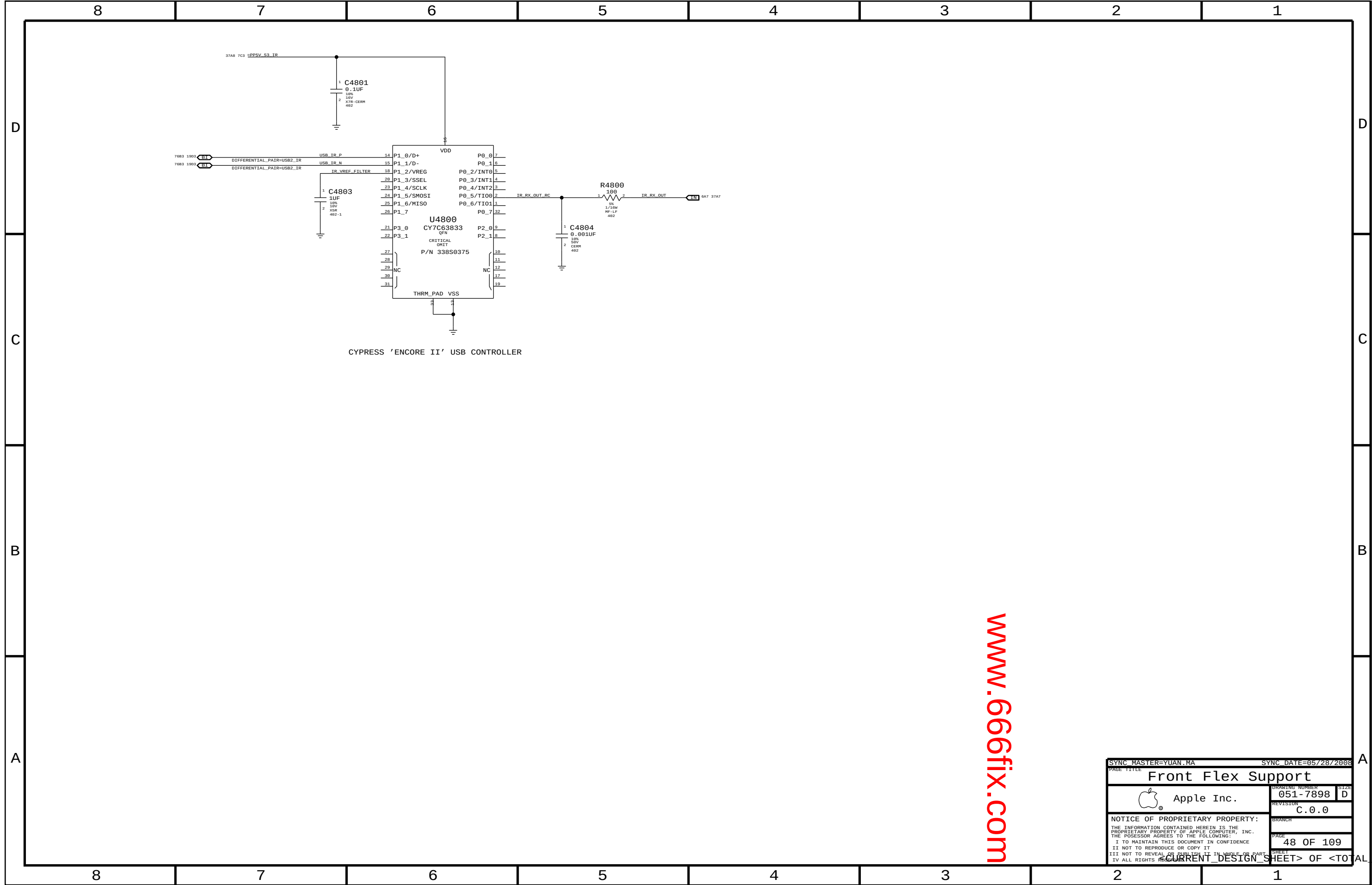
ESD and late-VG rail
for snap-back diodes
(Common to all ports)



SYNCH MASTER=K19 MLB		SYNCH DATE=11/02/2005	
PAGE TITLE			
FireWire Ports			
 Apple Inc.	DRAWING NUMBER	SIZE	
	051-7898	D	
	REVISION		
	C.0.0		
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		43	OF 109
SHEET			
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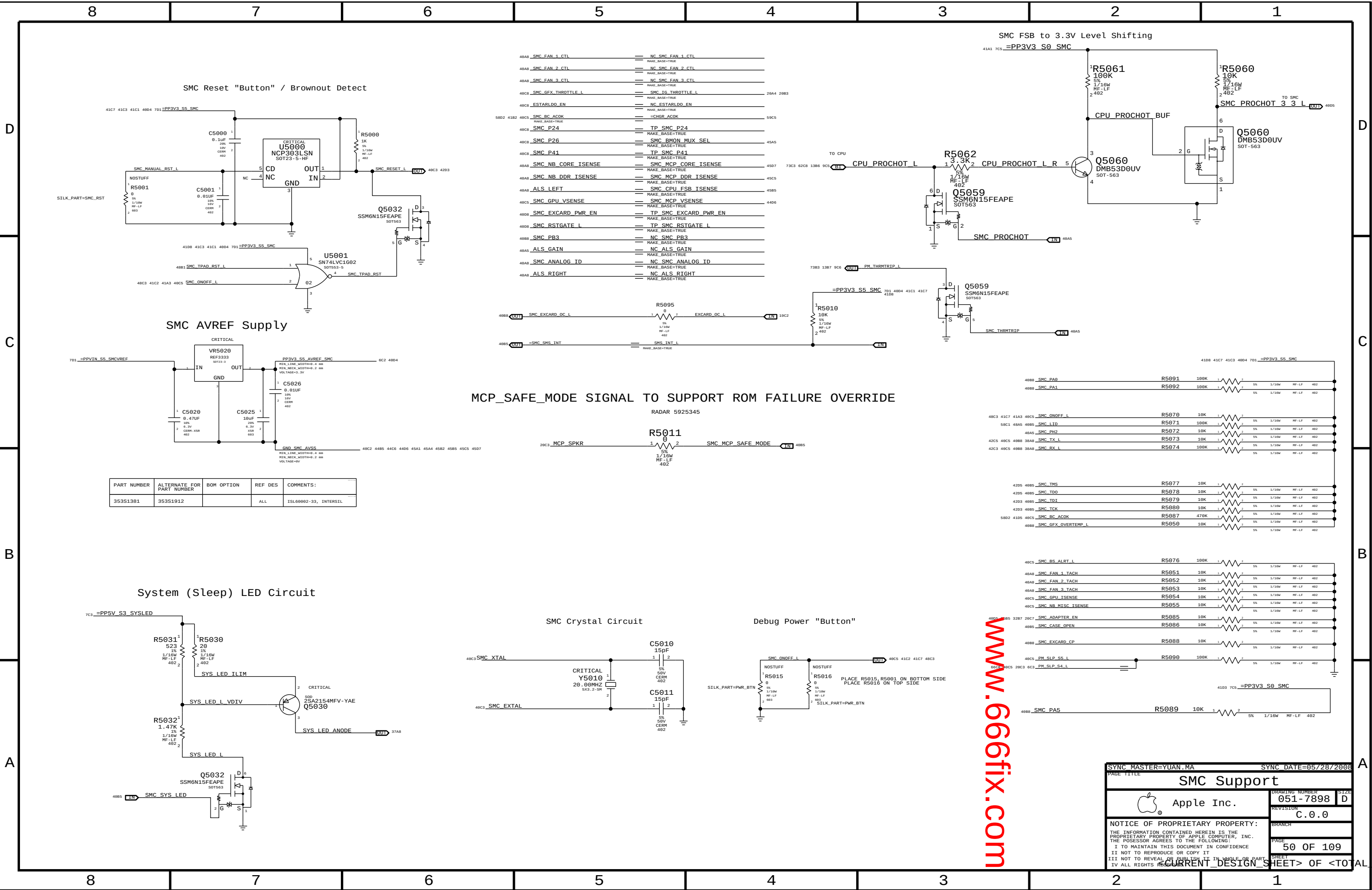


SYNC MASTER=YUAN.MA		SYNC DATE=01/18/2008	
PAGE TITLE		External USB Connectors	
DRAWING NUMBER		051-7898	D
REVISION		C.0.0	
BRANCH			
PAGE		46 OF 109	
SHEET			
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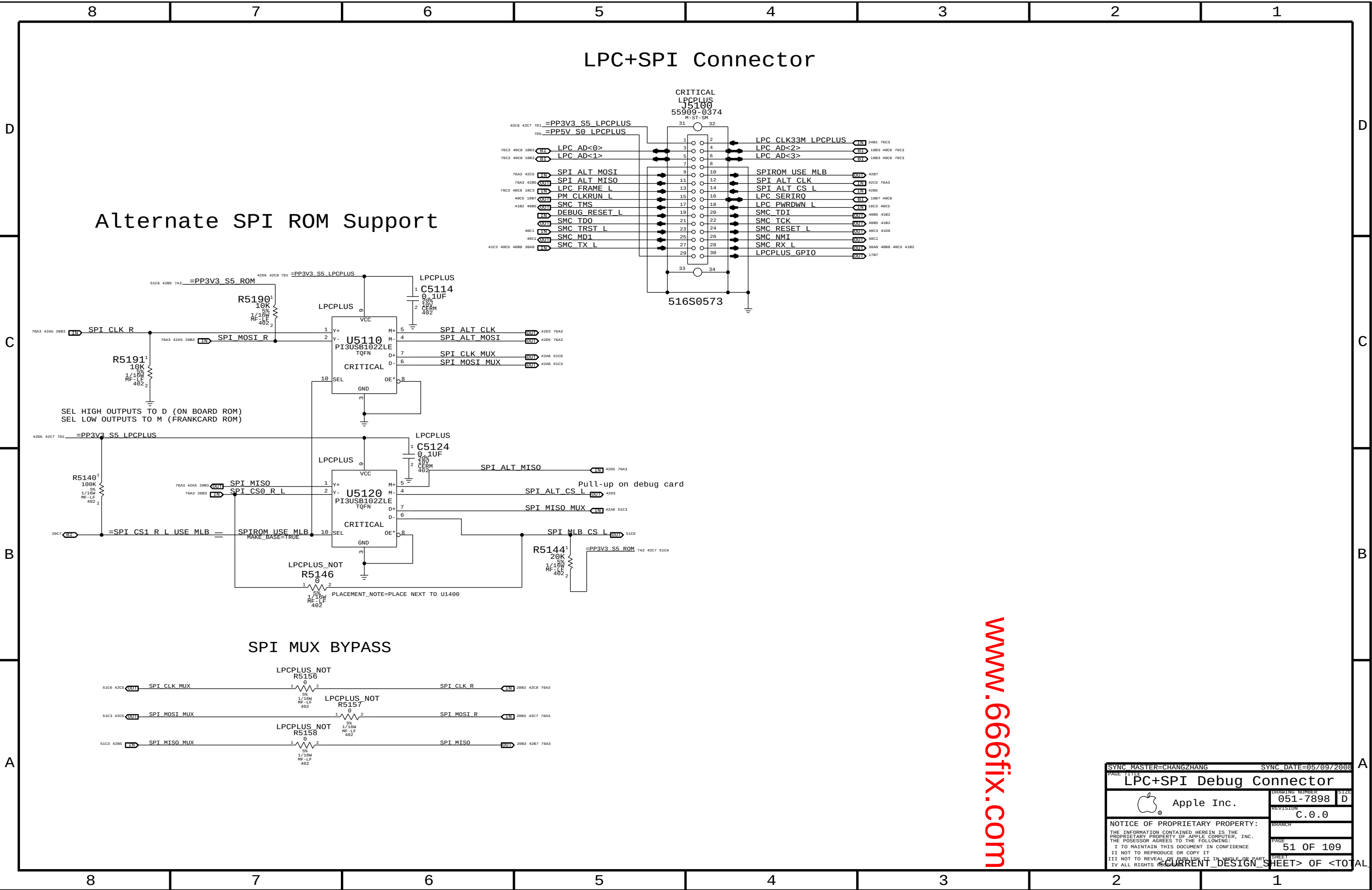
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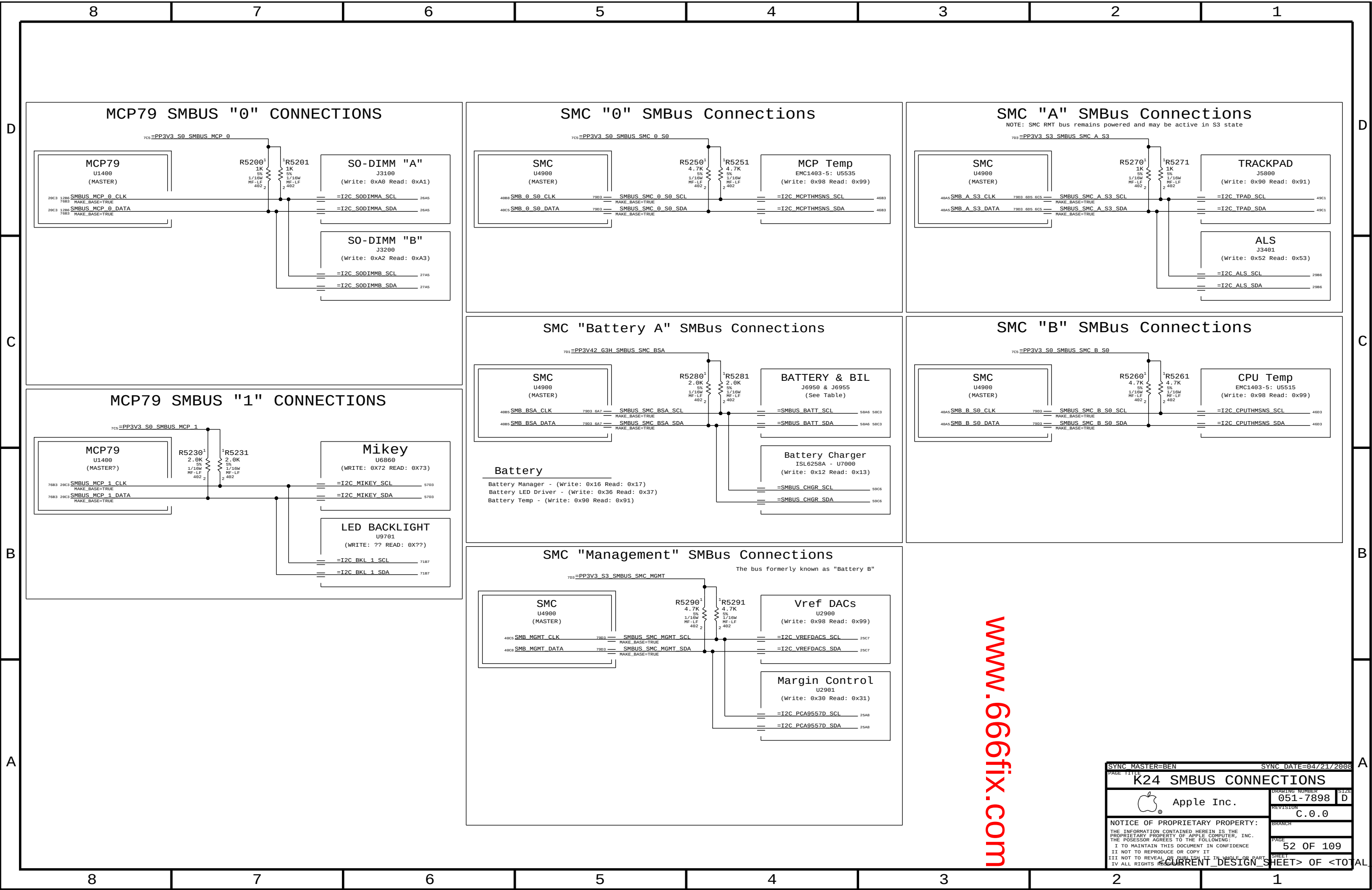
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PAGE TITLE			
Front Flex Support			
	Apple Inc.		
	DRAWING NUMBER		SIZE
		051-7898	D
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		BRANCH	
		PAGE	
		48 OF 109	
		SHEET	
CURRENT DESIGN SHEET		OF <TOTAL>	

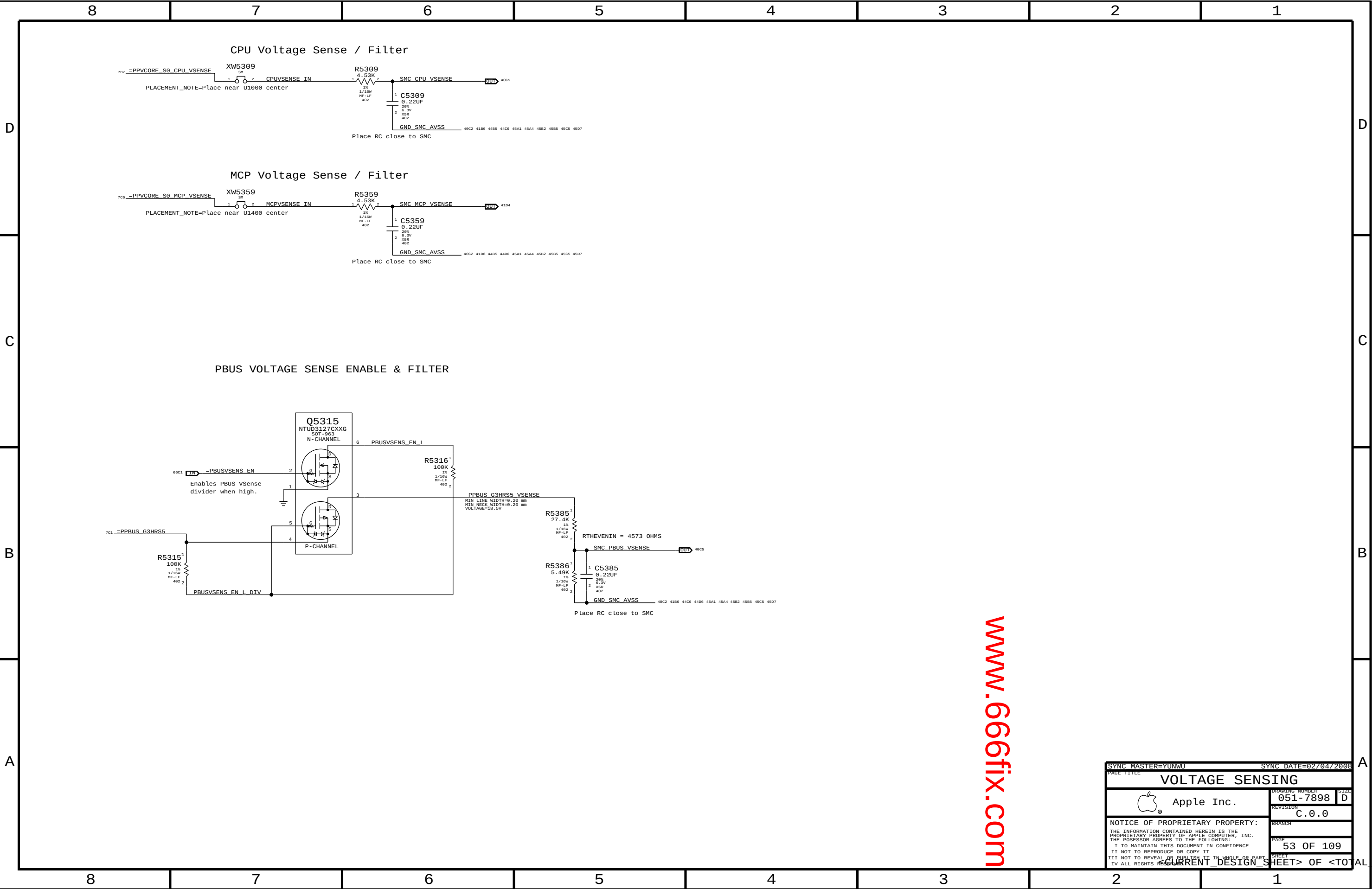


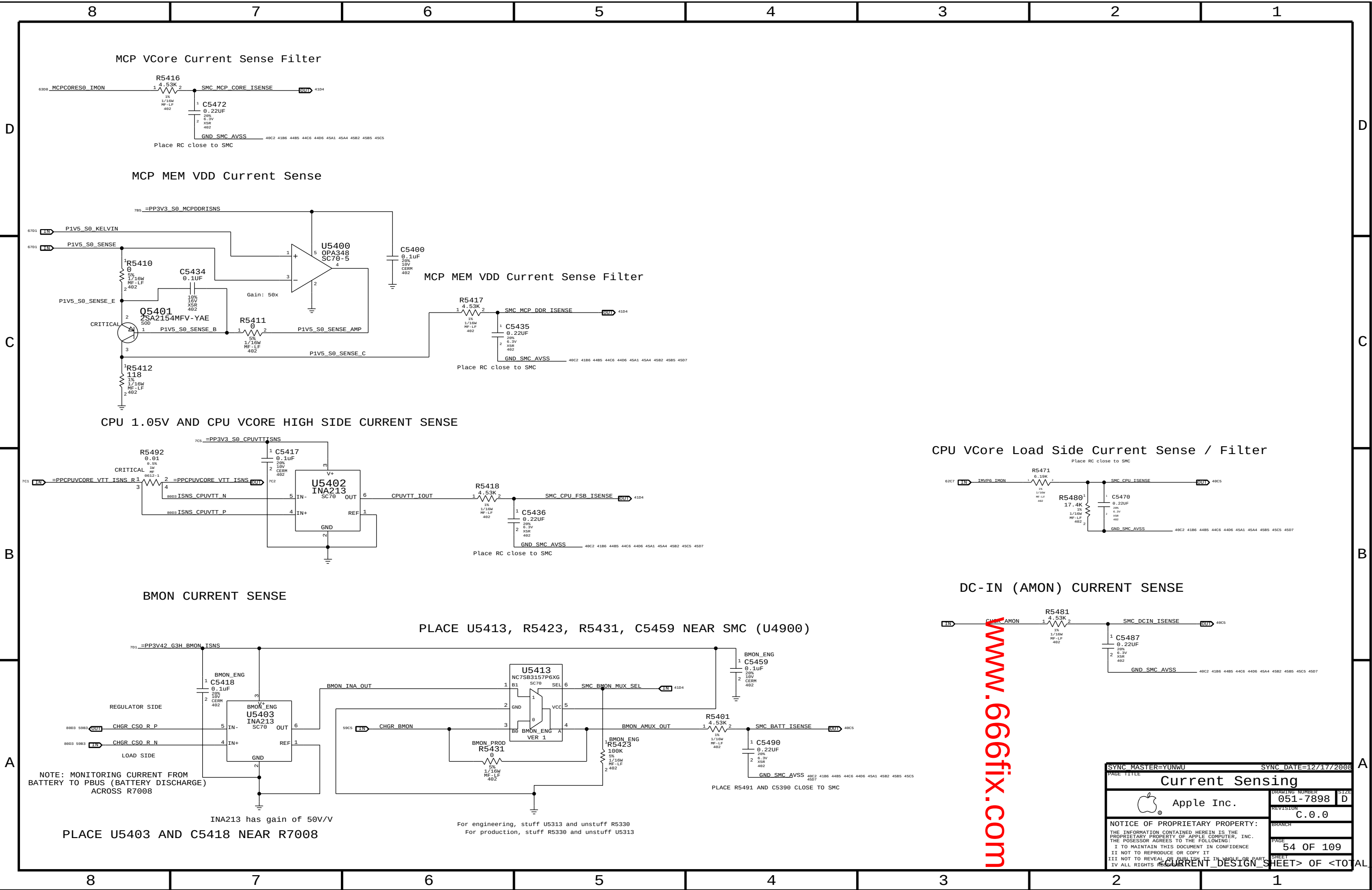
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S1381	353S1912		ALL	ISL60002-33, INTERSIL

SYNC MASTER=YUAN.MA		SYNC DATE=05/28/2008			
PAGE TITLE					
SMC Support					
DRAWING NUMBER		051-7898	SIZE D		
REVISION		C.0.0			
BRANCH					
PAGE		50 OF 109	SHEET		
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D



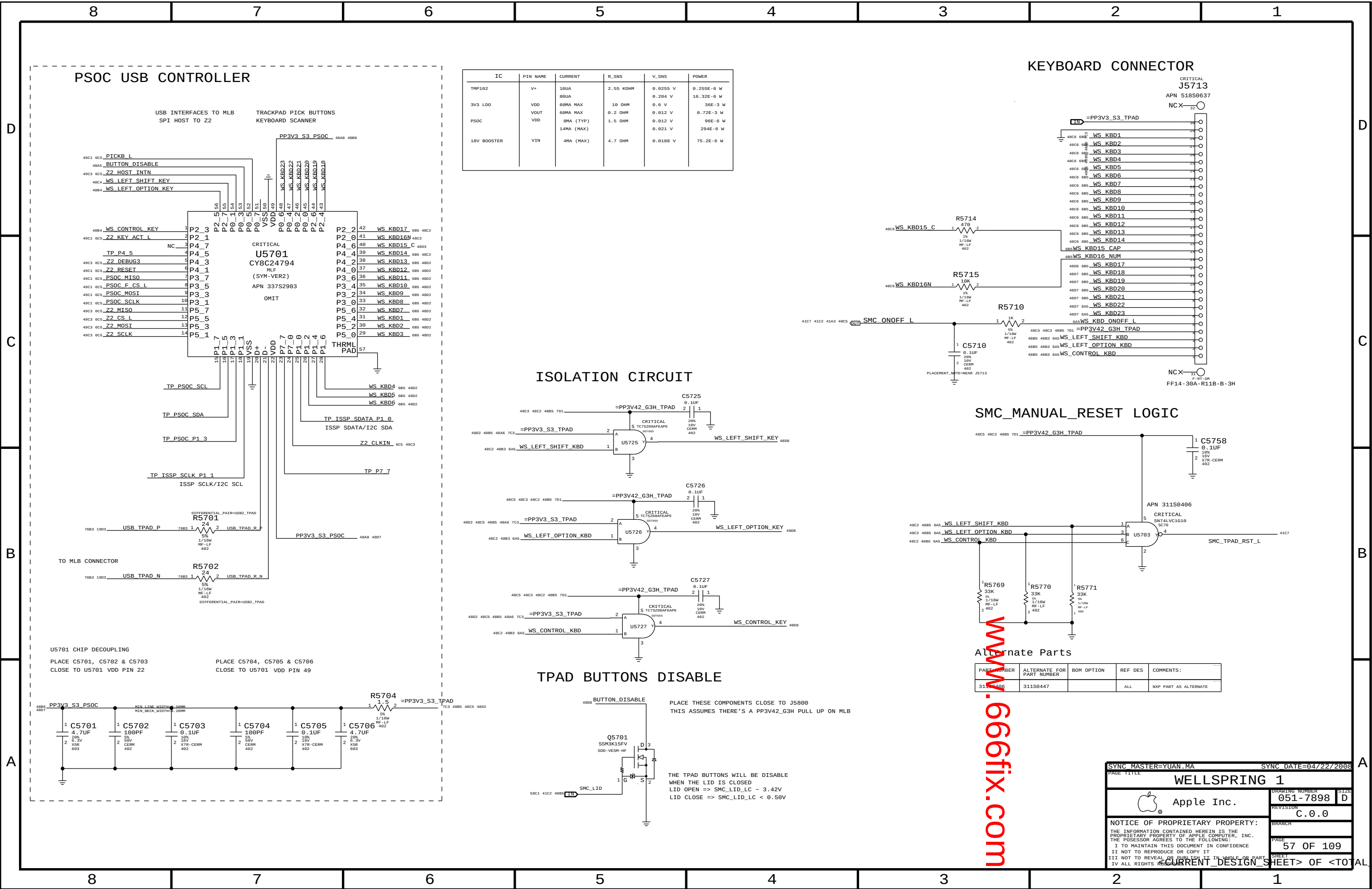
C

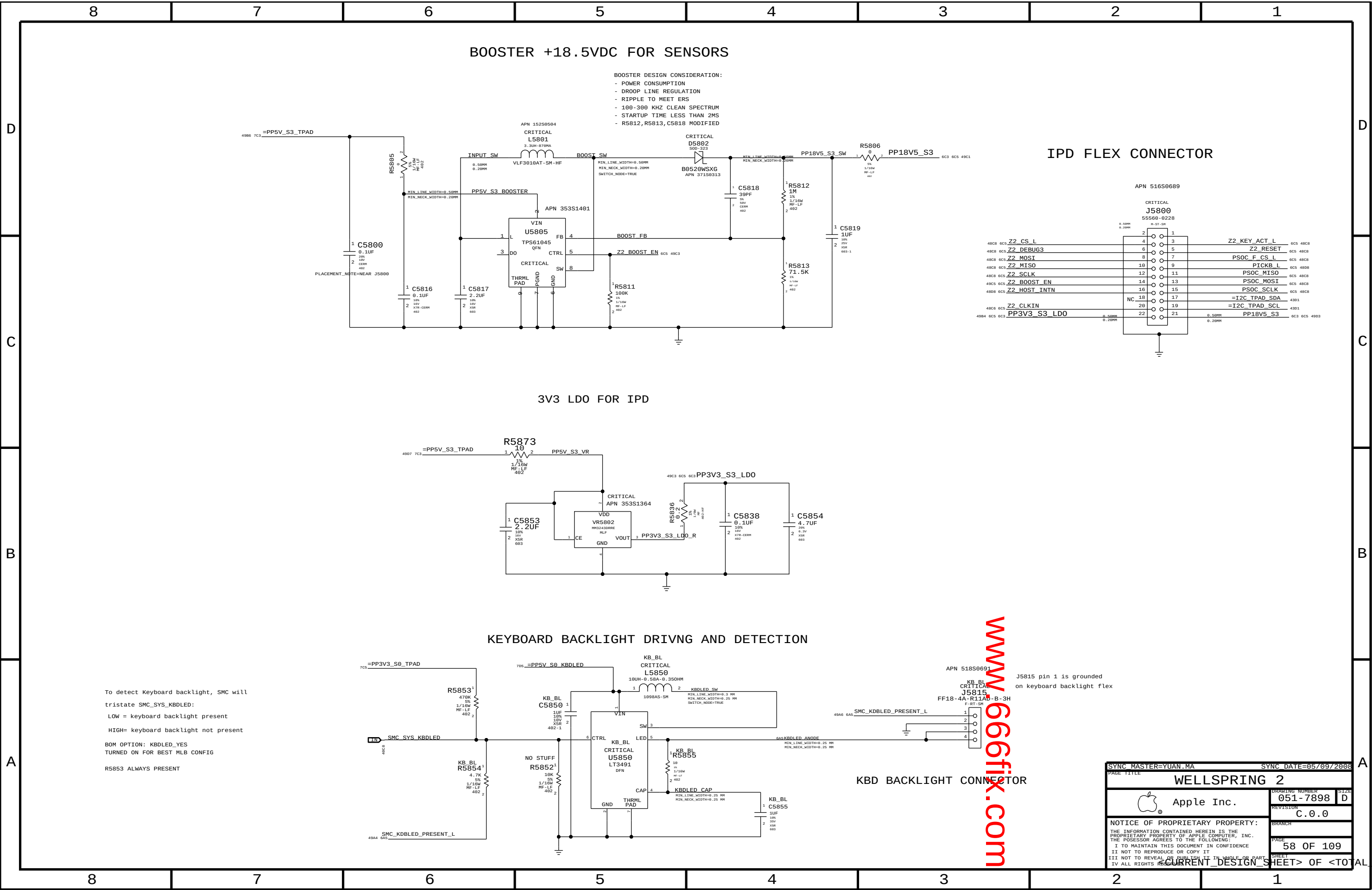


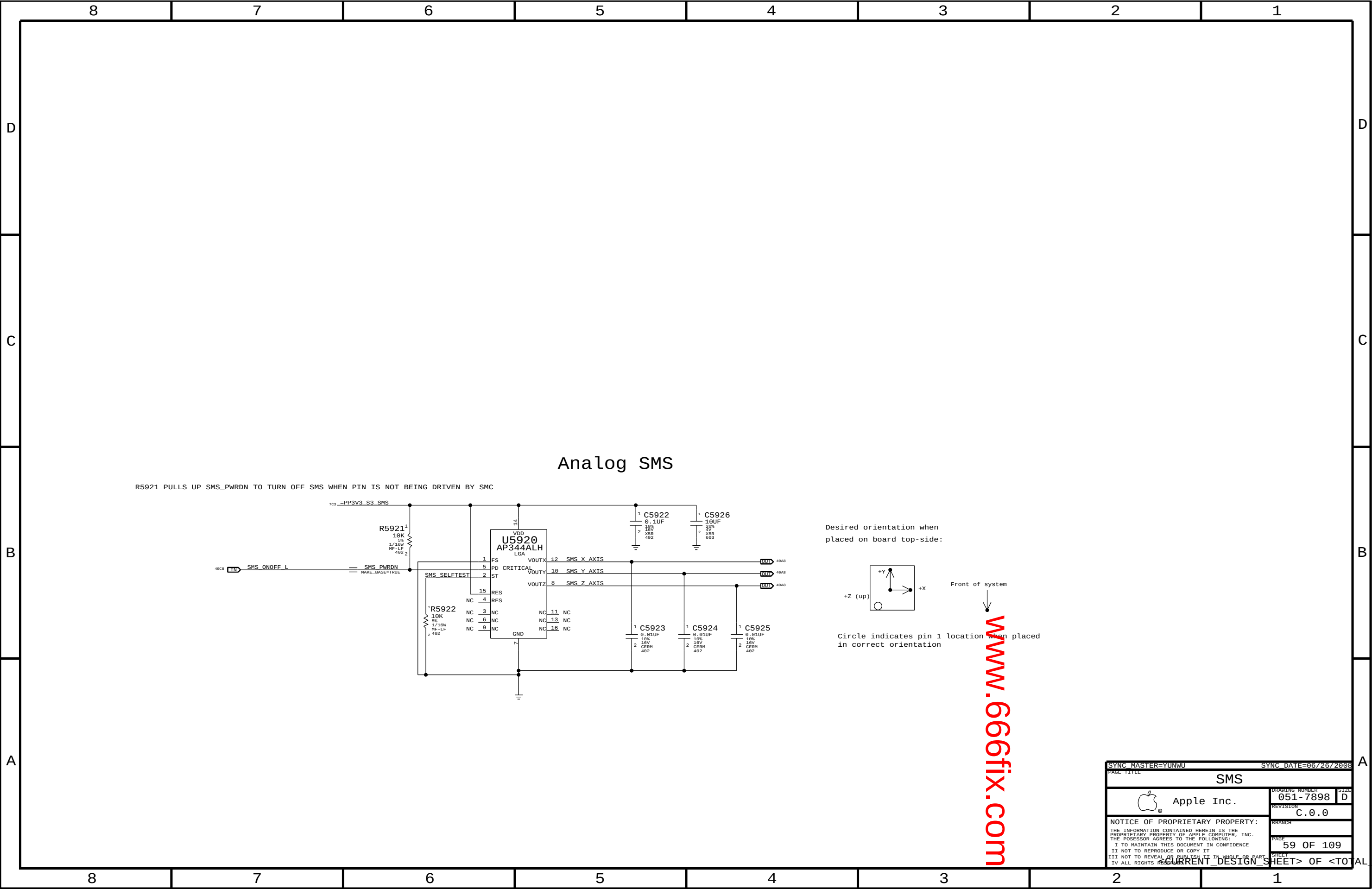
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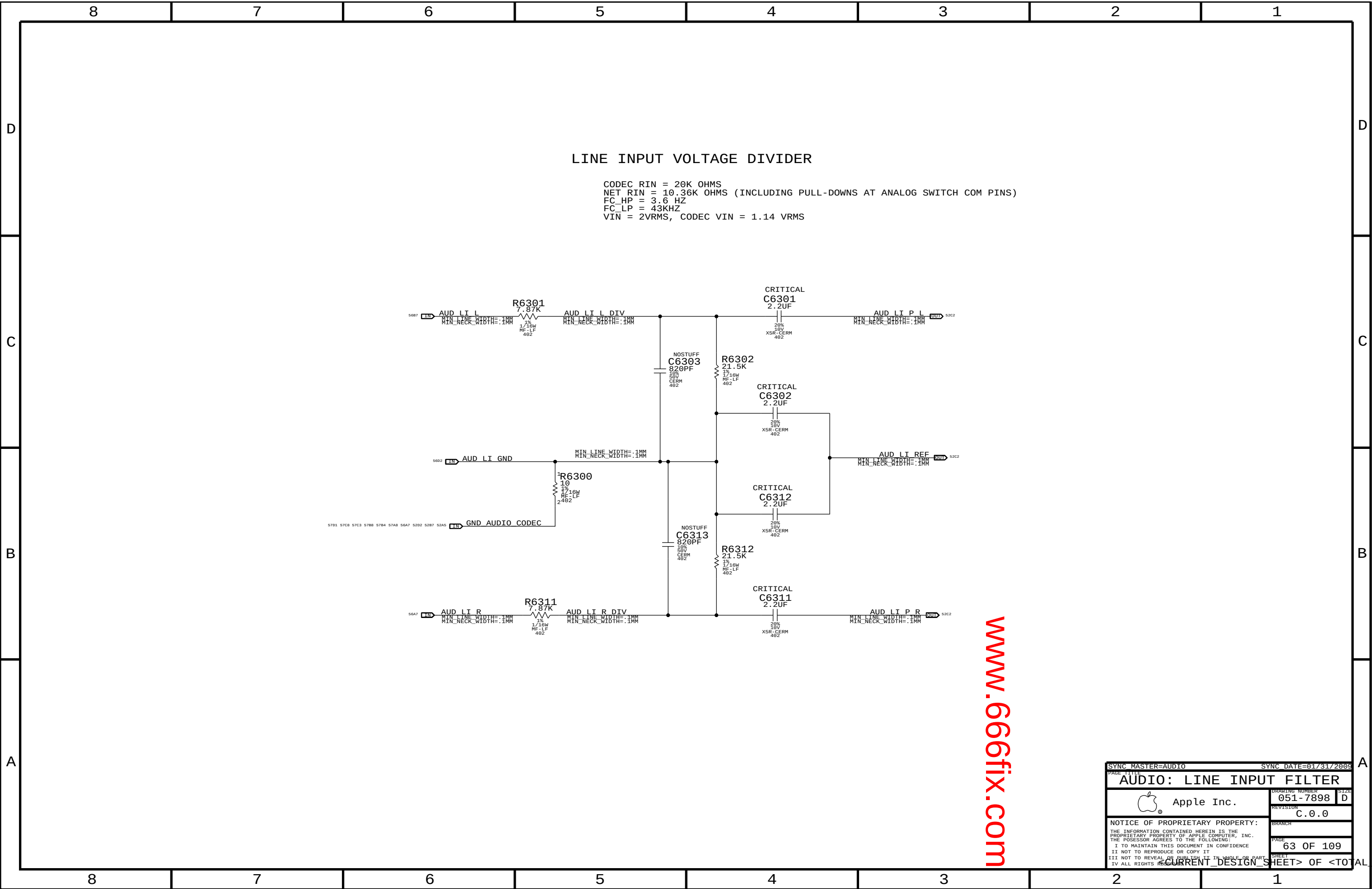
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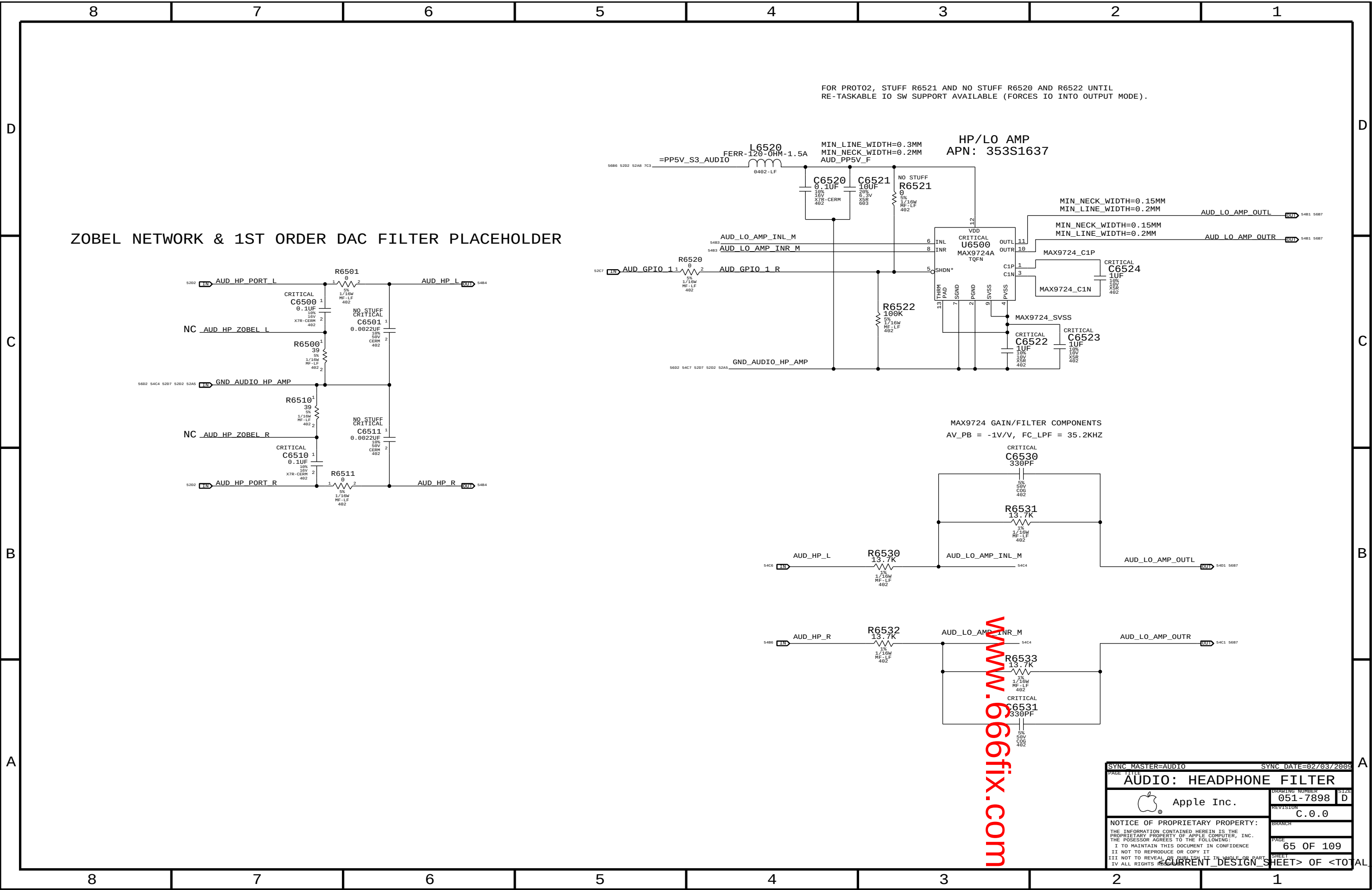
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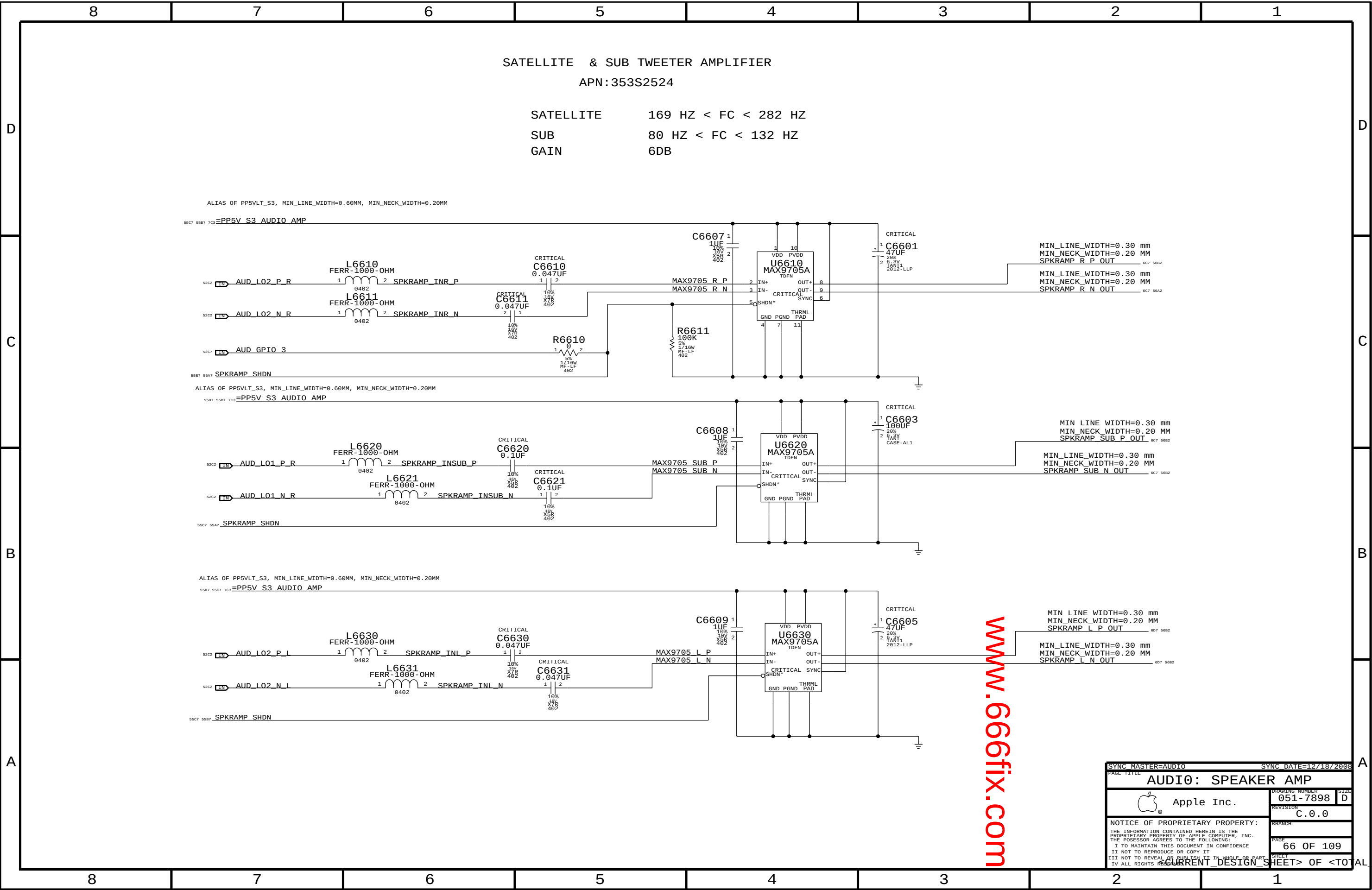




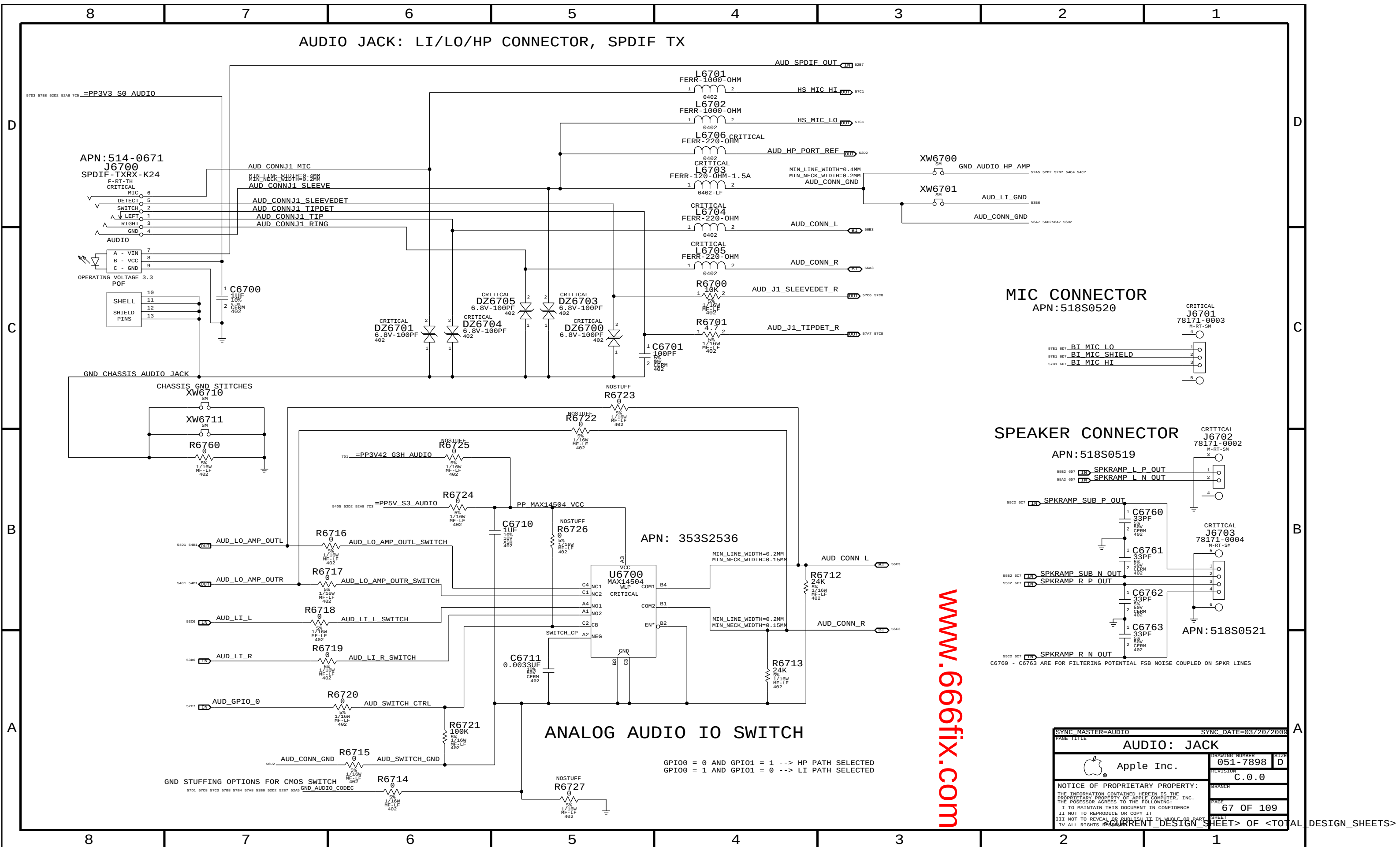


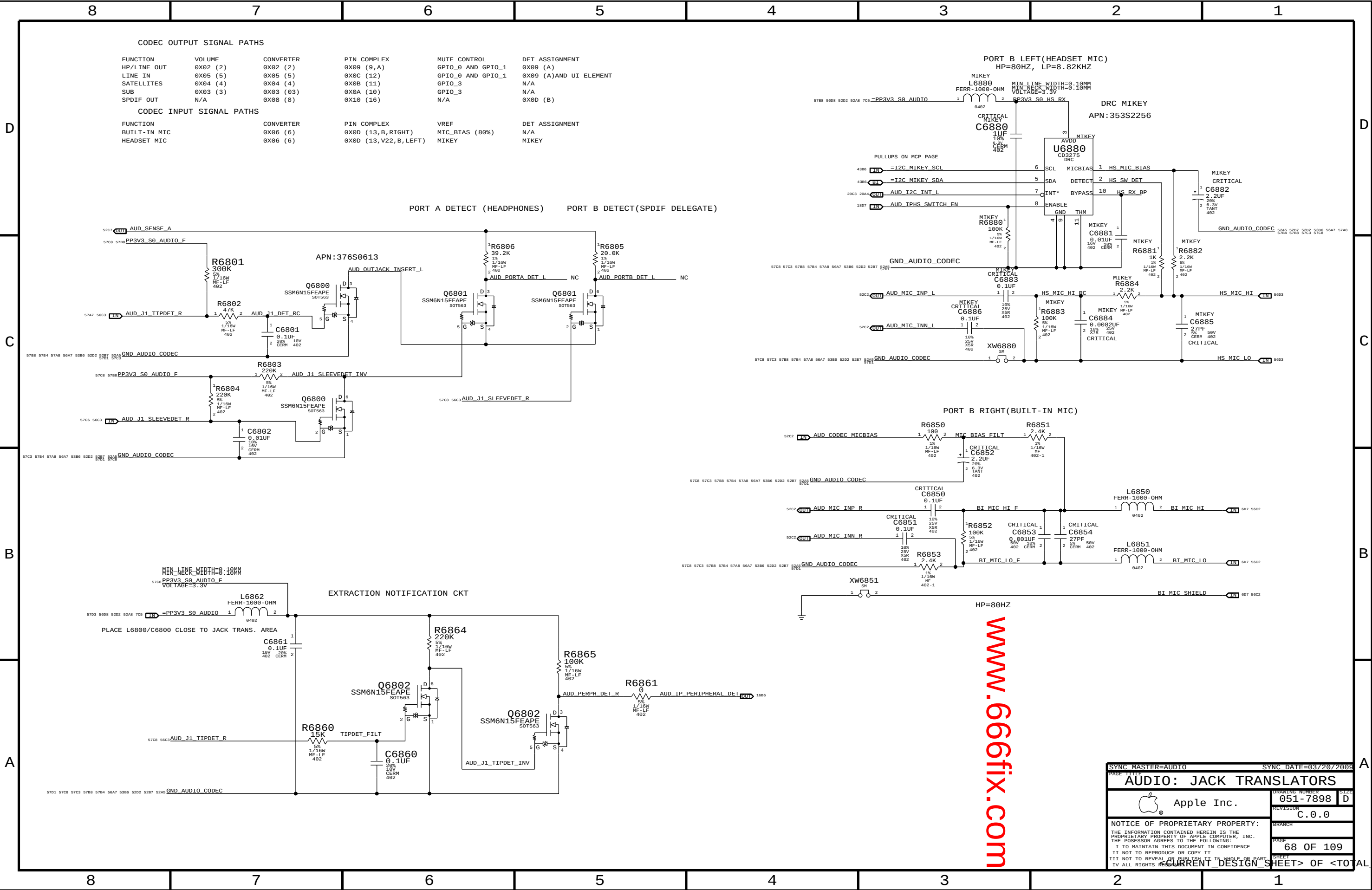




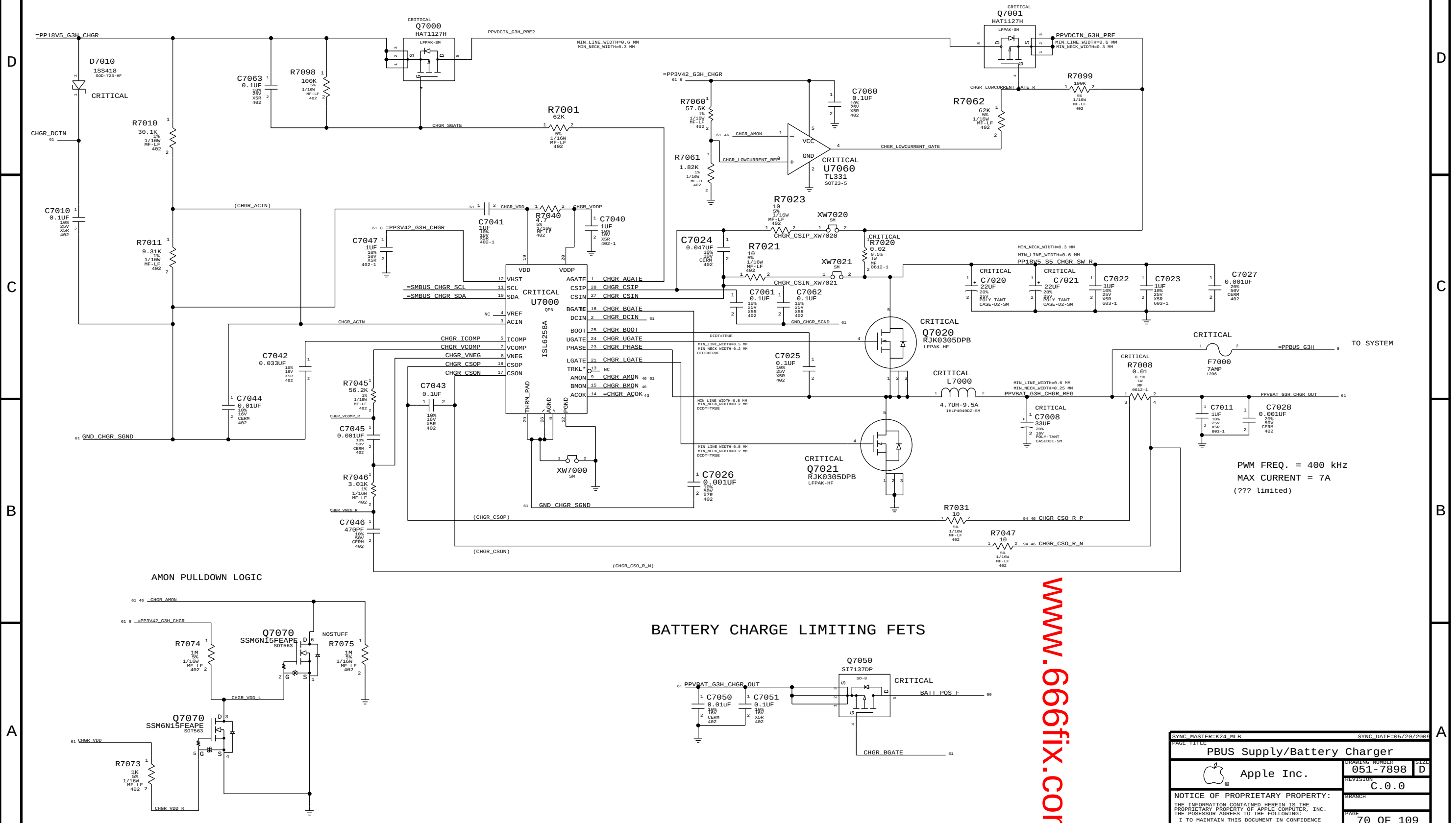


SYNC MASTER=AUDIO0		SYNC DATE=12/18/2008	
PAGE TITLE			
AUDIO0: SPEAKER AMP			
	Apple Inc.		DRAWING NUMBER
			051-7898
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PBUS SUPPLY / BATTERY CHARGER



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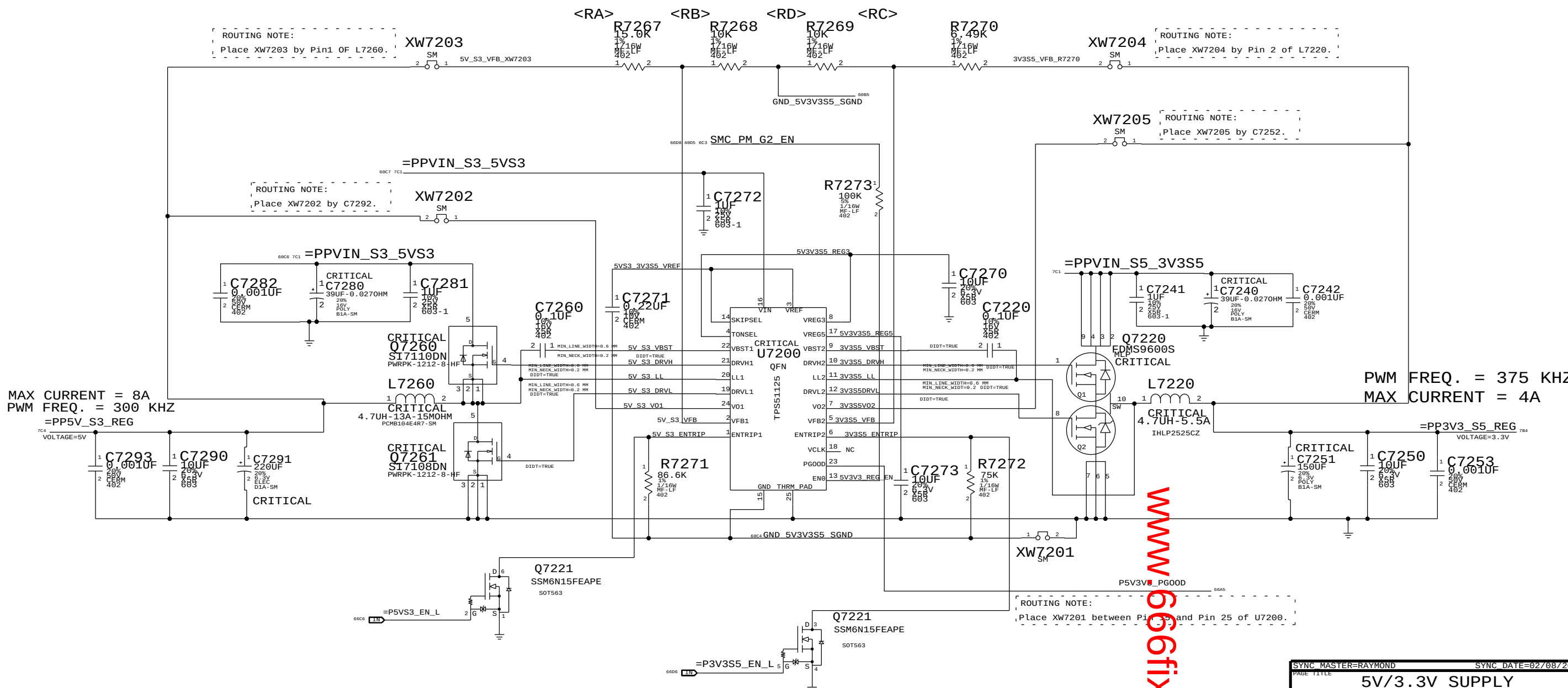
SYNC_MASTER=K24_MLB		SYNC_DATE=05/20/2003	
PAGE TITLE		PBUS Supply/Battery Charger	
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5V_S3/3.3V_S5 POWER SUPPLY

$$V_{OUT} = (2 * RA / RB) + 2$$

$$V_{OUT} = (2 * RC / RD) + 2$$



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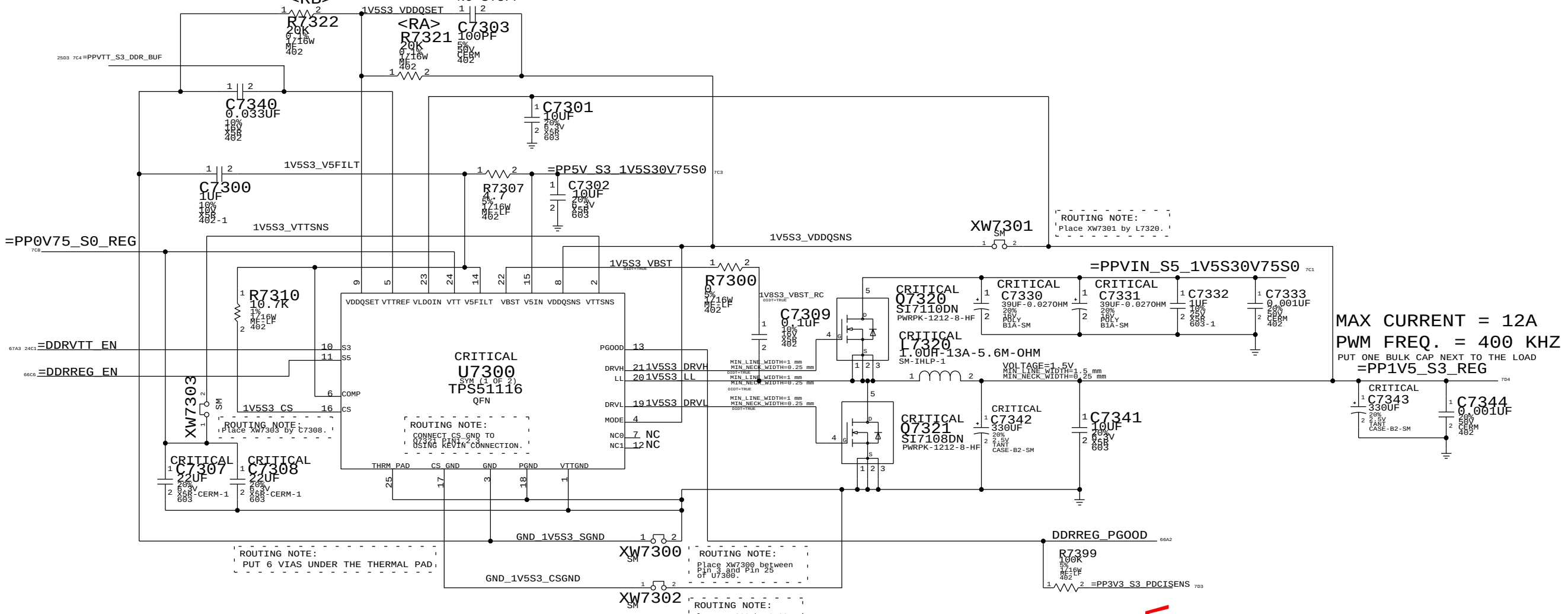
SYNC MASTER=RAYMOND		SYNC DATE=02/08/2008	
PAGE TITLE		5V/3.3V SUPPLY	
DRAWING NUMBER		051-7898	D
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SEPERATED MASTER PG000 FOR BOTH 5V AND 3V3.

<CURRENT DESIGN SHEET> OF <TOTAL DESIGN SHEETS>

1.5V/0.75V(DDR3) POWER SUPPLY

$$V_{OUT} = 0.75V * (1 + R_A / R_B)$$



MAX CURRENT = 12A
PWM FREQ. = 400 KHZ
PUT ONE BULK CAP NEXT TO THE LOAD
=PP1V5_S3_REG

STATE	PM_SLP_S4_L	PM_SLP_S3_L	PP1V5_S3	PP0V75_S0
S0	HIGH	HIGH	1.5V	0.75V
S3	HIGH	LOW	1.5V	0.0V
S5/G3HOT	LOW	LOW	0.0V	0.0V

SYNC MASTER=RAYMOND

SYNC DATE=01/31/2008

1.5V/0.75V DDR3 SUPPLY

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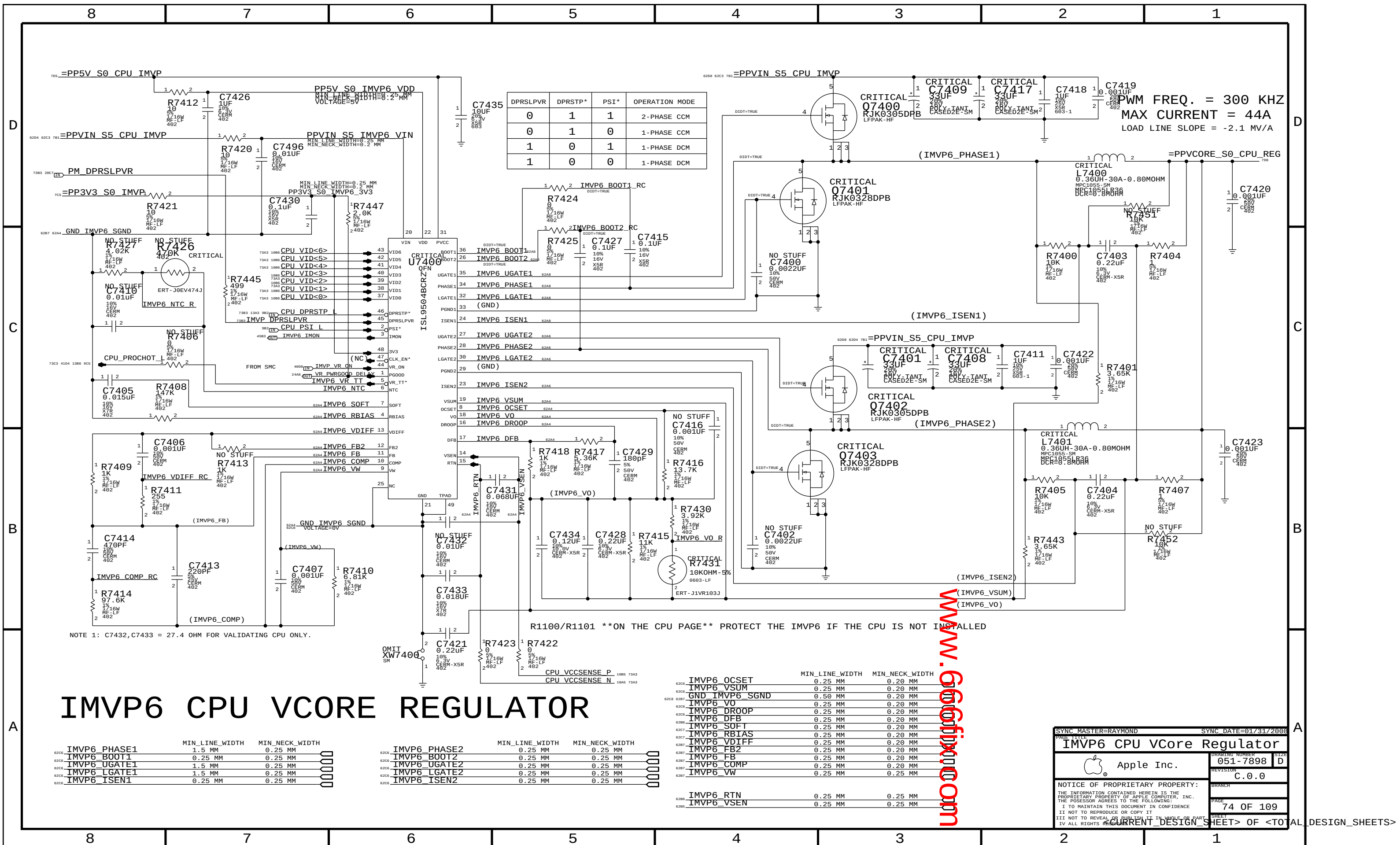
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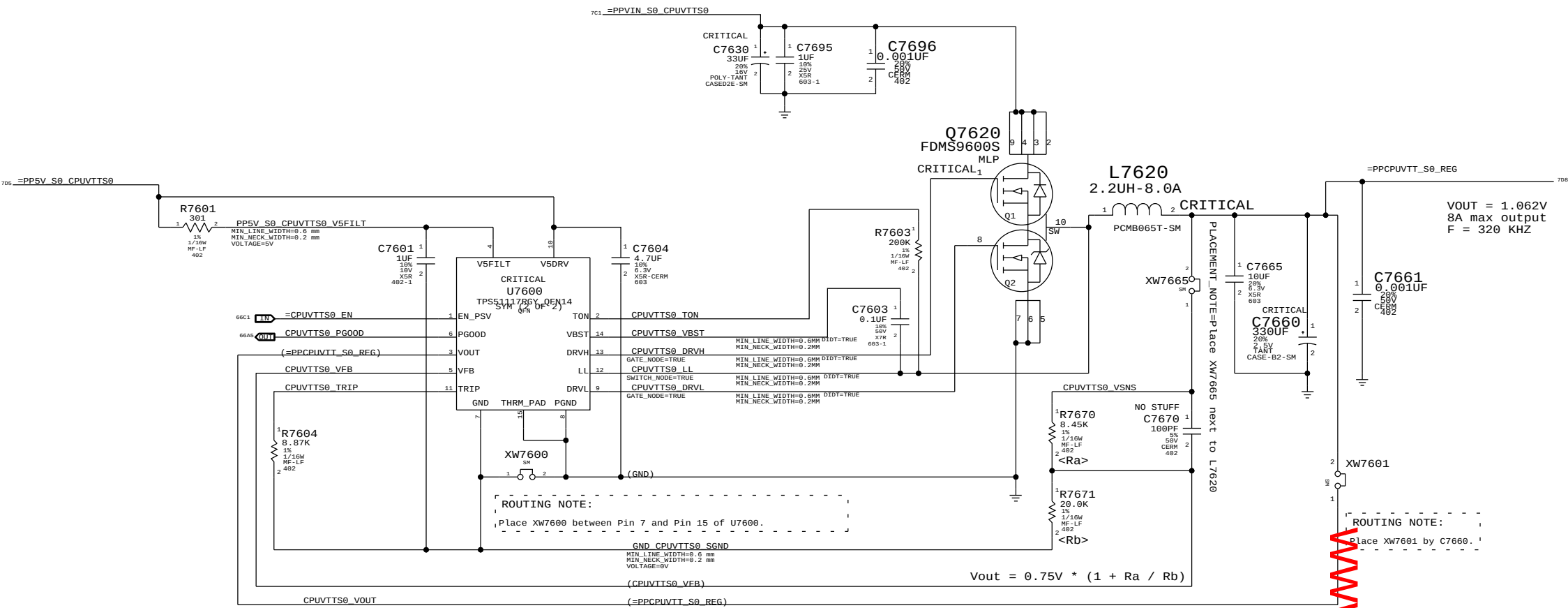
[illegible]

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VID<2:0>	MCP_TARGET
000	+1.05V
001	+1.00V
010	+0.95V
011	+0.90V
100	+0.85V
101	+0.80V
110	+0.75V
111	+0.70V

SYNCH MASTER=K19 MLB		SYNCH DATE=12/16/2008	
PAGE TITLE			
MCP CORE REGULATOR			
 Apple Inc.	DRAWING NUMBER	SIZE	
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CPUVTT POWER SUPPLY

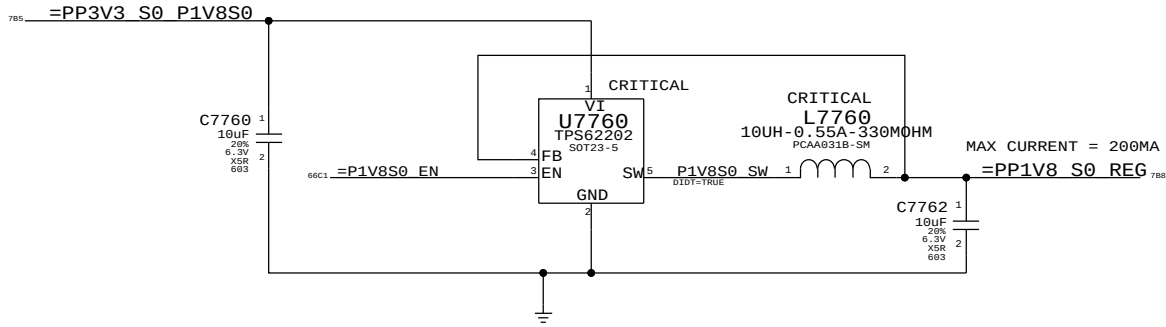


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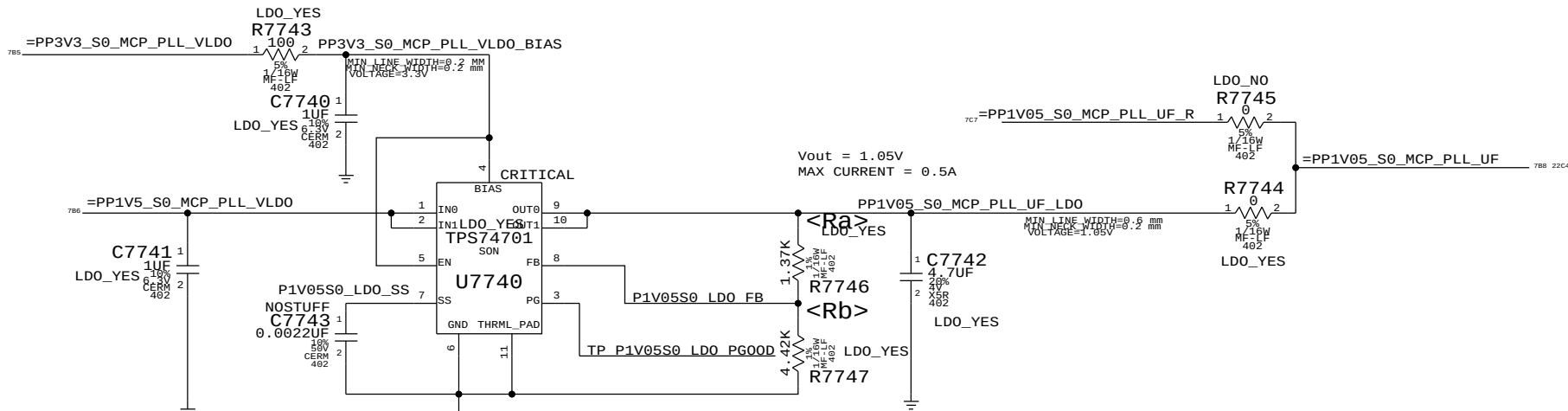
SYNC MASTER=RAYMOND		SYNC DATE=02/08/2008	
PAGE TITLE		CPU VTT(1.05V) SUPPLY	
DRAWING NUMBER		051-7898	D
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1.8V S0 SWITCHER

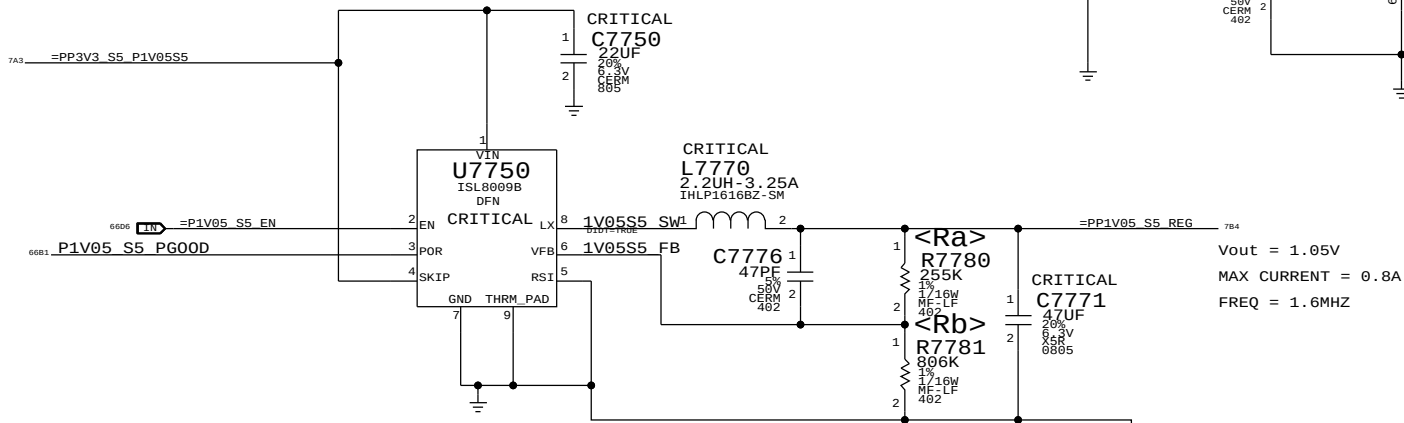


1.05V S0 PLL LDO



$$V_{OUT} = 0.8V * (1 + R_A / R_B)$$

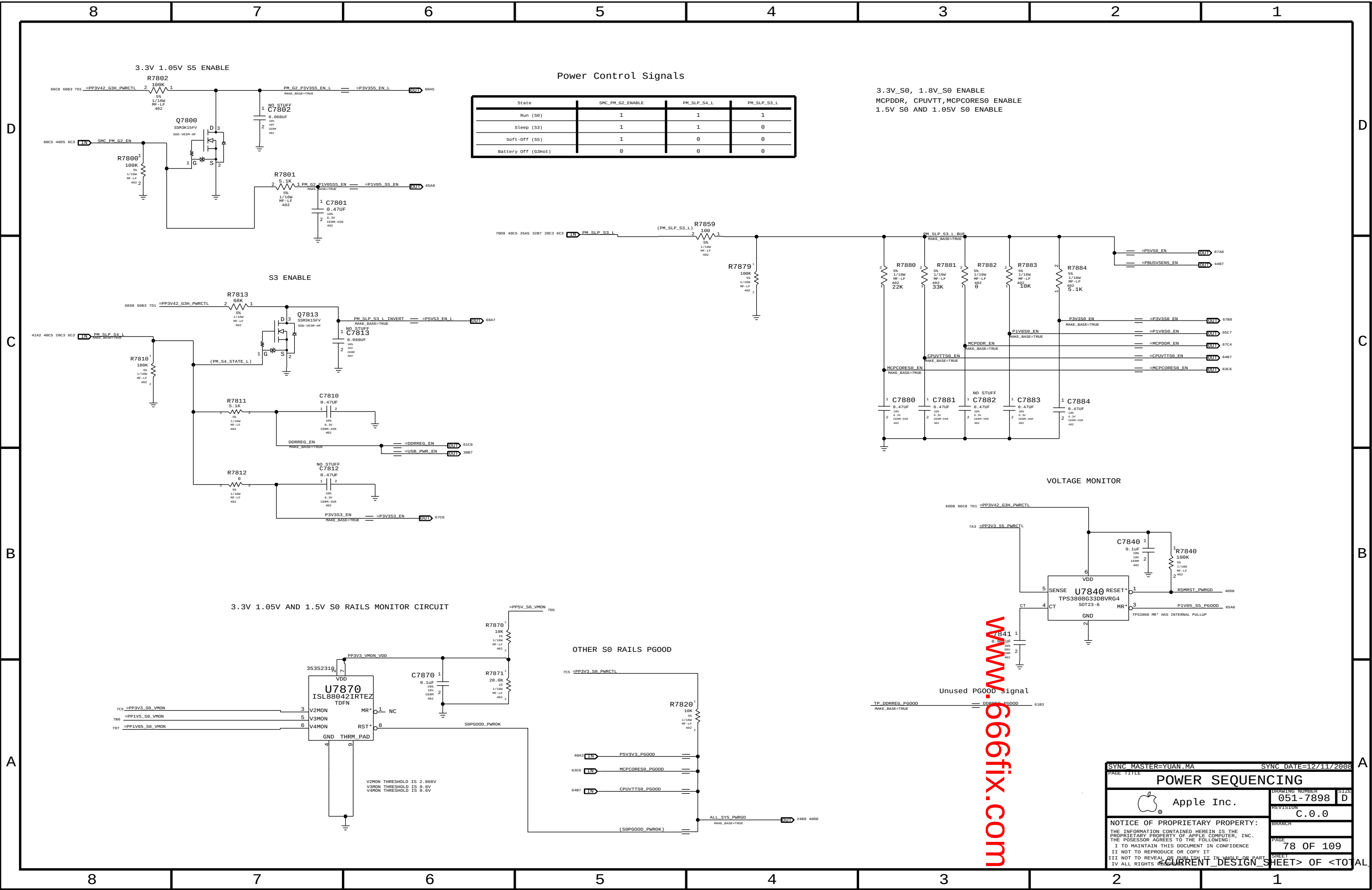
MCP 1.05V S5 (AUXC) SUPPLY

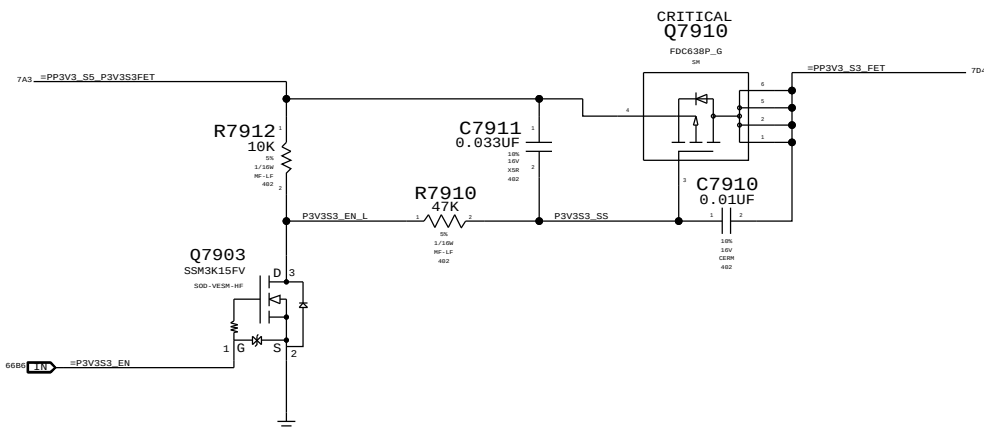


$$V_{OUT} = 0.8V * (1 + R_A / R_B)$$

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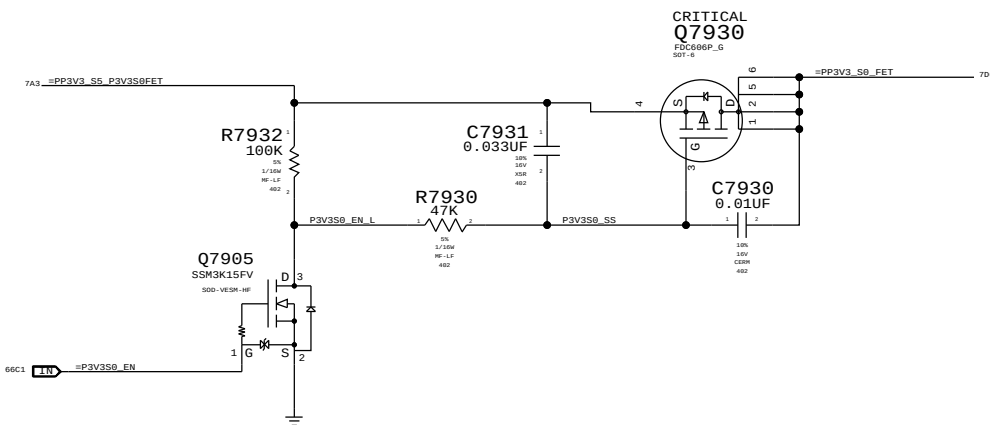
SYNC MASTER=RAYMOND		SYNC DATE=01/23/2008	
PAGE TITLE			
MISC POWER SUPPLIES			
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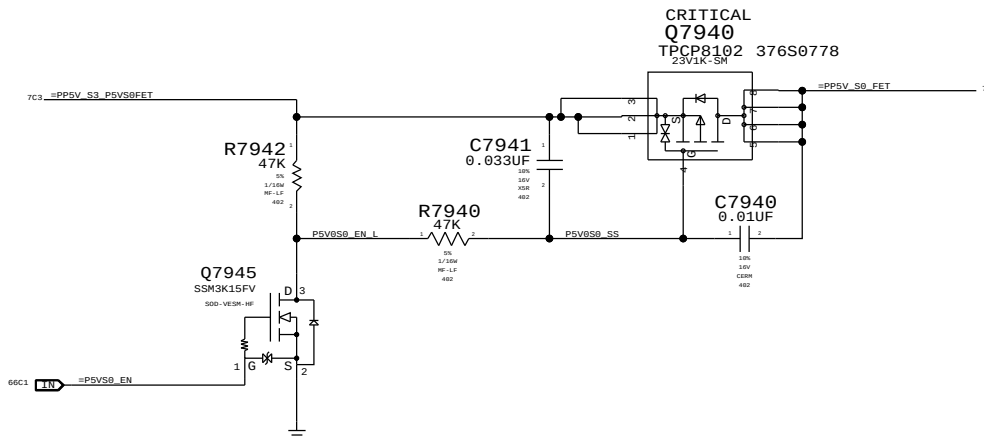
3.3V S3 FET

MOSFET	FDC638P
CHANNEL	P - TYPE
RDS(ON)	48 mOhm @4.5V
LOADING	0.182 A (EDP)



3.3V S0 FET

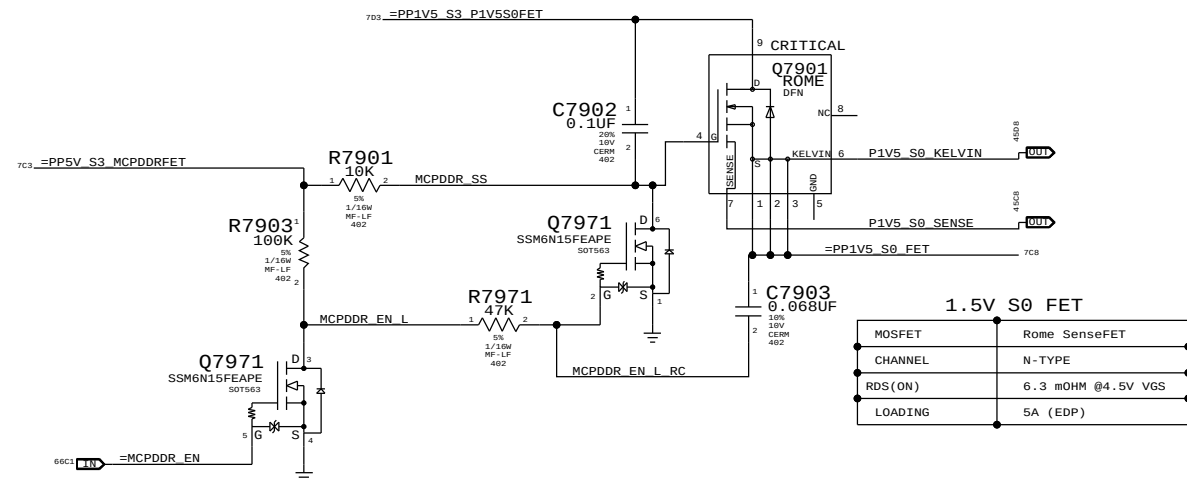
MOSFET	FDC606P
CHANNEL	P-TYPE
RDS(ON)	26 MOHM @4.5V
LOADING	1.431 A (EDP)



5.0V S0 FET

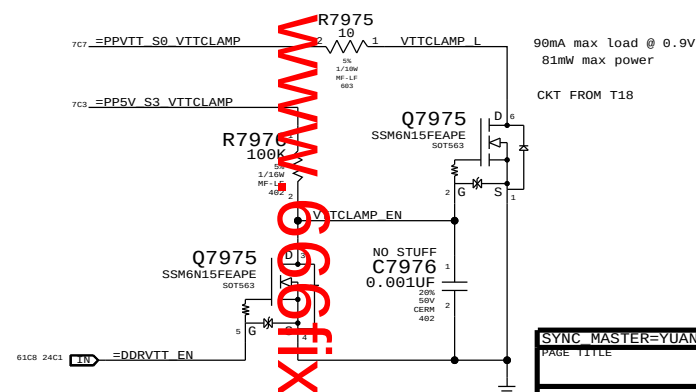
MOSFET	TPCP8102
CHANNEL	P- TYPE
RDS(ON)	13.5 MOHM @4.5V
LOADING	1.7 A (EDP)

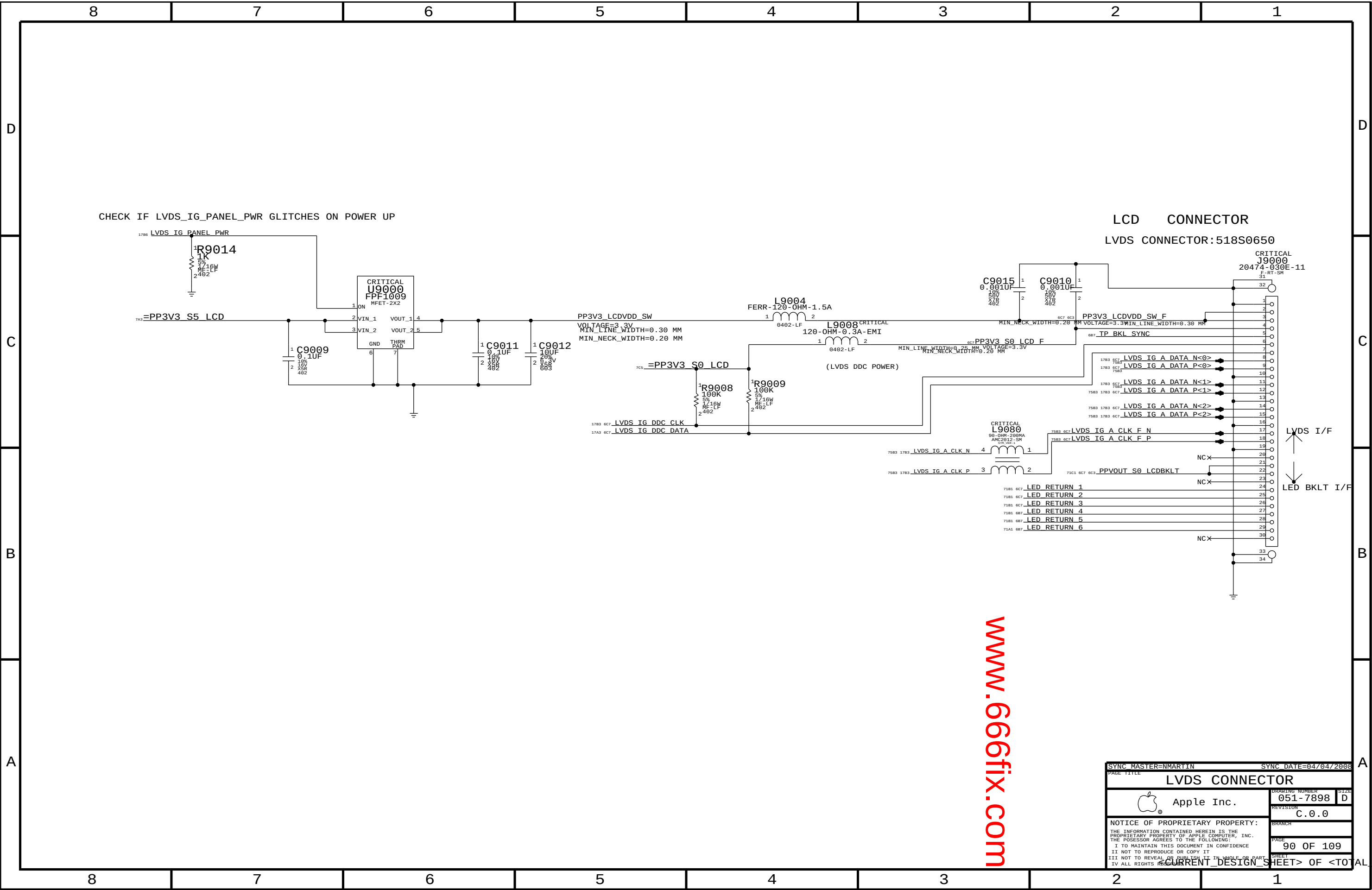
(1.5V S0 FET FOR DDR3 MEM, MCP79 AND CPU)



1.5V S0 FET	
MOSFET	Rome SenseFET
CHANNEL	N - TYPE
RDS(ON)	6.3 mOHM @4.5V VGS
LOADING	5A (EDP)

MCP79 DDR PAD LEAKAGE IS HIGH ENOUGH THAT NVIDIA RECOMMENDS UNPOWERING DURING SLEEP. IN ORDER TO SUPPORT UNPOWERING RAIL, HARDWARE MUST GUARANTEE MEM_CKE SIGNALS ARE LOW BEFORE RAIL IS TURNED OFF, AND REMAINS LOW UNTIL AFTER RAIL TURNS BACK ON OR DIMMS WILL EXIT SELF-REFRESH PREMATURELY. MEM_VTT_EN OUTPUT FROM MCP79 USED TO ENABLE CLAMP ON VTT RAIL, WHICH PULLS ALL CKE SIGNALS LOW THROUGH VTT TERMINATION RESISTORS.

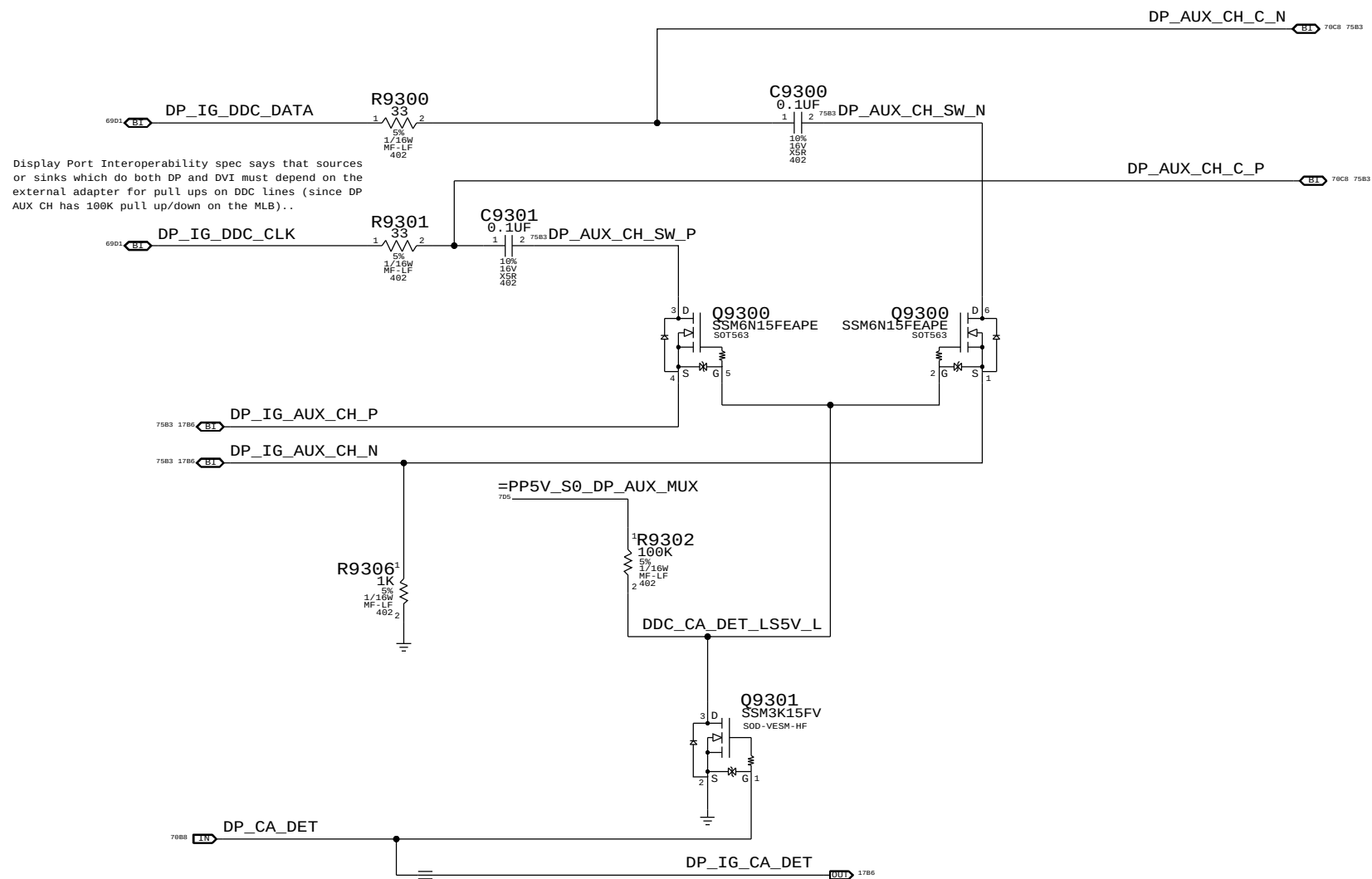




SYNC MASTER=NMARTIN		SYNC DATE=04/04/2008	
PAGE TITLE			
LVDS CONNECTOR			
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		REVISION	C.0.0
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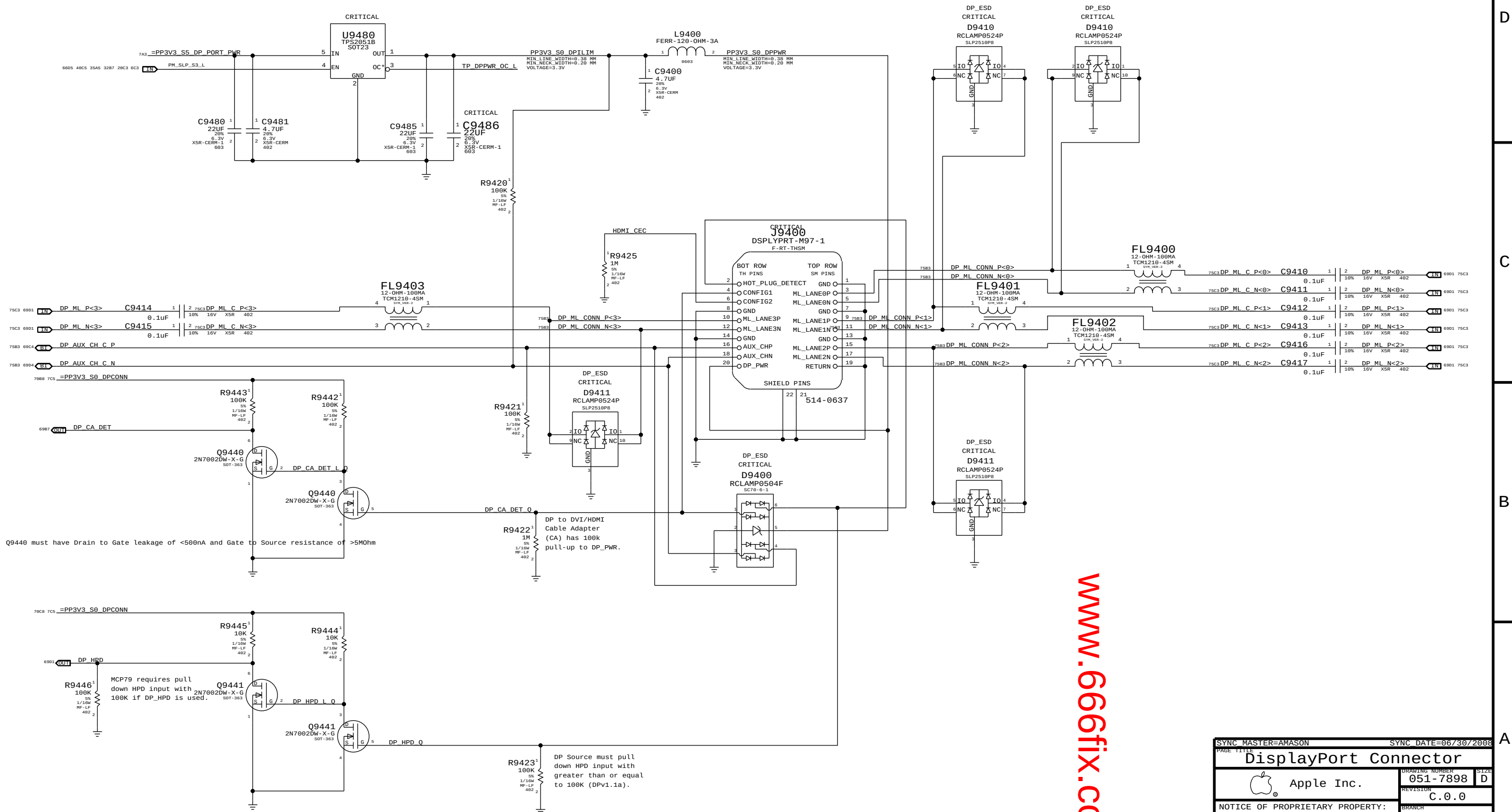
1766	=MCP_HDMI_TXC_P	DP_ML_P<3>	78C8 75C3
1766	=MCP_HDMI_TXC_N	DP_ML_N<3>	MAKE_BASE=TRUE 78C9 75C3
1766	=MCP_HDMI_TXD_P<0>	DP_ML_P<2>	MAKE_BASE=TRUE 78C9 75C3
1766	=MCP_HDMI_TXD_N<0>	DP_ML_N<2>	MAKE_BASE=TRUE 78C9 75C3
1766	=MCP_HDMI_TXD_P<1>	DP_ML_P<1>	MAKE_BASE=TRUE 78C9 75C3
1766	=MCP_HDMI_TXD_N<1>	DP_ML_N<1>	MAKE_BASE=TRUE 78C9 75C3
1766	=MCP_HDMI_TXD_P<2>	DP_ML_P<0>	MAKE_BASE=TRUE 78C9 75C3
1766	=MCP_HDMI_TXD_N<2>	DP_ML_N<0>	MAKE_BASE=TRUE 78C9 75C3
1766	=MCP_HDMI_HPD	DP_HPD	MAKE_BASE=TRUE 78A8
1766	=MCP_HDMI_DDC_CLK	DP_IG_DDC_CLK	MAKE_BASE=TRUE 69C8
17A3	=MCP_HDMI_DDC_CLK	DP_IG_DDC_CLK	MAKE_BASE=TRUE 69C8
17A3	=MCP_HDMI_DDC_DATA	DP_IG_DDC_DATA	MAKE_BASE=TRUE 69C8



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SYNC MASTER=AMASON		SYNC DATE=04/18/2008	
PAGE TITLE			
DISPLAYPORT SUPPORT			
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Port Power Switch



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DisplayPort Connector					
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	6	RES, THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402, SM	R9717, R9718, R9719, R9720, R9721, R9722		BKLT_ENG
116S0005	1	RES, 1/16W, 0.1 OHM, 1%, 0402, SM	R9700		BKLT_ENG

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S2670	1	IC, LP8543, WHT LED BKLT CTRLR, QFN24, PROD	U9701	CRITICAL	

SYNC MASTER=KIRAN

SYNC DATE=12/05/2008

PAGE TITLE

LCD BACKLIGHT DRIVER

Apple Inc.

DRAWING NUMBER

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FSB (Front-Side Bus) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FSB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
FSB_DSTB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	*	=2X_DIELECTRIC	?
FSB_DSTB	*	=3X_DIELECTRIC	?
FSB_ADDR	*	=STANDARD	?
FSB_ADSB	*	=2X_DIELECTRIC	?
FSB_1X	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FSB_DATA	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_DSTB	TOP,BOTTOM	=5x_DIELECTRIC	?
FSB_ADDR	TOP,BOTTOM	=3x_DIELECTRIC	?
FSB_ADSTB	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_1X	TOP,BOTTOM	=3x_DIELECTRIC	?

All 4x/2x/1x FSB signals with impedance requirements are 50-ohm single-ended.

FSB 4X signals / groups shown in signal table on right.

Signals within each 4x group should be matched within 5 ps of strobe.

DSTB# complementary pairs should be matched within 1 ps of each other, all DSTB#s matched to +/- 300 ps.

Spacing is 2x dielectric between DATA#, DINV# signals, with 3x dielectric spacing to the DSTB#s

DSTB# complementary pairs are spaced normally and are NOT routed as differential pairs.

FSB 2X signals / groups shown in signal table on right.

Signals within each 2x group should be matched within 20 ps. ADTSB#s should be matched +/- 300 ps.

Spacing is 1x dielectric between ADDR#, REQ# signals, with 2x dielectric spacing to ADSTB#.

FSB 1X signals shown in signal table on right.

Signals within each 1x group should be matched to CPU clock, +0/-1000 mils.

Design Guide recommends each strobe/signal group is routed on the same layer.

Intel Design Guide recommends FSB signals be routed only on internal layers.

NOTE: Intel Design Guide allows closer spacing if signal lengths can be shortened.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2

SOURCE: Santa Rosa Platform DG, Rev 1.5 (#22294), Sections 4.2 & 4.3

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_S0S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGT_L	*	=STANDARD	?
CPU_BMIL	*	8 MIL	?
CPU_COMP	*	25 MIL	?
CPU_GTLREF	*	25 MIL	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	25 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP,BOTTOM	=2x_DIELECTRIC	?

SR DG recommends at least 25 mils, >50 mils preferred

Most CPU signals with impedance requirements are 55-ohm single-ended.

Some signals require 27.4-ohm single-ended impedance.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2

SOURCE: Santa Rosa Platform DG, Rev 0.9 (#20517), Sections 4.4 & 5.8.2.4

MCP_ESB_COMP_Signal_Constraints

MCP FSB COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_FSB_COMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.4

FSB Clock Constraints

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_FSB	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_FSB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.5

CPU / FSB Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
FSB_DATA_GROUP0	FSB_50S	FSB_DATA	FSB D L<15..0>	9C4 1303
FSB_DATA_GROUP0	FSB_50S	FSB_DATA	FSB DINV L<0>	9C4 1306
FSB_D5TB0	FSB_D5TB 50S	FSB_D5TB	FSB_D5TB L P<0>	9C4 1306
FSB_D5TB0	FSB_D5TB 50S	FSB_D5TB	FSB_D5TB L N<0>	9C4 1306
FSB_DATA_GROUP1	FSB_50S	FSB_DATA	FSB D L<31..16>	984 9C4 13C3 1303
FSB_DATA_GROUP1	FSB_50S	FSB_DATA	FSB DINV L<1>	984 1306
FSB_D5TB1	FSB_D5TB 50S	FSB_D5TB	FSB_D5TB L P<1>	984 1306
FSB_D5TB1	FSB_D5TB 50S	FSB_D5TB	FSB_D5TB L N<1>	984 1306
FSB_DATA_GROUP2	FSB_50S	FSB_DATA	FSB D L<47..32>	9C2 1383 13C3
FSB_DATA_GROUP2	FSB_50S	FSB_DATA	FSB DINV L<2>	9C2 1306
FSB_D5TB2	FSB_D5TB 50S	FSB_D5TB	FSB_D5TB L P<2>	9C2 1306
FSB_D5TB2	FSB_D5TB 50S	FSB_D5TB	FSB_D5TB L N<2>	9C2 1306
FSB_DATA_GROUP3	FSB_50S	FSB_DATA	FSB D L<63..48>	982 9C2 1383
FSB_DATA_GROUP3	FSB_50S	FSB_DATA	FSB DINV L<3>	982 1306
FSB_D5TB3	FSB_D5TB 50S	FSB_D5TB	FSB_D5TB L P<3>	982 1306
FSB_D5TB3	FSB_D5TB 50S	FSB_D5TB	FSB_D5TB L N<3>	982 1306
FSB_ADDR_GROUP0	FSB_50S	FSB_ADDR	FSB A L<16..3>	908 13C6 1306
FSB_ADDR_GROUP0	FSB_50S	FSB_ADDR	FSB REQ L<4..0>	906 1386
FSB_AD5TB0	FSB_50S	FSB_AD5TB	FSB_AD5TB L<0>	906 1386
FSB_ADDR_GROUP1	FSB_50S	FSB_ADDR	FSB A L<35..17>	9C8 908 13C6
FSB_AD5TB1	FSB_50S	FSB_AD5TB	FSB_AD5TB L<1>	9C8 1386
FSB_1X	FSB_50S	FSB_1X	FSB ADS L	906 1386
FSB_BREQ0_I	FSB_50S	FSB_1X	FSB_BREQ0 L	906 1386
FSB_BREQ1_I	FSB_50S	FSB_1X	FSB_BREQ1 L	1386
FSB_1X	FSB_50S	FSB_1X	FSB_BNR L	906 1386
FSB_1X	FSB_50S	FSB_1X	FSB_BPRT L	906 1383
FSB_1X	FSB_50S	FSB_1X	FSB_OBSY L	906 1386
FSB_1X	FSB_50S	FSB_1X	FSB_DEFER L	906 1383
FSB_1X	FSB_50S	FSB_1X	FSB_DRDY L	906 1386
FSB_1X	FSB_50S	FSB_1X	FSB_HIT L	906 1386
FSB_1X	FSB_50S	FSB_1X	FSB_HITM L	906 1386
FSB_1X	FSB_50S	FSB_1X	FSB_LOCK L	906 1386
FSB_CPURST_I	FSB_50S	FSB_1X	FSB_CPURST L	906 12C2 13A3
FSB_1X	FSB_50S	FSB_1X	FSB_RS L<2..0>	906 13A6
FSB_1X	FSB_50S	FSB_1X	FSB_TRDY L	906 1386
CPU_ASYNC	CPU_50S	CPU_AGTL	CPU_A20M L	9C8 13A3
CPU_BSEL_I	CPU_50S	CPU_AGTL	CPU_BSEL<2..0>	882 984
CPU_FERR_I	CPU_50S	CPU_BMTL	CPU_FERR L	9C8 1387
CPU_ASYNC	CPU_50S	CPU_AGTL	CPU_IGNNE L	9C8 13A3
CPU_INIT_I	CPU_50S	CPU_AGTL	CPU_INIT L	906 13A3
CPU_ASYNC_R	CPU_50S	CPU_AGTL	CPU_INTR	9C8 13A3
CPU_ASYNC_R	CPU_50S	CPU_AGTL	CPU_NMI	988 13A3
CPU_PROCHOT_I	CPU_50S	CPU_AGTL	CPU_PROCHOT L	9C5 1386 4104 82C6
CPU_PWRGD	CPU_50S	CPU_AGTL	CPU_PWRGD	982 12C7 13A3
CPU_ASYNC	CPU_50S	CPU_AGTL	CPU_SMT L	988 13A3
CPU_ASYNC	CPU_50S	CPU_AGTL	CPU_STPCLK L	9C8 13A3
PM_THRMTRIP_I	CPU_50S	CPU_BMTL	PM_THRMTRIP L	9C6 1387 41C4
FSB_CPUSLP_P_I	CPU_50S	CPU_AGTL	FSB_CPUSLP L	982 13A3
CPU_FERR_SB	CPU_50S	CPU_AGTL	CPU_DP5SLP L	982 13A3
CPU_DPRSTP_I	CPU_50S	CPU_AGTL	CPU_DPRSTP L	982 13A3 62C7
CPU_ASYNC	CPU_50S	CPU_AGTL	FSB_DPWR L	982 13A3
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_BCLK_VML_COMP_VDD	13A6
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_BCLK_VML_COMP_GND	13A6
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_CPU_COMP_VCC	13A6
MCP_CPU_COMP	MCP_50S	MCP_FSB_COMP	MCP_CPU_COMP_GND	13A6
FSB_CLK_CPU	CLK_FSB_108D	CLK_FSB	FSB_CLK_CPU_P	986 1383
FSB_CLK_CPU	CLK_FSB_108D	CLK_FSB	FSB_CLK_CPU_N	986 1383
FSB_CLK_TTP	CLK_FSB_108D	CLK_FSB	FSB_CLK_TTP_P	12C3 1383
FSB_CLK_TTP	CLK_FSB_108D	CLK_FSB	FSB_CLK_TTP_N	12C3 1383

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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_40S_VDD	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF
MEM_70D_VDD	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_20THER	*	25 MIL	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_DATA	*	*	MEM_20THER
MEM_DQS	*	*	MEM_20THER

Need to support MEM_*-style wildcards!

DDR2:

DQ signals should be matched within 20 ps of associated DQS pair.

DQS intra-pair matching should be within 1 ps, no inter-pair matching requirement.

All DQS pairs should be matched within 100 ps of clocks.

CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 140 ps.

A/BA/cmd signals should be matched within 75 ps, no CLK matching requirement.

All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate).

DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.

DDR3:

DQ signals should be matched within 5 ps of associated DQS pair.

DQS intra-pair matching should be within 1 ps, inter-pair matching should be within 180 ps

No DQS to clock matching requirement.

CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 2 ps.

A/BA/cmd signals should be matched within 5 ps of CLK pairs.

All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate).

DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.

SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.3

SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Section 6.2

MCP MEM COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MCP_MEM_COMP	*	Y	7 MIL	7 MIL	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MCP_MEM_COMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.3.4

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_70D_VDD	MEM_CLK	MEM A CLK P<5..0>	1485 26C5 26C7
MEM_A_CLK	MEM_70D_VDD	MEM_CLK	MEM A CLK N<5..0>	1485 26C5 26C7
MEM_A_CMTL	MEM_40S_VDD	MEM_CTRL	MEM A CKE<3..0>	1445 26D5 26D7
MEM_A_CMTL	MEM_40S_VDD	MEM_CTRL	MEM A CS L<3..0>	1485 26C5 26C7
MEM_A_CMTL	MEM_40S_VDD	MEM_CTRL	MEM A ODT<3..0>	1485 26C5
MEM_A_CMD	MEM_40S_VDD	MEM_CMD	MEM A A<14..0>	1485 14C5 26C5 26C7
MEM_A_CMD	MEM_40S_VDD	MEM_CMD	MEM A BA<2..0>	14C5 26C5 26C7
MEM_A_CMD	MEM_40S_VDD	MEM_CMD	MEM A RAS L	14C5 26C5
MEM_A_CMD	MEM_40S_VDD	MEM_CMD	MEM A CAS L	14C5 26C7
MEM_A_CMD	MEM_40S_VDD	MEM_CMD	MEM A WE L	14C5 26C7
MEM_A_DQ_BYTE0	MEM_40S	MEM_DATA	MEM A DQ<7..0>	1487 26C2 26C4 26D2 26D4
MEM_A_DQ_BYTE1	MEM_40S	MEM_DATA	MEM A DQ<15..8>	1487 26C2 26C4
MEM_A_DQ_BYTE2	MEM_40S	MEM_DATA	MEM A DQ<23..16>	1487 14C7 26B2 26B4 26C2 26C4
MEM_A_DQ_BYTE3	MEM_40S	MEM_DATA	MEM A DQ<31..24>	14C7 26C2 26C4
MEM_A_DQ_BYTE4	MEM_40S	MEM_DATA	MEM A DQ<39..32>	14C7 26B5 26B7 26C5 26C7
MEM_A_DQ_BYTE5	MEM_40S	MEM_DATA	MEM A DQ<47..40>	14C7 14D7 26B5 26B7
MEM_A_DQ_BYTE6	MEM_40S	MEM_DATA	MEM A DQ<55..48>	14D7 26B5 26B7
MEM_A_DQ_BYTE7	MEM_40S	MEM_DATA	MEM A DQ<63..56>	14D7 26A5 26A7 26B5 26B7
MEM_A_DQ_BYTE8	MEM_40S	MEM_DATA	MEM A DM<0>	14A7 26C4
MEM_A_DQ_BYTE1	MEM_40S	MEM_DATA	MEM A DM<1>	14A7 26C2
MEM_A_DQ_BYTE2	MEM_40S	MEM_DATA	MEM A DM<2>	14B7 26B4
MEM_A_DQ_BYTE3	MEM_40S	MEM_DATA	MEM A DM<3>	14B7 26C2
MEM_A_DQ_BYTE4	MEM_40S	MEM_DATA	MEM A DM<4>	14B7 26B5
MEM_A_DQ_BYTE5	MEM_40S	MEM_DATA	MEM A DM<5>	14B7 26B7
MEM_A_DQ_BYTE6	MEM_40S	MEM_DATA	MEM A DM<6>	14B7 26B5
MEM_A_DQ_BYTE7	MEM_40S	MEM_DATA	MEM A DM<7>	14B7 26A7
MEM_A_DQS0	MEM_70D	MEM_DQS	MEM A DQS P<0>	14D5 26C2
MEM_A_DQS0	MEM_70D	MEM_DQS	MEM A DQS N<0>	14D5 26D2
MEM_A_DQS1	MEM_70D	MEM_DQS	MEM A DQS P<1>	14D5 26C4
MEM_A_DQS1	MEM_70D	MEM_DQS	MEM A DQS N<1>	14D5 26C4
MEM_A_DQS2	MEM_70D	MEM_DQS	MEM A DQS P<2>	14D5 26B2
MEM_A_DQS2	MEM_70D	MEM_DQS	MEM A DQS N<2>	14D5 26C2
MEM_A_DQS3	MEM_70D	MEM_DQS	MEM A DQS P<3>	14D5 26C4
MEM_A_DQS3	MEM_70D	MEM_DQS	MEM A DQS N<3>	14D5 26C4
MEM_A_DQS4	MEM_70D	MEM_DQS	MEM A DQS P<4>	14D5 26B7
MEM_A_DQS4	MEM_70D	MEM_DQS	MEM A DQS N<4>	14D5 26B7
MEM_A_DQS5	MEM_70D	MEM_DQS	MEM A DQS P<5>	14D5 26B5
MEM_A_DQS5	MEM_70D	MEM_DQS	MEM A DQS N<5>	14D5 26B5
MEM_A_DQS6	MEM_70D	MEM_DQS	MEM A DQS P<6>	14D5 26B7
MEM_A_DQS6	MEM_70D	MEM_DQS	MEM A DQS N<6>	14D5 26B7
MEM_A_DQS7	MEM_70D	MEM_DQS	MEM A DQS P<7>	14D5 26A5
MEM_A_DQS7	MEM_70D	MEM_DQS	MEM A DQS N<7>	14D5 26A5
MEM_B_CLK	MEM_70D_VDD	MEM_CLK	MEM B CLK P<5..0>	14B5 27C5 27C7
MEM_B_CLK	MEM_70D_VDD	MEM_CLK	MEM B CLK N<5..0>	14B5 27C5 27C7
MEM_B_CMTL	MEM_40S_VDD	MEM_CTRL	MEM B CKE<3..0>	14A1 27D5 27D7
MEM_B_CMTL	MEM_40S_VDD	MEM_CTRL	MEM B CS L<3..0>	14B5 27C5 27C7
MEM_B_CMTL	MEM_40S_VDD	MEM_CTRL	MEM B ODT<3..0>	14B5 27C5
MEM_B_CMD	MEM_40S_VDD	MEM_CMD	MEM B A<14..0>	14B5 14C1 27C5 27C7
MEM_B_CMD	MEM_40S_VDD	MEM_CMD	MEM B BA<2..0>	14C1 27C5 27C7
MEM_B_CMD	MEM_40S_VDD	MEM_CMD	MEM B RAS L	14C1 27C5
MEM_B_CMD	MEM_40S_VDD	MEM_CMD	MEM B CAS L	14C1 27C7
MEM_B_CMD	MEM_40S_VDD	MEM_CMD	MEM B WE L	14C1 27C7
MEM_B_DQ_BYTE0	MEM_40S	MEM_DATA	MEM B DQ<7..0>	14B3 27C2 27C4 27D2 27D4
MEM_B_DQ_BYTE1	MEM_40S	MEM_DATA	MEM B DQ<15..8>	14B3 27C2 27C4
MEM_B_DQ_BYTE2	MEM_40S	MEM_DATA	MEM B DQ<23..16>	14B3 14C3 27C2 27C4
MEM_B_DQ_BYTE3	MEM_40S	MEM_DATA	MEM B DQ<31..24>	14C3 27B2 27B4 27C2 27C4
MEM_B_DQ_BYTE4	MEM_40S	MEM_DATA	MEM B DQ<39..32>	14C3 27B5 27B7 27C5 27C7
MEM_B_DQ_BYTE5	MEM_40S	MEM_DATA	MEM B DQ<47..40>	14C3 14D3 27B5 27B7
MEM_B_DQ_BYTE6	MEM_40S	MEM_DATA	MEM B DQ<55..48>	14D3 27B5 27B7
MEM_B_DQ_BYTE7	MEM_40S	MEM_DATA	MEM B DQ<63..56>	14D3 27A5 27A7 27B5 27B7
MEM_B_DQ_BYTE8	MEM_40S	MEM_DATA	MEM B DM<0>	14A3 27C4
MEM_B_DQ_BYTE1	MEM_40S	MEM_DATA	MEM B DM<1>	14A3 27C2
MEM_B_DQ_BYTE2	MEM_40S	MEM_DATA	MEM B DM<2>	14B3 27C2
MEM_B_DQ_BYTE3	MEM_40S	MEM_DATA	MEM B DM<3>	14B3 27B4
MEM_B_DQ_BYTE4	MEM_40S	MEM_DATA	MEM B DM<4>	14B3 27B5
MEM_B_DQ_BYTE5	MEM_40S	MEM_DATA	MEM B DM<5>	14B3 27B7
MEM_B_DQ_BYTE6	MEM_40S	MEM_DATA	MEM B DM<6>	14B3 27B5
MEM_B_DQ_BYTE7	MEM_40S	MEM_DATA	MEM B DM<7>	14B3 27A7
MEM_B_DQS0	MEM_70D	MEM_DQS	MEM B DQS P<0>	14D3 27C2
MEM_B_DQS0	MEM_70D	MEM_DQS	MEM B DQS N<0>	14D3 27D2
MEM_B_DQS1	MEM_70D	MEM_DQS	MEM B DQS P<1>	14D3 27C4
MEM_B_DQS1	MEM_70D	MEM_DQS	MEM B DQS N<1>	14D3 27C4
MEM_B_DQS2	MEM_70D	MEM_DQS	MEM B DQS P<2>	14D3 27C4
MEM_B_DQS2	MEM_70D	MEM_DQS	MEM B DQS N<2>	14D3 27C4
MEM_B_DQS3	MEM_70D	MEM_DQS	MEM B DQS P<3>	14D3 27B2
MEM_B_DQS3	MEM_70D	MEM_DQS	MEM B DQS N<3>	14D3 27C2
MEM_B_DQS4	MEM_70D	MEM_DQS	MEM B DQS P<4>	14D3 27B7
MEM_B_DQS4	MEM_70D	MEM_DQS	MEM B DQS N<4>	14D3 27B7
MEM_B_DQS5	MEM_70D	MEM_DQS	MEM B DQS P<5>	14D3 27B5
MEM_B_DQS5	MEM_70D	MEM_DQS	MEM B DQS N<5>	14D3 27B5
MEM_B_DQS6	MEM_70D	MEM_DQS	MEM B DQS P<6>	14D3 27B7
MEM_B_DQS6	MEM_70D	MEM_DQS	MEM B DQS N<6>	14D3 27B7
MEM_B_DQS7	MEM_70D	MEM_DQS	MEM B DQS P<7>	14D3 27A5
MEM_B_DQS7	MEM_70D	MEM_DQS	MEM B DQS N<7>	14D3 27A5
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP_VDD	15C6
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP MEM COMP_GND	15C6

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Memory Constraints

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1807</td></tr><tr><td>☐</td><td>PCT_AD</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_AD<23..8></td></tr><tr><td>☐</td><td>PCT_AD24</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_AD<24></td></tr><tr><td>☐</td><td>PCT_AD</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_AD<31..25></td></tr><tr><td>☐</td><td>PCT_AD</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_PAR</td></tr><tr><td>☐</td><td>PCT_C_BE_L</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_C_BE_L<3..0></td></tr><tr><td>☐</td><td>PCT_CNTL</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_TRDY_L</td></tr><tr><td>☐</td><td>PCT_CNTL</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_DEVSEL_L</td></tr><tr><td>☐</td><td>PCT_CNTL</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_PERR_L</td></tr><tr><td>☐</td><td>PCT_CNTL</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_SERR_L</td></tr><tr><td>☐</td><td>PCT_CNTL</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_STOP_L</td></tr><tr><td>☐</td><td>PCT_CNTL</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_TRDY_L</td></tr><tr><td>☐</td><td>PCT_CNTL</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_FRAME_L</td></tr><tr><td>☐</td><td>PCT_REQ0_L</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_REQ0_L 1802 1807</td></tr><tr><td>☐</td><td>PCT_GNT0_L</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_GNT0_L</td></tr><tr><td>☐</td><td>PCT_REQ1_L</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_REQ1_L 1802 1807</td></tr><tr><td>☐</td><td>PCT_GNT1_L</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_GNT1_L</td></tr><tr><td>☐</td><td>PCT_INTW_L</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_INTW_L</td></tr><tr><td>☐</td><td>PCT_INTX_L</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_INTX_L</td></tr><tr><td>☐</td><td>PCT_INTY_L</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_INTY_L</td></tr><tr><td>☐</td><td>PCT_INTZ_L</td><td>PCT_55S</td><td>PCT</td><td></td><td>PCI_INTZ_L</td></tr><tr><td>☐</td><td>MCP_PCT_CLK2</td><td>CLK_PCT_55S</td><td>CLK_PCT</td><td></td><td>PCI_CLK33M_MCP_R 18C5</td></tr><tr><td>☐</td><td></td><td>CLK_PCT_55S</td><td>CLK_PCT</td><td></td><td>PCI_CLK33M_MCP 18C5</td></tr><tr><td>☐</td><td>LPC_AD</td><td>LPC_55S</td><td>LPC</td><td></td><td>LPC_AD<3..0> 18B3 40C8 4203 4205</td></tr><tr><td>☐</td><td>LPC_FRAME_L</td><td>LPC_55S</td><td>LPC</td><td></td><td>LPC_FRAME_L 18C3 40C8 4205</td></tr><tr><td>☐</td><td>LPC_RESET_L</td><td>LPC_55S</td><td>LPC</td><td></td><td>LPC_RESET_L 18C3 24D4</td></tr><tr><td>☐</td><td>MCP_LPC_CLK0</td><td>CLK_LPC_55S</td><td>CLK_LPC</td><td></td><td>LPC_CLK33M_SMC_R 18B3 24B4</td></tr><tr><td>☐</td><td></td><td>CLK_LPC_55S</td><td>CLK_LPC</td><td></td><td>LPC_CLK33M_SMC 24B1 40C8</td></tr><tr><td>☐</td><td></td><td>CLK_LPC_55S</td><td>CLK_LPC</td><td></td><td>LPC_CLK33M_LPCPLUS 24B1 42D3</td></tr><tr><td>☐</td><td>USB_EXTN</td><td>USB_900</td><td>USB</td><td></td><td>USB_EXTN_P 1903 38A8</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_EXTN_N 1903 38A8</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_EXTN_MUXED_P 38C4</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_EXTN_MUXED_N 38C4</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>CONN_USB_EXTN_P 38C3</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>CONN_USB_EXTN_N 38C3</td></tr><tr><td>☐</td><td>USB_CAMERA</td><td>USB_900</td><td>USB</td><td></td><td>USB_CAMERA_P 1903 29B5</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_CAMERA_N 1903 29B5</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_CAMERA_CONN_P 605 29B7</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_CAMERA_CONN_N 605 29B7</td></tr><tr><td>☐</td><td>USB_BT</td><td>USB_900</td><td>USB</td><td></td><td>USB_BT_P 1903 29B5</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_BT_N 19C3 29B5</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>CONN_USB2_BT_P 605 29B7</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>CONN_USB2_BT_N 605 29B7</td></tr><tr><td>☐</td><td>USB_TPAB</td><td>USB_900</td><td>USB</td><td></td><td>USB_TPAB_P 1903 48B8</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_TPAB_N 1903 48B8</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_TPAB_R_P 48B7</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_TPAB_R_N 48B7</td></tr><tr><td>☐</td><td>USB_IR</td><td>USB_900</td><td>USB</td><td></td><td>USB_IR_P 1903 39D7</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_IR_N 1903 39D7</td></tr><tr><td>☐</td><td>USB_EXTB</td><td>USB_900</td><td>USB</td><td></td><td>USB_EXTB_P 19C3 38A4</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_EXTB_N 19C3 38B4</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>CONN_USB_EXTB_P 38B3</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>CONN_USB_EXTB_N 38B3</td></tr><tr><td>☐</td><td>USB_SD</td><td>USB_900</td><td>USB</td><td></td><td>USB_CARDREADER_P 19C3 30C7</td></tr><tr><td>☐</td><td></td><td>USB_900</td><td>USB</td><td></td><td>USB_CARDREADER_N 19C3 30C7</td></tr><tr><td>☐</td><td>MCP_USB_RBIA5</td><td>MCP_USB_RBIA5</td><td></td><td></td><td>MCP_USB_RBIA5_GND 19C4</td></tr><tr><td>☐</td><td>SMBUS_MCP_0_CLK</td><td>SMB_55S</td><td>SMB</td><td></td><td>SMBUS_MCP_0_CLK 12B6 29C3 43D8</td></tr><tr><td>☐</td><td>SMBUS_MCP_0_DATA</td><td>SMB_55S</td><td>SMB</td><td></td><td>SMBUS_MCP_0_DATA 12B6 29C3 43D8</td></tr><tr><td>☐</td><td>SMBUS_MCP_1_CLK</td><td>SMB_55S</td><td>SMB</td><td></td><td>SMBUS_MCP_1_CLK 29C3 43B8</td></tr><tr><td>☐</td><td>SMBUS_MCP_1_DATA</td><td>SMB_55S</td><td>SMB</td><td></td><td>SMBUS_MCP_1_DATA 29C3 43B8</td></tr><tr><td>☐</td><td>HDA_BIT_CLK</td><td>HDA_55S</td><td>HDA</td><td></td><td>HDA_BIT_CLK 28D2 52C7</td></tr><tr><td>☐</td><td></td><td>HDA_55S</td><td>HDA</td><td></td><td>HDA_BIT_CLK_R 28A7 28D4</td></tr><tr><td>☐</td><td>HDA_SYNC</td><td>HDA_55S</td><td>HDA</td><td></td><td>HDA_SYNC 28D2 52C7</td></tr><tr><td>☐</td><td></td><td>HDA_55S</td><td>HDA</td><td></td><td>HDA_SYNC_R 28A7 28D4</td></tr><tr><td>☐</td><td>HDA_RST_L</td><td>HDA_55S</td><td>HDA</td><td></td><td>HDA_RST_R_L 28A7 28D4</td></tr><tr><td>☐</td><td></td><td>HDA_55S</td><td>HDA</td><td></td><td>HDA_RST_L 28D2 52C7</td></tr><tr><td>☐</td><td>HDA_SDIN0</td><td>HDA_55S</td><td>HDA</td><td></td><td>HDA_SDIN0 28D7 52C7</td></tr><tr><td>☐</td><td></td><td>HDA_55S</td><td>HDA</td><td></td><td>HDA_SDIN_CODEC</td></tr><tr><td>☐</td><td>HDA_SDOUT</td><td>HDA_55S</td><td>HDA</td><td></td><td>HDA_SDOUT 28D2 52C7</td></tr><tr><td>☐</td><td></td><td>HDA_55S</td><td>HDA</td><td></td><td>HDA_SDOUT_R 28A7 28D4</td></tr><tr><td>☐</td><td>MCP_HDA_PULLDN_COMP</td><td>MCP_HDA_COMP</td><td></td><td></td><td>MCP_HDA_PULLDN_COMP 12C7</td></tr><tr><td>☐</td><td>MCP_SUS_CLK</td><td>CLK_SLOW_55S</td><td>CLK_SLOW</td><td></td><td>PM_CLK32K_SUSCLK_R 19B3 24B4</td></tr><tr><td>☐</td><td></td><td>CLK_SLOW_55S</td><td>CLK_SLOW</td><td></td><td>PM_CLK32K_SUSCLK 19B1 49C5</td></tr><tr><td>☐</td><td>SPT_CLK</td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_CLK_R 19B3 42A5 42C8</td></tr><tr><td>☐</td><td></td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_CLK 12C5</td></tr><tr><td>☐</td><td></td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_ALT_CLK 12C5 42D3</td></tr><tr><td>☐</td><td>SPT_MOSI</td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_MOSI_R 19B3 42A5 42C7</td></tr><tr><td>☐</td><td></td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_MOSI 51C4</td></tr><tr><td>☐</td><td></td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_ALT_MOSI 12C5 42D5</td></tr><tr><td>☐</td><td>SPT_MISO</td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_MISO 19B3 42A5 42B7</td></tr><tr><td>☐</td><td></td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_MISO_R 51C4</td></tr><tr><td>☐</td><td></td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_ALT_MISO 12B6 42D5</td></tr><tr><td>☐</td><td>SPT_CS0</td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_CS0_R_L 19B3 42B7</td></tr><tr><td>☐</td><td></td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_CS0_L</td></tr><tr><td>☐</td><td></td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_CS1_R_L</td></tr><tr><td>☐</td><td></td><td>SPT_55S</td><td>SPT</td><td></td><td>SPT_CS1_R_L_USE_MLB</td></tr></table>									ELECTRICAL_CONSTRAINT_SET			NET_TYPE						PHYSICAL	SPACING		☐	MCP_DEBUG	PCT_55S	PCT		MCP_DEBUG<7..0> 12C3 1807	☐	PCT_AD	PCT_55S	PCT		PCI_AD<23..8>	☐	PCT_AD24	PCT_55S	PCT		PCI_AD<24>	☐	PCT_AD	PCT_55S	PCT		PCI_AD<31..25>	☐	PCT_AD	PCT_55S	PCT		PCI_PAR	☐	PCT_C_BE_L	PCT_55S	PCT		PCI_C_BE_L<3..0>	☐	PCT_CNTL	PCT_55S	PCT		PCI_TRDY_L	☐	PCT_CNTL	PCT_55S	PCT		PCI_DEVSEL_L	☐	PCT_CNTL	PCT_55S	PCT		PCI_PERR_L	☐	PCT_CNTL	PCT_55S	PCT		PCI_SERR_L	☐	PCT_CNTL	PCT_55S	PCT		PCI_STOP_L	☐	PCT_CNTL	PCT_55S	PCT		PCI_TRDY_L	☐	PCT_CNTL	PCT_55S	PCT		PCI_FRAME_L	☐	PCT_REQ0_L	PCT_55S	PCT		PCI_REQ0_L 1802 1807	☐	PCT_GNT0_L	PCT_55S	PCT		PCI_GNT0_L	☐	PCT_REQ1_L	PCT_55S	PCT		PCI_REQ1_L 1802 1807	☐	PCT_GNT1_L	PCT_55S	PCT		PCI_GNT1_L	☐	PCT_INTW_L	PCT_55S	PCT		PCI_INTW_L	☐	PCT_INTX_L	PCT_55S	PCT		PCI_INTX_L	☐	PCT_INTY_L	PCT_55S	PCT		PCI_INTY_L	☐	PCT_INTZ_L	PCT_55S	PCT		PCI_INTZ_L	☐	MCP_PCT_CLK2	CLK_PCT_55S	CLK_PCT		PCI_CLK33M_MCP_R 18C5	☐		CLK_PCT_55S	CLK_PCT		PCI_CLK33M_MCP 18C5	☐	LPC_AD	LPC_55S	LPC		LPC_AD<3..0> 18B3 40C8 4203 4205	☐	LPC_FRAME_L	LPC_55S	LPC		LPC_FRAME_L 18C3 40C8 4205	☐	LPC_RESET_L	LPC_55S	LPC		LPC_RESET_L 18C3 24D4	☐	MCP_LPC_CLK0	CLK_LPC_55S	CLK_LPC		LPC_CLK33M_SMC_R 18B3 24B4	☐		CLK_LPC_55S	CLK_LPC		LPC_CLK33M_SMC 24B1 40C8	☐		CLK_LPC_55S	CLK_LPC		LPC_CLK33M_LPCPLUS 24B1 42D3	☐	USB_EXTN	USB_900	USB		USB_EXTN_P 1903 38A8	☐		USB_900	USB		USB_EXTN_N 1903 38A8	☐		USB_900	USB		USB_EXTN_MUXED_P 38C4	☐		USB_900	USB		USB_EXTN_MUXED_N 38C4	☐		USB_900	USB		CONN_USB_EXTN_P 38C3	☐		USB_900	USB		CONN_USB_EXTN_N 38C3	☐	USB_CAMERA	USB_900	USB		USB_CAMERA_P 1903 29B5	☐		USB_900	USB		USB_CAMERA_N 1903 29B5	☐		USB_900	USB		USB_CAMERA_CONN_P 605 29B7	☐		USB_900	USB		USB_CAMERA_CONN_N 605 29B7	☐	USB_BT	USB_900	USB		USB_BT_P 1903 29B5	☐		USB_900	USB		USB_BT_N 19C3 29B5	☐		USB_900	USB		CONN_USB2_BT_P 605 29B7	☐		USB_900	USB		CONN_USB2_BT_N 605 29B7	☐	USB_TPAB	USB_900	USB		USB_TPAB_P 1903 48B8	☐		USB_900	USB		USB_TPAB_N 1903 48B8	☐		USB_900	USB		USB_TPAB_R_P 48B7	☐		USB_900	USB		USB_TPAB_R_N 48B7	☐	USB_IR	USB_900	USB		USB_IR_P 1903 39D7	☐		USB_900	USB		USB_IR_N 1903 39D7	☐	USB_EXTB	USB_900	USB		USB_EXTB_P 19C3 38A4	☐		USB_900	USB		USB_EXTB_N 19C3 38B4	☐		USB_900	USB		CONN_USB_EXTB_P 38B3	☐		USB_900	USB		CONN_USB_EXTB_N 38B3	☐	USB_SD	USB_900	USB		USB_CARDREADER_P 19C3 30C7	☐		USB_900	USB		USB_CARDREADER_N 19C3 30C7	☐	MCP_USB_RBIA5	MCP_USB_RBIA5			MCP_USB_RBIA5_GND 19C4	☐	SMBUS_MCP_0_CLK	SMB_55S	SMB		SMBUS_MCP_0_CLK 12B6 29C3 43D8	☐	SMBUS_MCP_0_DATA	SMB_55S	SMB		SMBUS_MCP_0_DATA 12B6 29C3 43D8	☐	SMBUS_MCP_1_CLK	SMB_55S	SMB		SMBUS_MCP_1_CLK 29C3 43B8	☐	SMBUS_MCP_1_DATA	SMB_55S	SMB		SMBUS_MCP_1_DATA 29C3 43B8	☐	HDA_BIT_CLK	HDA_55S	HDA		HDA_BIT_CLK 28D2 52C7	☐		HDA_55S	HDA		HDA_BIT_CLK_R 28A7 28D4	☐	HDA_SYNC	HDA_55S	HDA		HDA_SYNC 28D2 52C7	☐		HDA_55S	HDA		HDA_SYNC_R 28A7 28D4	☐	HDA_RST_L	HDA_55S	HDA		HDA_RST_R_L 28A7 28D4	☐		HDA_55S	HDA		HDA_RST_L 28D2 52C7	☐	HDA_SDIN0	HDA_55S	HDA		HDA_SDIN0 28D7 52C7	☐		HDA_55S	HDA		HDA_SDIN_CODEC	☐	HDA_SDOUT	HDA_55S	HDA		HDA_SDOUT 28D2 52C7	☐		HDA_55S	HDA		HDA_SDOUT_R 28A7 28D4	☐	MCP_HDA_PULLDN_COMP	MCP_HDA_COMP			MCP_HDA_PULLDN_COMP 12C7	☐	MCP_SUS_CLK	CLK_SLOW_55S	CLK_SLOW		PM_CLK32K_SUSCLK_R 19B3 24B4	☐		CLK_SLOW_55S	CLK_SLOW		PM_CLK32K_SUSCLK 19B1 49C5	☐	SPT_CLK	SPT_55S	SPT		SPT_CLK_R 19B3 42A5 42C8	☐		SPT_55S	SPT		SPT_CLK 12C5	☐		SPT_55S	SPT		SPT_ALT_CLK 12C5 42D3	☐	SPT_MOSI	SPT_55S	SPT		SPT_MOSI_R 19B3 42A5 42C7	☐		SPT_55S	SPT		SPT_MOSI 51C4	☐		SPT_55S	SPT		SPT_ALT_MOSI 12C5 42D5	☐	SPT_MISO	SPT_55S	SPT		SPT_MISO 19B3 42A5 42B7	☐		SPT_55S	SPT		SPT_MISO_R 51C4	☐		SPT_55S	SPT		SPT_ALT_MISO 12B6 42D5	☐	SPT_CS0	SPT_55S	SPT		SPT_CS0_R_L 19B3 42B7	☐		SPT_55S	SPT		SPT_CS0_L	☐		SPT_55S	SPT		SPT_CS1_R_L	☐		SPT_55S	SPT		SPT_CS1_R_L_USE_MLB
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